Exact Replica Mechanics and Born-Record Large Deviations in Monitored Quantum Circuits

Amin Abaee[1,*]

[1] *Independent Researcher*

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We develop an exact finite-dimensional theory of the replica mechanics underlying measurement-induced entanglement transitions in monitored quantum circuits, separating three structures that are often conflated: linear outcome-summed replica channels, normalized conditional trajectories, and asymptotic large deviations. For fixed-basis projective measurements we determine the minimal invariant unital algebra of the one-wire feature space and prove that independently Haar-averaged two-site gates on a perfect matching close the n-replica permutation space exactly for every replica number, with an explicit Weingarten bond channel. The resulting one-period transfer operators have real, positive, semisimple nonzero spectra that compress to bond-label space; at $n = 2$ the model is exactly the anisotropic triangular-lattice Ising model of Bao, Choi and Altman, with a closed-form critical point and a closed free-fermion spectrum. Carrying the same computation through $n = 3, 4, 5$ locates the annealed critical points $p_c^{(2)} = 0.2338 < p_c^{(3)} = 0.305(3) < p_c^{(4)} \approx 0.383 < p_c^{(5)} \approx 0.47$ with the Potts-class exponents at $n = 3, 4$, and supports, at the three-size level, the first-order prediction of the $q = n$-Potts identification at $n = 5$: an exact three-size test ($L = 4, 6, 8$) on a symmetry-reduced orbit basis of dimension 3,865 over the 2.1×10^8 bond-label space shows the two-phase gap closing faster than the continuous baseline at every size, with the scaled gap falling below the continuous envelope at the matched sizes and dual effective closing rates ordered consistently with the continuous, marginal and first-order behaviours at matched sizes—a directional trend, with the registered conditional that completes it: if the closing is genuinely exponential, a fourth size ($L = 10$) will show the effective closing rate stabilizing at a positive interface tension, and if the asymptotic is a slowly decaying power law, that rate will continue to decay—and at local dimension $d = 3$ the same three-size test keeps the gap level below the $d = 2$ values at every size. We further prove a no-freezing theorem for finite-reachable record processes, establish the worst-case complexity of conditional-trajectory decision, derive the exact replica interpolation identity with its measurable self-averaging criterion, and give the first numerical quenched record scaled cumulant generating functions for Haar circuits, which exhibit genuine record multifractality; the Clifford record-count disorder statistics are closed exactly through a verified conjugation 3-design.

I. INTRODUCTION, SCOPE, AND CONVENTIONS

Monitored random quantum circuits exhibit measurement-induced entanglement transitions: in the thermodynamic limit the entanglement of a typical Born trajectory scales volumetrically or logarithmically in the system size depending on the measurement rate [1–8]. The theoretical descriptions of these transitions—replica field theories of the annealed and quenched kinds, transfer-matrix contractions, and large-deviation functions of the measurement record—all rest on replica objects whose finite-dimensional structure has largely been treated perturbatively or accessed only numerically. Exact statements about the replica transfer operators themselves, their spectra, ranks, symmetries, and the critical points they encode, have been scarce; this paper supplies them.

The first contribution is an exact finite-dimensional mechanics of the replica channels of a canonical monitored-circuit period. For fixed-basis projective measurements we determine the minimal invariant unital algebra of the one-wire entanglement-feature space and quantify its leakage; a subsequent layer of independently Haar-averaged two-site gates on a perfect matching closes the leaked directions exactly, for every replica number, through an explicit Weingarten bond channel. The period map is of the form $\mathsf{A}\mathsf{A}^\dagger$ in the Hilbert–Schmidt inner product, so its nonzero spectrum is real, positive and semisimple and compresses to bond-label space; at $n = 2$ the compressed operator is the row-to-row transfer matrix of the anisotropic triangular-lattice Ising model of Ref. [8], so the annealed two-replica problem is exactly solvable and its critical point is known in closed form.

The second contribution is the annealed replica ladder built on this foundation. Exact symmetry-resolved computations locate the annealed critical points at $n = 3, 4, 5$ and identify their universality classes: Potts exponents at $n = 3$ (continuous) and marginal $q = 4$ behaviour at $n = 4$, while at $n = 5$—where the $q > 4$ criterion demands a first-order transition—an exact three-size test ($L = 4, 6, 8$) on a symmetry-reduced orbit basis of dimension 3,865 over 2.1×10^8 bond labels provides the first three-size evidence in the first-order direction on the gap closing in this family, with a registered prediction for its completion. The third contribution separates the Born-weighted objects from the symmetric unnormalized ones—an information-theoretic no-go with a bivariate

* ORCID: 0000-0002-0019-1842; Email: abaee@ut.ac.ir

recovery, and the worst-case complexity of conditional-trajectory decision—and develops the large deviations of the measurement record: a no-freezing theorem, the exact replica interpolation identity with its measurable self-averaging criterion, the first numerical quenched record SCGFs for Haar circuits, and an exact closure of the Clifford record-count disorder statistics through a verified conjugation 3-design.

A delimitation is stated at the outset. The exactly solved and exactly computed objects of this paper are annealed averages; none of them determines by itself the quenched, Born-weighted transition that experiments and trajectory simulations target. The value of the annealed ladder is precisely that it is exactly computable at every replica number, providing a controlled calibration target for replica field theories of the quenched problem, with the interpolation identity of Sec. IV making the connection quantitative; the detailed delimitations are collected at the end of this section.

This work establishes exact finite-dimensional constraints on the replica mechanics underlying these transitions. The proved and computed statements serve as controlled inputs to later analytical and numerical studies: the fixed-basis replica leakage and its exact post-Haar closure, the two-replica transfer operator with its rank, its Ising structure and its closed-form annealed critical point, the spectral structure for every replica number and exact rank certificates for three replicas, Born-normalization identities, the worst-case complexity of conditional trajectories, and the large-deviation definitions for the Born record, together with the computed annealed replica chain—$p_c^{(2)} = 0.233810 < p_c^{(3)} = 0.305(3) < p_c^{(4)} \approx 0.383 < p_c^{(5)} \approx 0.47$–$0.48$ with the directional three-size evidence for the first-order prediction at $n = 5$, the no-freezing theorem for finite-reachable processes, and the first sequential-Monte-Carlo measurement of the quenched record SCGF. None of the exact results determines the quenched, Born-weighted transition: the critical point $p_c^{(2)}(d)$ of Sec. II E is the known annealed two-replica Ising point [8, 9], and no logarithmic conformal field theory or replica-symmetry-breaking phase is derived. Appendix B reports a separate, purely computational finite-size-scaling benchmark of the quenched transition for the Clifford realization of the same period (system sizes $16 \le L \le 512$, crossings, data collapse, bootstrap errors): $p_c = 0.1597(8)$, $\nu = 1.24(7)$, and critical half-chain entropy $S_{L/2} = \alpha \ln L + b$ with $\alpha = 1.55(7)$ bits, i.e. 0.54(2) nats per cut; the quenched point lies 0.074 below the exact annealed point $p_c^{(2)}(2) = 0.2338$. These numbers are consistent with the literature [10, 11, 14]; they are results of simulation, not theorems, and are quoted with systematic rather than statistical uncertainties. The logical spine of the paper: Secs. II D–II E establish the exact two-replica mechanics and its closed-form solution; Secs. II F–II H carry the spectral computation to $n = 3, 4, 5$ and test the $q = n$-Potts prediction; Sec. III separates the Born-weighted objects from the symmetric ones and summarizes the worst-case complexity statements, whose definitions and proofs are collected in Appendix C; Sec. IV develops the record large deviations, proves the no-freezing theorem, measures the first quenched record SCGFs for Haar circuits, and closes the Clifford record-count disorder statistics with exact anchors, self-averaging, and the annealed–quenched gap, and closes $\Lambda(2)$ exactly through the verified Clifford conjugation 3-design; the $n = 4$ chain is re-read through the marginal $q = 4$ log corrections (the amplitude ratio converging to the corrected target 1/4); Sec. V delimits what remains conditional; Sec. VII states the standing metaphysics, locates the readings of quantum mechanics at the level of appearance, classifies the paper's results by their relation to necessity, and draws no structural parallel between the display and the Absolute; Sec. VIII concludes.

A local wire has Hilbert space $H \cong \mathbb{C}^d$, $d \ge 2$. The two-replica one-wire space is $\mathcal{K} = H^{\otimes 2}$. Superoperators act on column coordinate vectors, and products act from right to left. We write $I = I_{\mathcal{K}}$, S for the replica swap, $P_m = |m\rangle\langle m|$, and

$$C = \sum_{m=1}^{d} P_m^{\otimes 2} = \sum_{m=1}^{d} |mm\rangle\langle mm|. \tag{1}$$

The fixed-basis outcome-correlated channel relevant to a linear two-replica outcome sum is

$$\mathcal{D}(X) = \sum_{m=1}^{d} P_m^{\otimes 2} X P_m^{\otimes 2}. \tag{2}$$

The same outcome label appears in both replicas. Equation (2) is not independent-replica dephasing $\Delta^{\otimes 2}$, which contains two independent outcome labels. It is also not a normalized conditional update, which is nonlinear in the state. Throughout, $p \in [0, 1]$ is the per-site measurement probability, n the replica number, L the (even) number of sites, and

$$\vartheta = d - (d - 1)p \in [1, d] \tag{3}$$

the combination through which p enters every two-replica formula. The letter q is reserved for the multifractal moment order of Sec. IV; the sans-serif K_R denotes a Kraus operator, $K_{\mathrm{o/e}}$ the matching factors of Sec. II D, $K_{d,h}$ Ising couplings, and $K_\omega(x, dy)$ a Markov kernel.

II. EXACT TWO-REPLICA LATTICE MECHANICS

The coordinates of a replicated operator in the basis $\{I, S\}^{\otimes L}$ are the "entanglement features" of Ref. [15]; we use that name for the span $\mathrm{span}\{I, S\}^{\otimes L}$. Relative to that reference, the present analysis supplies the two-sided statement the feature space requires: Theorem 1

proves that the span is not closed under fixed-basis projective measurement—the leakage of Eq. (8)—and Theorem 2, with its general-n form (Eq. (14)), proves that a subsequent layer of independently Haar-averaged two-site gates on a perfect matching closes the leaked directions exactly, for every replica number.

A. Fixed-basis leakage and its minimal algebra

Theorem 1 (Fixed-basis algebra and failure of local feature closure)**.** *The map $\mathcal{D}$ is the Hilbert–Schmidt orthogonal projector onto*

$$\mathfrak{A}_d = \mathrm{span}\{E_m = |mm\rangle\langle mm| : 1 \le m \le d\}. \tag{4}$$

Thus $\mathrm{Ran}\,\mathcal{D} = \mathrm{Fix}\,\mathcal{D} = \mathfrak{A}_d$*, an abelian d-dimensional $*$-algebra with unit C on its support. Moreover,*

$$\mathcal{D}(I) = \mathcal{D}(S) = \mathcal{D}(C) = C. \tag{5}$$

For $d \ge 2$, $\mathrm{span}\{I, S\}$ is not invariant. The smallest unital $*$-subalgebra containing I, S and invariant under $\mathcal{D}$ is

$$\mathfrak{B}_d = \mathrm{span}\{I, S, C\}. \tag{6}$$

Proof. Since E_m is rank one, $\mathcal{D}(X) = \sum_m \langle mm|X|mm\rangle E_m$. The E_m are pairwise Hilbert–Schmidt orthonormal idempotents, proving the projector, range, fixed-point, and algebra statements. Since $E_m S = S E_m = E_m$, Eq. (5) follows. The Gram matrix of (I, S) is

$$G_{IS} = \begin{pmatrix} d^2 & d \\ d & d^2 \end{pmatrix}, \qquad G_{IS}^{-1} = \frac{1}{d(d^2-1)} \begin{pmatrix} d & -1 \\ -1 & d \end{pmatrix}. \tag{7}$$

It gives

$$\Pi_{IS} C = \frac{I+S}{d+1}, \qquad \|C - \Pi_{IS} C\|_{\mathrm{HS}}^2 = \frac{d(d-1)}{d+1} > 0. \tag{8}$$

Hence the often-used equality $C = (I+S)/(d+1)$ is false; the right-hand side is only the orthogonal projection. Any invariant algebra containing I, S contains $\mathcal{D}(I) = C$. Conversely $S^2 = I$, $C^2 = C$, and $SC = CS = C$, so Eq. (6) is a unital invariant $*$-algebra and is minimal. □

For computations, the ordered nonorthogonal basis (I, S, C) has

$$G_{ISC} = \begin{pmatrix} d^2 & d & d \\ d & d^2 & d \\ d & d & d \end{pmatrix}, \quad G_{ISC}^{-1} = \tfrac{1}{d(d-1)} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & d+1 \end{pmatrix}. \tag{9}$$

For $\mathcal{M}_p = (1-p)\,\mathrm{id} + p\mathcal{D}$, column coordinates give

$$[\mathcal{D}]_{ISC} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad [\mathcal{M}_p]_{ISC} = \begin{pmatrix} 1-p & 0 & 0 \\ 0 & 1-p & 0 \\ p & p & 1 \end{pmatrix}. \tag{10}$$

For arbitrary X, the orthogonal feature projection is

$$\Pi_{IS} X = \alpha I + \beta S, \quad \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \tfrac{1}{d(d^2-1)} \begin{pmatrix} d & -1 \\ -1 & d \end{pmatrix} \begin{pmatrix} \mathrm{Tr}\,X \\ \mathrm{Tr}(SX) \end{pmatrix}. \tag{11}$$

Applying Eq. (11) after each measurement is an approximation, not an exact representation of Eq. (2). Its size as an operator is exact: on $\mathrm{span}\{I, S\}$ with the Hilbert–Schmidt norm, the one-site defect $\Delta_p = \mathcal{M}_p - \Pi_{IS}\mathcal{M}_p = p(\mathrm{id} - \Pi_{IS})\mathcal{D}$ has operator norm

$$\|\Delta_p\|_{\mathrm{span}\{I,S\} \to \mathcal{K}\otimes\mathcal{K}} = p\,\frac{\sqrt{2(d-1)}}{d+1}, \tag{12}$$

because $\Delta_p(\alpha I + \beta S) = p(\alpha+\beta)(C - \Pi_{IS} C)$ and $\sup |\alpha+\beta|/\|\alpha I + \beta S\|_{\mathrm{HS}} = \sqrt{2/(d(d+1))}$; the number $p\sqrt{d(d-1)/(d+1)}$ sometimes quoted is $\|\Delta_p(I)\|_{\mathrm{HS}}$, not a norm bound.

The statements above extend to every replica number. Let $n \ge 2$, let U_π ($\pi \in S_n$) permute the replicas of $H^{\otimes n}$, $\mathcal{G}_n = \mathrm{span}\{U_\pi\}$, $C_n = \sum_m P_m^{\otimes n}$, and $\mathcal{D}_n(X) = \sum_m P_m^{\otimes n} X P_m^{\otimes n}$.

Proposition 1 (General replica number)**.** *(i) $\mathcal{D}_n(U_\pi) = C_n$ for every π, and the smallest unital $*$-algebra containing $\mathcal{G}_n$ and invariant under $\mathcal{D}_n$ is $\mathcal{G}_n + \mathbb{C}C_n$. (ii) The Hilbert–Schmidt projection of C_n onto $\mathcal{G}_n$ and the leakage norm are*

$$\Pi_n C_n = \frac{\sum_{\pi\in S_n} U_\pi}{(d+1)(d+2)\cdots(d+n-1)},$$
$$\|C_n - \Pi_n C_n\|_{\mathrm{HS}}^2 = d\Big[1 - \frac{n!}{(d+1)\cdots(d+n-1)}\Big], \tag{13}$$

valid also for $d < n$, where the U_π are linearly dependent. (iii) For a gate bond, the n-fold Haar twirl $\mathcal{E}_{ij}^{(n)}$ is the Hilbert–Schmidt projector onto $\mathrm{span}\{U_\sigma \otimes U_\sigma : \sigma \in S_n\}$; hence $\mathcal{G}_n^{\otimes L}$ is invariant from one post-gate strobe to the next for every n, with bond coordinates

$$W_{p,n}(\pi \mid \mu, \nu) = \sum_{\sigma\in S_n} \mathrm{Wg}_{d^2}(\pi^{-1}\sigma)\left[(1-p)d^{c(\sigma^{-1}\mu)} + pd\right] \times \left[(1-p)d^{c(\sigma^{-1}\nu)} + pd\right], \tag{14}$$

where $c(\cdot)$ counts cycles, Wg_D is the Weingarten function of $U(D)$ [16, 17] (the inverse of the Gram matrix $D^{c(\sigma^{-1}\tau)}$ for $D \ge n$, its pseudo-inverse otherwise), and μ, ν, π label the input and output permutation operators on the two wires.

Proof. (i) $U_\pi |m\cdots m\rangle = |m\cdots m\rangle$ gives $\mathcal{D}_n(U_\pi) = \sum_m \langle m\cdots m|U_\pi|m\cdots m\rangle P_m^{\otimes n} = C_n$; also $U_\pi C_n = C_n U_\pi = C_n$ and $C_n^2 = C_n$, so $\mathcal{G}_n + \mathbb{C}C_n$ is a unital $*$-algebra invariant under $\mathcal{D}_n$, and any invariant algebra containing I contains $\mathcal{D}_n(I) = C_n$. (ii) By Schur–Weyl duality the orthogonal projector onto the commutant $\mathcal{G}_n$ is the Haar twirl $X \mapsto \int dU\, U^{\otimes n} X U^{\dagger\otimes n}$. Applied to $P_m^{\otimes n}$ it gives $\int d\psi\, (|\psi\rangle\langle\psi|)^{\otimes n} = P_{\mathrm{sym}}/\binom{d+n-1}{n}$

with $P_{\rm sym}=\frac{1}{n!}\sum_\pi U_\pi$; summing over m gives the first formula. Since Π_n is an orthogonal projection, $\|C_n-\Pi_nC_n\|^2=\|C_n\|^2-\langle C_n,\Pi_nC_n\rangle=d-\sum_\pi \mathrm{Tr}(C_nU_\pi)/[(d+1)\cdots(d+n-1)]$ and $\mathrm{Tr}(C_nU_\pi)=d$. (iii) Under the identification $(H_i\otimes H_j)^{\otimes n}\cong(\mathbb{C}^{d^2})^{\otimes n}$, Schur–Weyl duality for $U(d^2)$ says that the commutant of $\{U^{\otimes n}:U\in U(d^2)\}$ is spanned by the replica permutations of $\mathbb{C}^{d^2}$, which are the operators $U_\sigma\otimes U_\sigma$, $\sigma\in S_n$. The twirl $\mathcal{E}_{ij}^{(n)}$ is an average of Hilbert–Schmidt-unitary conjugations, hence self-adjoint, and it is idempotent by the invariance of Haar measure; its fixed points are exactly the operators commuting with every $U^{\otimes n}$. It is therefore the Hilbert–Schmidt orthogonal projector onto $\mathcal{V}_n^{\rm b}=\mathrm{span}\{U_\sigma\otimes U_\sigma\}\subset\mathcal{G}_n\otimes\mathcal{G}_n$. Closure: start from $X\in\mathcal{G}_n^{\otimes L}$; the site layer maps it into $\mathrm{End}(H^{\otimes n})^{\otimes L}$, and the gate layer applies $\mathcal{E}_{ij}^{(n)}$ on every bond of a perfect matching, so each bond factor of the output lies in $\mathcal{V}_n^{\rm b}$; since every site belongs to exactly one bond, the output lies in $\mathcal{G}_n^{\otimes L}$. Coordinates: the orthogonal projection of a bond operator X onto $\mathcal{V}_n^{\rm b}$ is $\sum_\pi y_\pi U_\pi\otimes U_\pi$ with $\mathsf{G}y=t$, where $\mathsf{G}(\pi,\sigma)=\mathrm{Tr}[(U_\pi\otimes U_\pi)^\dagger(U_\sigma\otimes U_\sigma)]=(d^2)^{c(\pi^{-1}\sigma)}$ is the Gram matrix and $t_\sigma=\mathrm{Tr}[(U_\sigma\otimes U_\sigma)^\dagger X]$; for $d^2\ge n$, G is invertible with inverse $\mathrm{Wg}_{d^2}(\pi^{-1}\sigma)$ [16, 17], and otherwise t lies in the range of G and $y=\mathsf{G}^+t$ is a solution. For $X=\mathcal{M}_p^{(n)}(U_\mu)\otimes\mathcal{M}_p^{(n)}(U_\nu)$ the trace factors over the two wires, $t_\sigma=\mathrm{Tr}[U_\sigma^\dagger\mathcal{M}_p^{(n)}(U_\mu)]\ \mathrm{Tr}[U_\sigma^\dagger\mathcal{M}_p^{(n)}(U_\nu)]$, and by (i)

$$\begin{aligned}\mathrm{Tr}[U_\sigma^\dagger\mathcal{M}_p^{(n)}(U_\mu)]&=(1-p)\,\mathrm{Tr}\,U_{\sigma^{-1}\mu}+p\,\mathrm{Tr}(U_\sigma^\dagger C_n)\\&=(1-p)d^{c(\sigma^{-1}\mu)}+pd,\end{aligned}$$

because $\mathrm{Tr}\,U_\pi=d^{c(\pi)}$ and $\mathrm{Tr}(U_\sigma^\dagger C_n)=\sum_m\langle m\cdots m|U_{\sigma^{-1}}|m\cdots m\rangle=d$. Substituting gives Eq. (14). □

For $n=3$, Eq. (13) gives $\|C_3-\Pi_3C_3\|^2=d(d-1)(d+4)/[(d+1)(d+2)]$; for $n=2$ it reproduces Eq. (8) and Eq. (14) reproduces Theorem 2 below. The n-replica transfer operator acts on $\mathcal{G}_n^{\otimes L}$, of dimension $(\dim\mathcal{G}_n)^L$ ($=n!^L$ for $d\ge n$; $\dim\mathcal{G}_3=5$ and $\dim\mathcal{G}_4=14$ at $d=2$); its general structure—real positive semisimple nonzero spectrum, strict contraction, symmetry, and compression—is established in Theorem 5 below, after the $n=2$ case has been worked out in full.

B. Exact post-gate stroboscopic closure

Let a Haar gate $U\in U(d^2)$ act on neighboring physical wires i,j and define its two-replica twirl

$$\mathcal{E}_{ij}(X)=\int dU\,U^{\otimes2}X(U^\dagger)^{\otimes2}. \tag{15}$$

Here the replica ordering is understood consistently so that the bond identity and global replica swap are II and SS.

Theorem 2 (Exact gate-to-gate feature dynamics)**.** *Suppose that (i) each site is independently subjected to $\mathcal{M}_p$, (ii) the next layer consists of independent Haar gates on a disjoint perfect matching, and (iii) the replicated operator is sampled immediately after the Haar average of that gate layer. Then $\{I,S\}^{\otimes L}$ is invariant from one post-gate strobe to the next, although it is not invariant at the intermediate post-measurement time.*

For one gate bond define

$$\mathcal{B}_p=\mathcal{E}_{ij}\circ(\mathcal{M}_{p,i}\otimes\mathcal{M}_{p,j}). \tag{16}$$

In the ordered bond feature basis (II,IS,SI,SS),

$$[\mathcal{B}_p]=\begin{pmatrix}a&b&b&c\\0&0&0&0\\0&0&0&0\\c&b&b&a\end{pmatrix}, \tag{17}$$

where, with $\vartheta=d-(d-1)p$,

$$a=\frac{d^2\vartheta^2-1}{d^4-1},\qquad b=\frac{\vartheta}{d^2+1},\qquad c=\frac{d^2-\vartheta^2}{d^4-1}. \tag{18}$$

At $p=0$, $a=1,c=0,b=d/(d^2+1)$; at $p=1$, every column is $(II+SS)/(d^2+1)$. For $0<p<1$ all three entries are positive, and

$$a-c=\frac{\vartheta^2-1}{d^2-1},\qquad a+c=\frac{\vartheta^2+1}{d^2+1},\qquad \frac{b}{a+c}=\frac{\vartheta}{\vartheta^2+1}. \tag{19}$$

Proof. By the two-copy Schur–Weyl commutant, or equivalently the second Haar moment [16], $\mathcal{E}_{ij}$ is the Hilbert–Schmidt orthogonal projector onto $\mathrm{span}\{II,SS\}$. Hence, with the bond dimension d^2,

$$\mathcal{E}_{ij}(X)=\alpha_X II+\beta_X SS, \tag{20}$$

with

$$\begin{pmatrix}\alpha_X\\\beta_X\end{pmatrix}=\frac{1}{d^2(d^4-1)}\begin{pmatrix}d^2&-1\\-1&d^2\end{pmatrix}\begin{pmatrix}\mathrm{Tr}\,X\\\mathrm{Tr}(SSX)\end{pmatrix}. \tag{21}$$

For $Y_A=\mathcal{M}_p(A)$, Eq. (5) yields

	$\mathrm{Tr}\,Y_A$	$\mathrm{Tr}(SY_A)$
$A=I$	$d\vartheta$	d
$A=S$	d	$d\vartheta$

. (22)

For $X=Y_A\otimes Y_B$, both traces in Eq. (21) factor. Substitution gives Eqs. (17) and (18). The image on each bond lies in $\mathrm{span}\{II,SS\}$, and a perfect matching covers every site. Tensoring the bond channels therefore maps the global feature space into itself. Alternating the two matchings preserves the assertion at each post-gate strobe. □

The entries (a,b,c) are the $n=2$ Weingarten data of the replica statistical models of Refs. [7, 8], here organised as a strobed transfer matrix; Sec. II E makes the identification with the Ising model of Ref. [8] exact.

For a periodic even chain, let

$$\mathbb{B}_{\rm odd}(p) = \bigotimes_{(ij)\in\mathcal{G}_{\rm odd}} \mathcal{B}_p^{(ij)}, \qquad \mathbb{B}_{\rm even}(p) = \bigotimes_{(ij)\in\mathcal{G}_{\rm even}} \mathcal{B}_p^{(ij)}, \tag{23}$$

with $\mathcal{G}_{\rm odd} = \{(1,2),(3,4),\dots,(L-1,L)\}$ and $\mathcal{G}_{\rm even} = \{(2,3),\dots,(L,1)\}$. If a period begins just after an even gate layer, the exact column update is

$$v_{t+1} = \mathbb{T}_{\rm period}(p)v_t, \qquad \mathbb{T}_{\rm period}(p) = \mathbb{B}_{\rm even}(p)\mathbb{B}_{\rm odd}(p). \tag{24}$$

Starting immediately after an even gate layer, the corresponding chronological sequence is "measurements, odd gates, measurements, even gates." Equation (24) is therefore a gate-to-gate stroboscopic map; at either intermediate post-measurement time the replicated operator generally belongs to $\mathfrak{B}_d^{\otimes L}$, not the feature space.

An enlarged-basis implementation need not store a 9×9 gate table. In the order (I,S,C) define $\mathtt{t} = (d^2,d,d)$ and $\mathtt{s} = (d,d^2,d)$. For $A,B\in\{I,S,C\}$,

$$\mathcal{E}_{ij}(A\otimes B) = \alpha_{AB} II + \beta_{AB} SS, \tag{25}$$

with

$$\alpha_{AB} = \frac{d^2\mathtt{t}_A\mathtt{t}_B - \mathtt{s}_A\mathtt{s}_B}{d^2(d^4-1)}, \qquad \beta_{AB} = \frac{-\mathtt{t}_A\mathtt{t}_B + d^2\mathtt{s}_A\mathtt{s}_B}{d^2(d^4-1)}. \tag{26}$$

Equations (10) and (26) provide a full-space check of Eq. (17).

Remark 1 (Initial and terminal vectors; record-collision interpretation). For a computational-basis product initial state, or for any pure product state if the period begins with a gate layer, the post-even-gate strobe vector is

$$v_0 = \bigotimes_{(ij)\in\mathcal{G}_{\rm even}} \frac{II+SS}{d^2(d^2+1)}, \tag{27}$$

because a pure product state has unit trace and unit purity on each bond; for a non-basis product state preceded by a measurement layer the coefficients differ. Let $\langle{\rm tr}| = (d^2,d)^{\otimes L}$ be the trace functional in the (I,S) basis and $\langle{\rm tr}_A|$ the functional with (d,d^2) on the sites of a region A and (d^2,d) elsewhere. Since $v_t = \mathbb{E}_U\sum_R \tilde\rho_R^{\otimes 2}$ with unnormalized conditional states $\tilde\rho_R$, ${\rm Tr}\,\tilde\rho_R = p_R$,

$$\begin{aligned}\bar Z_2(t) &= \mathbb{E}_U\sum_R p_R(t)^2 = \langle{\rm tr}|\mathbb{T}_{\rm period}^t v_0\rangle,\\ \bar P_{2,A}(t) &= \mathbb{E}_U\sum_R p_R(t)^2\,{\rm Tr}\,\rho_{A,R}^2 = \langle{\rm tr}_A|\mathbb{T}_{\rm period}^t v_0\rangle.\end{aligned} \tag{28}$$

$\bar Z_2(t)$ is the probability that two independent runs of the same circuit produce the same record. The leading eigenvalue λ_1 of $\mathbb{T}_{\rm period}$ is therefore the decay rate of the record-collision probability, and $\lambda_1 < 1$ for $p > 0$ expresses the growth of the record entropy, not an entanglement property; equivalently $\log\lambda_1$ is the annealed record scaled-cumulant generating function of Sec. IV at $k=-1$ ($q=2$). At $p=0$, $\lambda_1 = 1$ is doubly degenerate (all-I and all-S are both fixed).

C. Proof-to-computation contract

Let Π_{IS} be the one-wire Hilbert–Schmidt projector in Eq. (11). On the ordered feature basis (I,S), the projected one-site map $\widehat{\mathcal{M}}_p = \Pi_{IS}\mathcal{M}_p$ has the symmetric coordinate matrix

$$[\widehat{\mathcal{M}}_p]_{IS} = (1-p)\begin{pmatrix}1&0\\0&1\end{pmatrix} + \frac{p}{d+1}\begin{pmatrix}1&1\\1&1\end{pmatrix}. \tag{29}$$

This projection is exact in one restricted composition. Indeed, $\mathcal{E}_{ij}(\Pi_{IS}\otimes\Pi_{IS}) = \mathcal{E}_{ij}$ because ${\rm Ran}\,\mathcal{E}_{ij} = {\rm span}\{II, SS\}$ is contained in ${\rm Ran}(\Pi_{IS}\otimes\Pi_{IS})$. Consequently

$$\mathcal{E}_{ij}(\mathcal{M}_{p,i}\otimes\mathcal{M}_{p,j}) = \mathcal{E}_{ij}(\widehat{\mathcal{M}}_{p,i}\otimes\widehat{\mathcal{M}}_{p,j}) \tag{30}$$

on feature-space inputs. The projected map is therefore exact when the complete site layer is immediately followed by the independently Haar-averaged gate layer on a perfect matching, and it is not an exact representation at the intermediate post-measurement time, after a partial matching, or without the subsequent Haar average. Two objects are excluded by name below. An *exposed projected measurement layer* is an application of $\widehat{\mathcal{M}}_p^{\otimes L}$ whose output is read out, or acted on by anything other than the complete Haar-averaged matching, before the next gate average (for instance a terminal post-measurement strobe); its error per site is Eq. (12). A *triangular one-site surrogate* is any one-site (I,S) update whose coordinate matrix is triangular, such as the heuristic "measurement fixes I and maps $S\mapsto(1-p)S + (p/d)I$"; no such matrix equals Eq. (10) or its symmetric projection Eq. (29).

Definition 1 (Authoritative spectral operator)**.** *For even L, periodic boundary conditions, column basis $\{I,S\}^{\otimes L}$ in lexicographic bit order, and a post-even-gate initial strobe, the authoritative operator is Eq.* (24)*, whose bond action is exactly Eq.* (17)*. It may be implemented either directly from that bond action or by the symmetric projected factorization in Eqs.* (29)–(30)*, provided every projected site layer is immediately followed by the complete independently Haar-averaged gate matching. No triangular one-site surrogate or exposed projected measurement layer is part of this definition.*

The definition fixes basis, boundary condition, multiplication order, and observation time. A matrix-free implementation is accepted as an implementation of $\mathbb{T}_{\rm period}$ only after the following finite tests.

Proposition 2 (Necessary implementation checks)**.** *An exact-arithmetic or tolerance-controlled implementation of Definition 1 must satisfy:*

1. *every local column agrees with Eqs.* (17)–(18)*;*
2. *a direct bond implementation agrees with the symmetric projected factorization of Eq.* (30)*;*

3. *the feature-space result agrees, for random input vectors at several small L, with the enlarged $(I, S, C)^{\otimes L}$ construction from Eqs.* (10) *and* (26)*;*

4. *the matrix-free action and its transpose agree with an explicitly assembled dense matrix and its transpose for small L;*

5. *$p = 0$ and $p = 1$ reproduce the limiting columns stated in Theorem 2, and the rank and spectral structure agree with Theorem 4 of Sec. II D;*

6. *every reported eigenpair (λ, r) has the scale-free right residual*

$$\eta_R = \frac{\|\mathbb{T}r - \lambda r\|_2}{\|\mathbb{T}r\|_2 + |\lambda|\,\|r\|_2} \tag{31}$$

below a declared tolerance (with $0/0$ defined as zero only for the exact zero pair). For a non-normal spectral analysis, the analogous left residual η_L, the biorthogonal condition number $\kappa_\lambda = \|\ell\|_2\|r\|_2/|\ell^\dagger r|$ [18], and the matching convention are also reported. A small residual certifies a returned pair, not its claimed global modulus rank; that rank requires a full dense spectrum or a separately validated enclosure/selection argument.

Proof. Item 1: by Definition 1 the bond action of the authoritative operator is Eq. (17) with the entries (18); an implementation whose local columns differ implements a different operator, and the comparison of four columns is a finite test. Item 2: Eq. (30) is an identity of maps on feature-space inputs, so a direct bond implementation and the symmetric projected factorization must agree on every input; a discrepancy identifies an exposed projected layer or a triangular surrogate. Item 3: the enlarged construction represents the site channel exactly, by Eq. (10), inside the invariant algebra $\mathfrak{B}_d^{\otimes L}$ of Theorem 1, and applies the exact twirl table (26); it never uses the bond matrix (17), and on feature-space inputs its output equals $\mathbb{T}_{\text{period}}v$ by Theorem 2. Agreement therefore checks the derivation of Eq. (17) as well as the code. Item 4: the columns of the dense matrix are the images of the basis vectors under the matrix-free action, and an independent dense assembly from Kronecker products of Eq. (17) must reproduce them; the transpose is a separate code path and is needed for the left eigenvectors of item 6. Item 5: the limiting columns are the $p = 0$ and $p = 1$ statements of Theorem 2; the rank $3 \cdot 2^{L/2-2}$, the reality and non-negativity of the spectrum, the simplicity of λ_1 and the multiplicity $2^L - 3\cdot 2^{L/2-2}$ of the zero eigenvalue are Theorem 4, each a finite test with a known answer. Item 6: for a computed pair (λ, r) the smallest perturbation $\delta\mathbb{T}$ (in the spectral norm) with $(\mathbb{T}+\delta\mathbb{T})r = \lambda r$ exactly has norm $\|\mathbb{T}r - \lambda r\|_2/\|r\|_2$, attained by $\delta\mathbb{T} = -(\mathbb{T}r - \lambda r)r^\dagger/\|r\|_2^2$; η_R is this backward error in scale-free form. For a simple eigenvalue with left and right eigenvectors ℓ, r the first-order change under a perturbation is $\delta\lambda = \ell^\dagger\,\delta\mathbb{T}\,r/\ell^\dagger r$, so $|\delta\lambda| \le \kappa_\lambda\|\delta\mathbb{T}\|_2$ [18]; the left residual certifies ℓ in the same way as η_R certifies r. A backward error bounds the distance of a returned pair from an exact pair of a nearby matrix and says nothing about eigenvalues that were not returned, which is why the global modulus rank requires the full spectrum or a separate enclosure. Consequently, failure of any item prevents identification of the numerical object with the proved operator or of a numerical vector with its claimed eigenpair. □

Proposition 3 (Approximate designs)**.** *Let every gate be drawn independently from a distribution ν on $U(d^2)$ whose two-replica twirl satisfies $\|\mathcal{E}_\nu - \mathcal{E}_{ij}\|_\diamond \le \varepsilon$ (ε-approximate unitary 2-design in diamond norm [19]), and let $\mathbb{T}_\nu$ be the resulting one-period superoperator on two-replica operators. Then*

$$\|\mathbb{T}_\nu - \mathbb{T}_{\text{period}}\|_\diamond \le L\varepsilon, \tag{32}$$

with $(L/2)\varepsilon$ per matching layer and no higher-order term. For $d = 2$ and ν uniform on the two-qubit Clifford group, $\varepsilon = 0$ and $\mathbb{T}_\nu = \mathbb{T}_{\text{period}}$ exactly [20].

Proof. Write $\mathsf{F}_b = \mathcal{E}_{ij} \circ (\mathcal{M}_{p,i} \otimes \mathcal{M}_{p,j})$ and $\mathsf{F}_b^\nu = \mathcal{E}_\nu \circ (\mathcal{M}_{p,i} \otimes \mathcal{M}_{p,j})$ for the bond $b = (ij)$, each extended by the identity on the other sites, and number the L bonds so that the odd matching comes first: $\mathbb{T}_{\text{period}} = \mathsf{F}_L \cdots \mathsf{F}_1$ and $\mathbb{T}_\nu = \mathsf{F}_L^\nu \cdots \mathsf{F}_1^\nu$, because factors within one matching act on disjoint sites and commute. The telescoping identity

$$\begin{aligned}&\mathsf{F}_L \cdots \mathsf{F}_1 - \mathsf{F}_L^\nu \cdots \mathsf{F}_1^\nu \\ &\qquad = \sum_{k=1}^{L} \mathsf{F}_L \cdots \mathsf{F}_{k+1}\,(\mathsf{F}_k - \mathsf{F}_k^\nu)\,\mathsf{F}_{k-1}^\nu \cdots \mathsf{F}_1^\nu\end{aligned}$$

is exact. The diamond norm is submultiplicative, unchanged by tensoring with an identity map, and equal to $\|\Phi^*(\mathbb{1})\|_\infty$ for a completely positive map Φ [21]. The twirls $\mathcal{E}_{ij}$ and $\mathcal{E}_\nu$ are mixtures of unitary conjugations, hence of diamond norm one; $\mathcal{M}_p$ is completely positive with $\mathcal{M}_p^*(\mathbb{1}) = (1-p)\mathbb{1} + pC \le \mathbb{1}$ (the map $\mathcal{D}$ discards the off-diagonal replica blocks), so $\|\mathcal{M}_p \otimes \mathcal{M}_p\|_\diamond = \|\mathcal{M}_p^*(\mathbb{1}) \otimes \mathcal{M}_p^*(\mathbb{1})\|_\infty \le 1$. Hence every F_b and F_b^ν has diamond norm at most one, while $\|\mathsf{F}_b - \mathsf{F}_b^\nu\|_\diamond \le \|\mathcal{E}_{ij} - \mathcal{E}_\nu\|_\diamond\,\|\mathcal{M}_p \otimes \mathcal{M}_p\|_\diamond \le \varepsilon$. The triangle inequality applied to the L terms gives Eq. (32), with $L/2$ terms, hence $(L/2)\varepsilon$, per matching layer; the identity is exact, so no higher-order term arises. For the two-qubit Clifford group the second-moment operator equals the Haar one [20], i.e. $\mathcal{E}_\nu = \mathcal{E}_{ij}$ and $\varepsilon = 0$. Note that $\mathbb{T}_\nu$ need not preserve the feature space; the bound is a statement on the full operator space, and the induced first-order shift of a simple eigenvalue, $|\delta\lambda| \le \kappa_\lambda\|\delta\mathbb{T}\| + O(\|\delta\mathbb{T}\|^2)$, must be evaluated with κ_λ in a norm compatible with the one used for $\delta\mathbb{T}$. □

Remark 2 (Clifford ensembles). Because two-qubit Cliffords form an exact 2-design, all annealed two-replica quantities of this section coincide for Haar and Clifford gates. The quenched record statistics of Sec. IV, however,

are trivial for Clifford circuits with stabilizer initial states and Pauli-basis measurements: whether a measurement outcome is deterministic or unbiased depends only on the unsigned stabilizer group, which evolves identically on every branch, so every record with positive probability has the same probability $2^{-N_{\mathrm{rand}}(\omega)}$, the surprisal $\mathcal{A}_T$ of Sec. IV is deterministic, and $\psi_{L,\omega}(k)$ is exactly linear in k. Clifford circuits are therefore useless as a test bed for record multifractality, but they supply a free exact identity: $\mathbb{E}_\omega 2^{-N_{\mathrm{rand}}(\omega)} = \bar{Z}_2(t)$ of Eq. (28), a cross-check between a stabilizer simulator and the transfer matrix (verified at $L = 4$ to Monte Carlo accuracy, and at $L = 16$, $p = 0.16$, $t = 2$ to four digits in Appendix B, Table IX). The quenched entanglement transition of the Clifford ensemble is nontrivial, and Appendix B locates it numerically.

D. Nonzero-spectrum compression, rank, and spectral structure

Since every local bond map has rank at most two, the first matching layer obeys $\operatorname{rank}\mathbb{B}_{\mathrm{odd}} \le 2^{L/2}$, hence $\operatorname{rank}\mathbb{T}_{\mathrm{period}} \le 2^{L/2}$. Theorem 4 below shows that this bound is never attained for $L \ge 4$ and identifies the exact rank with a p- and d-independent combinatorial quantity.

The rank structure yields an exact spectral reduction. Write the local bond matrix as

$$[\mathcal{B}_p] = JK, \qquad J = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad K = \begin{pmatrix} a & b & b & c \\ c & b & b & a \end{pmatrix}, \tag{33}$$

and let $J_{\mathrm{o/e}} : \mathbb{C}^{2^{L/2}} \to \mathbb{C}^{2^L}$ and $K_{\mathrm{o/e}} : \mathbb{C}^{2^L} \to \mathbb{C}^{2^{L/2}}$ be the tensor products on the odd/even matchings. Define

$$\mathsf{C}_{\mathrm{comp}} = K_{\mathrm{e}} J_{\mathrm{o}} K_{\mathrm{o}} J_{\mathrm{e}} = M_1 M_2, \;\; M_1 = K_{\mathrm{e}} J_{\mathrm{o}}, \; M_2 = K_{\mathrm{o}} J_{\mathrm{e}}. \tag{34}$$

$J_{\mathrm{o/e}}$ are isometric embeddings and $K_{\mathrm{o/e}}$ are surjective whenever $a \ne c$, i.e. for $p < 1$.

Theorem 3 (Exact nonzero-spectrum compression)**.** *With $N = 2^L$ and $N_{\mathrm{c}} = 2^{L/2}$,*

$$\det(\lambda I_N - \mathbb{T}_{\mathrm{period}}) = \lambda^{N-N_{\mathrm{c}}} \det(\lambda I_{N_{\mathrm{c}}} - \mathsf{C}_{\mathrm{comp}}). \tag{35}$$

Consequently the nonzero eigenvalues, their algebraic multiplicities, and their nonzero Jordan elementary divisors agree. If $\mathsf{C}_{\mathrm{comp}} y = \lambda y$ with $\lambda \ne 0$, then $J_{\mathrm{e}} y$ is a right eigenvector of $\mathbb{T}_{\mathrm{period}}$.

Proof. The matching factorizations give $\mathbb{T}_{\mathrm{period}} = J_{\mathrm{e}} K_{\mathrm{e}} J_{\mathrm{o}} K_{\mathrm{o}} = \mathsf{XY}$ with $\mathsf{X} = J_{\mathrm{e}}$ and $\mathsf{Y} = K_{\mathrm{e}} J_{\mathrm{o}} K_{\mathrm{o}}$. For $\lambda \ne 0$ the block matrix $\begin{pmatrix} \lambda I_N & \mathsf{X} \\ \mathsf{Y} & I_{N_{\mathrm{c}}} \end{pmatrix}$ has the two Schur-complement determinants $\lambda^N \det(I_{N_{\mathrm{c}}} - \lambda^{-1}\mathsf{YX}) = \lambda^{N-N_{\mathrm{c}}} \det(\lambda I_{N_{\mathrm{c}}} - \mathsf{YX})$ and $\det(I_{N_{\mathrm{c}}}) \det(\lambda I_N - \mathsf{XY})$; both are polynomials in λ, so they agree identically (Sylvester's determinant identity [22]; Ref. [23], Theorem 1.3.22). With $\mathsf{YX} = \mathsf{C}_{\mathrm{comp}}$ this is Eq. (35), which identifies the nonzero eigenvalues together with their algebraic multiplicities. The eigenvector lift follows by multiplying $\mathsf{YX}y = \lambda y$ by X. For the Jordan data at $\lambda \ne 0$ let $G_\lambda(\mathsf{XY}) = \ker(\mathsf{XY} - \lambda)^N$ be the generalized eigenspace. If $x \in G_\lambda(\mathsf{XY})$ and $\mathsf{Y}x = 0$, then $(\mathsf{XY})^j x = 0$ for all $j \ge 1$ and $0 = (\mathsf{XY} - \lambda)^N x = (-\lambda)^N x$, so $x = 0$: Y is injective on $G_\lambda(\mathsf{XY})$, and $\mathsf{Y}(\mathsf{XY} - \lambda)^k = (\mathsf{YX} - \lambda)^k \mathsf{Y}$ shows that it maps $G_\lambda(\mathsf{XY})$ into $G_\lambda(\mathsf{YX})$ and intertwines the two restrictions. The same argument with X gives the reverse injection, so Y restricts to an isomorphism $G_\lambda(\mathsf{XY}) \to G_\lambda(\mathsf{YX})$ intertwining XY with YX, and the Jordan blocks at λ coincide in number and size (Flanders [24]; Ref. [23], Theorem 3.2.11.1). No similarity $\mathbb{T}_{\mathrm{period}} \sim \mathsf{C}_{\mathrm{comp}} \oplus 0$ is asserted at zero, where the block sizes of XY and YX may differ by one [24]. □

Write the bond labels as Ising spins, $I \mapsto s = +1$ and $S \mapsto s = -1$, and let $m = L/2$. The odd-bond labels $s = (s_1, \dots, s_m)$ and the even-bond labels $s' = (s'_1, \dots, s'_m)$ are cyclic, even bond k lying between odd bonds k and $k+1$. Reading Eq. (17) column by column, the bond weight is $\mathsf{w}(s' \mid s_1, s_2) = a$ if $s' = s_1 = s_2$, c if $s_1 = s_2 \ne s'$, and b if $s_1 \ne s_2$, so that

$$(M_1)_{s's} = \prod_{k=1}^{m} \mathsf{w}(s'_k \mid s_k, s_{k+1}),$$
$$(M_2)_{s's} = \prod_{k=1}^{m} \mathsf{w}(s'_k \mid s_{k-1}, s_k). \tag{36}$$

Call a bond k of a label configuration s a *wall* if $s_k \ne s_{k+1}$, and let $\#(s)$ be the number of walls (always even). Define the $2^m \times 2^m$ zero–one matrix

$$(V_m)_{s's} = \prod_{k=1}^{m} \big(\mathbf{1}[s_k \ne s_{k+1}] + \mathbf{1}[s_k = s_{k+1} = s'_k]\big), \tag{37}$$

which depends on neither p nor d, and, for $0 < p < 1$, the numbers (the notation anticipates the Ising identification of Sec. II E)

$$K_d = \tfrac{1}{4}\log\tfrac{a}{c} > 0, \quad K_h = \tfrac{1}{4}\log\tfrac{ac}{b^2} < 0, \quad \mathcal{W}_0 = (acb^2)^{1/4}. \tag{38}$$

The signs follow from $a - c > 0$, Eq. (19), and from $b^2 - ac = d^2(\vartheta^2 - 1)^2/[(d^2 - 1)^2(d^2 + 1)^2] > 0$ for $\vartheta > 1$.

Theorem 4 (Rank and spectral structure of the compressed operator)**.** *Let $L \ge 4$ be even, $d \ge 2$, and $m = L/2$.*

(i) (Two factorizations.) For $0 \le p < 1$, with $\mathsf{Q} = \begin{pmatrix} a & c \\ c & a \end{pmatrix}$ in the (I, S) basis and $\lambda_{\mathrm{w}} = b/(a + c) = \vartheta/(\vartheta^2 + 1)$,

$$M_1 = \mathsf{Q}^{\otimes m} V_m \operatorname{diag}\big(\lambda_{\mathrm{w}}^{\#(s)}\big). \tag{39}$$

For $0 < p < 1$, with $E_{s's} = \exp[K_d \sum_k s'_k(s_k + s_{k+1})]$ and $D_h = \mathrm{diag}(\exp[K_h \sum_k s_k s_{k+1}])$,

$$M_1 = \mathcal{W}_0^m\, E\, D_h, \qquad M_2 = \mathcal{W}_0^m\, E^{\mathsf{T}} D_h,$$
$$\mathsf{C}_{\mathrm{comp}} = \mathcal{W}_0^L\, E\, D_h E^{\mathsf{T}} D_h. \tag{40}$$

(ii) (Similarity to a positive semidefinite matrix.) For $0 < p < 1$, $\mathsf{C}_{\mathrm{comp}} = D_h^{-1/2}\big[\mathcal{W}_0^L\, \mathsf{S}\mathsf{S}^{\mathsf{T}}\big] D_h^{1/2}$ with $\mathsf{S} = D_h^{1/2} E D_h^{1/2}$. Hence the spectrum of $\mathsf{C}_{\mathrm{comp}}$, and therefore the nonzero spectrum of $\mathbb{T}_{\mathrm{period}}$, is real, nonnegative, and semisimple. Moreover $\operatorname{rank}\mathbb{T}^2_{\mathrm{period}} = \operatorname{rank}\mathbb{T}_{\mathrm{period}}$, so the zero eigenvalue of $\mathbb{T}_{\mathrm{period}}$ is semisimple as well: $\mathbb{T}_{\mathrm{period}}$ is diagonalizable with real nonnegative spectrum for $0 < p < 1$.

(iii) (Perron–Frobenius.) For $0 < p < 1$, $\mathsf{C}_{\mathrm{comp}}$ is entrywise positive. Its largest eigenvalue λ_1 is simple with strictly positive left and right eigenvectors, strictly larger than every other eigenvalue, and $\langle \mathrm{tr}|J_{\mathrm{e}} y_1\rangle\, \langle \ell_1|K_{\mathrm{e}} J_{\mathrm{o}} v_0\rangle > 0$ for the vectors of Remark 1. $\mathsf{C}_{\mathrm{comp}}$ commutes with the global relabeling $I \leftrightarrow S$; y_1 lies in the even sector.

(iv) (Exact rank.) For $0 \le p < 1$,

$$\operatorname{rank}\mathbb{T}_{\mathrm{period}} = \operatorname{rank} M_1 = \operatorname{rank} M_2 = \operatorname{rank} V_m = 3\cdot 2^{m-2}, \tag{41}$$

independent of p and d; for $0 < p < 1$ also $\operatorname{rank}\mathsf{C}_{\mathrm{comp}} = 3\cdot 2^{m-2}$. At $p = 1$ the rank is one.

(v) (Kernel.) $\ker M_1 = \mathrm{diag}(\lambda_{\mathrm{w}}^{-\#(s)}) \ker V_m$, so the null space is known for every p once it is known for V_m. $\dim\ker V_m = 2^{m-2}$; every null vector of V_m vanishes on the two uniform configurations and has parity $(-1)^{m+1}$ under the global flip $s \mapsto -s$. For even m the two all-wall configurations $s = (+,-,+,-,\dots)$ and $-s$ have identical columns; for odd $m \ge 3$, if $s^{(k)}$ denotes the configuration whose unique non-wall bond is k, then $e_{s^{(k)}} + e_{-s^{(k)}} - e_{s^{(j)}} - e_{-s^{(j)}} \in \ker V_m$ for $j \ne k$.

Proof. (i) If $s_k \ne s_{k+1}$, the column $\mathsf{w}(\cdot \mid s_k, s_{k+1}) = b(1,1)^{\mathsf{T}} = \lambda_{\mathrm{w}}\mathsf{Q}(1,1)^{\mathsf{T}}$ because $\mathsf{Q}(1,1)^{\mathsf{T}} = (a+c)(1,1)^{\mathsf{T}}$; if $s_k = s_{k+1} = s$, the column is $\mathsf{Q}e_s$. Taking the tensor product over k gives Eq. (39), the factor λ_{w} appearing once per wall. For the Ising form one checks the three cases: $s' = s_1 = s_2$ gives $\mathcal{W}_0 e^{2K_d + K_h} = a$, $s_1 = s_2 \ne s'$ gives $\mathcal{W}_0 e^{-2K_d + K_h} = c$, and $s_1 \ne s_2$ gives $\mathcal{W}_0 e^{-K_h} = b$, by Eq. (38); the horizontal factor $e^{K_h s_k s_{k+1}}$ depends only on the input row, which produces D_h, and the diagonal factors produce E. Since $E_{ss'} = \exp[K_d \sum_k s_k(s'_k + s'_{k+1})] = \exp[K_d \sum_k s'_k(s_{k-1} + s_k)]$, the transpose E^{T} has exactly the structure of M_2 in Eq. (36). (ii) D_h is a positive diagonal matrix for $0 < p < 1$, so the displayed similarity is an algebraic identity from Eq. (40), and $\mathsf{S}\mathsf{S}^{\mathsf{T}}$ is real symmetric positive semidefinite. Theorem 3 transfers the statement to the nonzero spectrum of $\mathbb{T}_{\mathrm{period}}$. For the zero eigenvalue, $\mathbb{T}_{\mathrm{period}} = J_{\mathrm{e}} M_1 K_{\mathrm{o}}$ and $\mathbb{T}^2_{\mathrm{period}} = J_{\mathrm{e}} M_1 M_2 M_1 K_{\mathrm{o}}$ with J_{e} injective and K_{o} surjective, while $M_1 M_2 M_1 = \mathcal{W}_0^{3m} D_h^{-1/2}\, \mathsf{S}\mathsf{S}^{\mathsf{T}}\mathsf{S}\, D_h^{1/2}$ and $\operatorname{rank}(\mathsf{S}\mathsf{S}^{\mathsf{T}}\mathsf{S}) = \operatorname{rank}\mathsf{S} = \operatorname{rank} M_1$; hence $\operatorname{rank}\mathbb{T}^2_{\mathrm{period}} = \operatorname{rank}\mathbb{T}_{\mathrm{period}}$, so the Jordan blocks at zero have size one. (iii) All products in Eq. (36) are positive because $a, b, c > 0$; the Perron–Frobenius theorem for positive matrices gives simplicity, positivity of the eigenvectors, and the strict gap. The overlaps are positive because v_0, $\langle \mathrm{tr}|$, J_{e}, $K_{\mathrm{e}} J_{\mathrm{o}}$ all have nonnegative and not identically vanishing coordinates against positive vectors. The relabeling symmetry is that of Eq. (17) under $(II, IS, SI, SS) \mapsto (SS, SI, IS, II)$; a positive vector is even. (iv) $\mathbb{T}_{\mathrm{period}} = J_{\mathrm{e}} M_1 K_{\mathrm{o}}$ with J_{e} injective and K_{o} surjective for $p < 1$, so $\operatorname{rank}\mathbb{T}_{\mathrm{period}} = \operatorname{rank} M_1$; Q is invertible because $a \ne c$ for $p < 1$, and $\lambda_{\mathrm{w}} > 0$, so $\operatorname{rank} M_1 = \operatorname{rank} V_m$ by Eq. (39); M_2 is M_1 with its rows permuted by the cyclic shift $s'_k \mapsto s'_{k+1}$, so $\operatorname{rank} M_2 = \operatorname{rank} M_1$. For $0 < p < 1$, $\operatorname{rank}\mathsf{C}_{\mathrm{comp}} = \operatorname{rank}(\mathsf{S}\mathsf{S}^{\mathsf{T}}) = \operatorname{rank}\mathsf{S} = \operatorname{rank} E = \operatorname{rank} M_1$. At $p = 1$, $a = c$ and every column of $[\mathcal{B}_p]$ coincides. The value $\operatorname{rank} V_m = 3\cdot 2^{m-2}$ is Lemma 1 below. (v) The first statement is Eq. (39). The parity statement and $\dim\ker V_m = 2^{m-2}$ are part of Lemma 1. For the uniform configurations, note that by Eq. (42) below a vector v lies in $\ker V_m$ iff the function $2^{\#(s)} v(s)$ is orthogonal to every f_Γ; the choices $\Gamma = \{1,\dots,m\}$ and $\Gamma = \{1,\dots,m\} \setminus \{j\}$ give $f_\Gamma = 2^{|\Gamma|}(\delta_{s_+} \pm \delta_{s_-})$ with both signs (a ring cannot have exactly one wall), so $\delta_{s_\pm}$ lie in the span and $v(s_\pm) = 0$. Finally, an all-wall configuration has column $(1,1)^{\otimes m}$ of V_m, whence the even case; for odd m, the column of $s^{(k)}$ is e_ϵ at position k ($\epsilon = s_k^{(k)}$) and $(1,1)^{\mathsf{T}}$ elsewhere, and summing over $\epsilon = \pm$ gives $(1,1)^{\otimes m}$ independently of k. □

Lemma 1 (Rank of V_m)**.** *For every $m \ge 2$, $\operatorname{rank} V_m = 3\cdot 2^{m-2}$, $\dim\ker V_m = 2^{m-2}$, and every vector in $\ker V_m$ has parity $(-1)^{m+1}$ under the global flip.*

Proof. Step 1 (character transform). For $\Gamma \subseteq \{1,\dots,m\}$ let $\chi_\Gamma(s') = \prod_{k\in\Gamma} s'_k$. Multiplying Eq. (37) by $\chi_\Gamma(s')$ and summing over s' bond by bond, using $\sum_{s'_k} s'_k = 0$ on wall bonds in Γ, $\sum_{s'_k} 1 = 2$ on wall bonds outside Γ, $\sum_{s'_k} s'_k \mathbf{1}[s'_k = s_k] = s_k$ and $\sum_{s'_k} \mathbf{1}[s'_k = s_k] = 1$ on non-wall bonds, gives

$$\sum_{s'} \chi_\Gamma(s')(V_m)_{s's} = 2^{\#(s) - |\Gamma|} f_\Gamma(s),$$
$$f_\Gamma(s) = \prod_{k\in\Gamma}(s_k + s_{k+1}). \tag{42}$$

The characters form an invertible transform of the row index and $2^{\#(s)}$ is an invertible column scaling, so $\operatorname{rank} V_m = \dim\mathrm{span}\{f_\Gamma\}$ in the 2^m-dimensional space of functions on $\{\pm 1\}^m$.

Step 2 (reduction to wall sets). Parametrize s by its wall set W, an arbitrary subset of even cardinality, and by $\sigma = s_1$; then $s_k = \sigma(-1)^{|W \cap \{1,\dots,k-1\}|}$. $f_\Gamma(s)$ vanishes

unless $\Gamma \cap W = \varnothing$, in which case $f_\Gamma(s) = 2^{|\Gamma|}\prod_{k\in\Gamma} s_k$, so

$$f_\Gamma(s) = 2^{|\Gamma|}\sigma^{|\Gamma|}g_\Gamma(W),$$
$$g_\Gamma(W) = \mathbf{1}[\Gamma \cap W = \varnothing]\,(-1)^{\sum_{w\in W}|\{k\in\Gamma : k>w\}|}. \quad (43)$$

Functions with $|\Gamma|$ even are flip-even and those with $|\Gamma|$ odd are flip-odd, so the span splits into the two sectors, and within a sector $\sigma^{|\Gamma|}$ is a common factor. Hence

$$\operatorname{rank} V_m = \dim\operatorname{span}\{g_\Gamma : |\Gamma| \equiv m\} + \dim\operatorname{span}\{g_\Gamma : |\Gamma| \not\equiv m\} \pmod 2, \quad (44)$$

both spans being taken in the 2^{m-1}-dimensional space of functions on even subsets W.

Step 3 (the sector $|\Gamma| \equiv m$ is full). Here $B = \Gamma^c$ is even, and $\Gamma \mapsto B$ is a bijection onto the even subsets. By Eq. (43), $g_\Gamma(W) \neq 0$ iff $W \subseteq B$, and $g_\Gamma(B) = \pm 1$. Ordering rows B and columns W by the same linear extension of inclusion (e.g. by cardinality), the $2^{m-1}\times 2^{m-1}$ matrix $(g_\Gamma(W))_{B,W}$ is triangular with unimodular diagonal, hence invertible: the sector has dimension 2^{m-1}.

Step 4 (the sector $|\Gamma| \not\equiv m$ has dimension 2^{m-2}). Here $|\Gamma^c|$ is odd. For every nonempty even set $\Lambda = \{c_1 < \cdots < c_{2j}\}$ we claim

$$\sum_{i=1}^{2j}(-1)^i\, g_{\Lambda^c\cup\{c_i\}} = 0. \quad (45)$$

Indeed, the ith term is nonzero at W only if $W \subseteq \Lambda\setminus\{c_i\}$, and there its sign is $(-1)^{\sum_{w\in W}|\{k\in\Lambda^c : k>w\}|}$, common to all i, times $(-1)^{|\{w\in W : w<c_i\}|}$. List the elements of $\Lambda \setminus W$ in increasing order; between two consecutive ones, $c_i < c_{i'}$, all $i'-i-1$ intermediate elements of Λ belong to W, so the exponent $i + |\{w \in W : w < c_i\}|$ changes by $2(i'-i)-1$, an odd number. The surviving terms therefore alternate in sign, and there is an even number of them ($|\Lambda \setminus W| = 2j - |W|$), so they cancel. Now let $|\Gamma| \not\equiv m$ with $m \in \Gamma$ and put $\Lambda = \Gamma^c \cup \{m\}$, an even set with $c_{2j} = m$ and $\Lambda^c \cup \{m\} = \Gamma$; relation (45) expresses g_Γ through the $g_{\Lambda^c\cup\{c_i\}}$, $i < 2j$, none of which contains m. Hence the sector is spanned by the 2^{m-2} functions $\{g_\Gamma : |\Gamma| \not\equiv m,\ m \notin \Gamma\}$. These are linearly independent: for such Γ the set Γ^c is odd and contains m, so $W_\Gamma = \Gamma^c \setminus \{m\}$ is an even set; $g_{\Gamma'}(W_\Gamma) \neq 0$ requires $W_\Gamma \subseteq \Gamma'^c$, i.e. $\Gamma^c \subseteq \Gamma'^c$ because both contain m, and $g_\Gamma(W_\Gamma) = \pm 1$. Ordering the Γ by a linear extension of inclusion of the complements, the square matrix $(g_{\Gamma'}(W_\Gamma))_{\Gamma,\Gamma'}$ is triangular with unimodular diagonal. So the sector has dimension exactly 2^{m-2}.

Step 5. By Eq. (44), $\operatorname{rank} V_m = 2^{m-1}+2^{m-2} = 3\cdot 2^{m-2}$ and $\dim\ker V_m = 2^{m-2}$. For the parity, V_m commutes with the flip, so it preserves the two flip eigenspaces $\mathcal{E}_\pm$, each of dimension 2^{m-1}. Rows of Eq. (42) with $(-1)^{|\Gamma|} \neq \epsilon$ annihilate $\mathcal{E}_\epsilon$, and the remaining rows are, up to the invertible column scaling, the functions $f_\Gamma \in \mathcal{E}_\epsilon$ themselves; hence $\operatorname{rank}(V_m|_{\mathcal{E}_\epsilon}) = \dim\operatorname{span}\{f_\Gamma : (-1)^{|\Gamma|} = \epsilon\}$, which by Steps 3–4 is 2^{m-1} for $\epsilon = (-1)^m$ and 2^{m-2} for $\epsilon = (-1)^{m+1}$. Thus V_m is injective on $\mathcal{E}_{(-1)^m}$ and $\ker V_m \subseteq \mathcal{E}_{(-1)^{m+1}}$. □

Theorem 4 corrects a natural but false expectation: entrywise positivity of C_{comp} gives Perron–Frobenius, not invertibility, and M_1 is a surjection followed by an injection, which need not be invertible (at $L = 4$ the columns $s = (+,-)$ and $(-,+)$ of M_1 coincide, so the rank is 3). For numerics the theorem is good news: the compressed operator can be restricted once and for all to a complement of the explicitly known, p-independent null space, and its spectrum can be obtained by a symmetric eigensolver applied to $\mathcal{W}_0^L\mathsf{S}\mathsf{S}^\mathsf{T}$. The forced zero eigenvalue of $\mathbb{T}_{\text{period}}$ has algebraic and geometric multiplicity $2^L - 3\cdot 2^{L/2-2}$. The rank values, the null-space parity, and the relations (45) were also verified in exact arithmetic for $m \le 11$.

E. The two-replica model is an exactly solved Ising model

Equation (40) says that the strobed two-replica model is, exactly, the anisotropic Ising model on the triangular lattice with ferromagnetic coupling K_d on the two diagonal directions (each even-bond spin couples to the two odd-bond spins below it, and vice versa) and antiferromagnetic coupling K_h on the horizontal direction. This is the $n = 2$ replica model of Bao, Choi and Altman [8] (their Sec. V.1; their weak-measurement angle corresponds to the projective rate through $\tan^4\alpha = p/(1-p)$, and their couplings are $J_d = -K_d \le 0$, $J_h = -K_h \ge 0$ in the convention $e^{-\beta H}$), whose critical point they located with Houtappel's solution [9]. Its consequences for the transfer matrix are the following.

Proposition 4 (Exact solution of the annealed two-replica model)**.** *For $0 < p < 1$:*

(a) The thermodynamic growth rate exists and equals $\lambda_\infty(p) = \lim_{L\to\infty}\lambda_1(L)^{1/L} = \mathcal{W}_0\, e^{f_{\text{H}}(K_d,K_d,K_h)}$, where f_{H} is Houtappel's free energy per site of the triangular lattice [9, 25],

$$f_{\rm H}(K_1,K_2,K_3)=\log 2+\frac{1}{8\pi^2}\int_0^{2\pi}\!\!\int_0^{2\pi} d\theta_1 d\theta_2\,\log\Big[\cosh 2K_1\cosh 2K_2\cosh 2K_3+\sinh 2K_1\sinh 2K_2\sinh 2K_3 - \sinh 2K_1\cos\theta_1-\sinh 2K_2\cos\theta_2-\sinh 2K_3\cos(\theta_1+\theta_2)\Big]. \quad (46)$$

(b) $\lambda_\infty(p)$ is real-analytic in $p\in(0,1)$ except at the unique point where the bracket in Eq. (46) vanishes at $\theta_1=\theta_2=0$, which in the variables of Theorem 2 is the condition

$$a-c=2b \iff \vartheta_c=\beta_d+\sqrt{\beta_d^2+1},\quad \beta_d=\frac{d^2-1}{d^2+1},\qquad p_c^{(2)}(d)=\frac{d-\vartheta_c}{d-1}. \quad (47)$$

Numerically $p_c^{(2)}=0.233810$ ($d=2$), 0.459688 ($d=3$), 0.594675 ($d=4$), 0.679004 ($d=5$), 0.846613 ($d=10$); Eq. (47) is Eq. (65) of Ref. [8] under $\tan^4\alpha_c=p_c/(1-p_c)$.

(c) The singularity is in the two-dimensional Ising universality class ($\nu=1$; scaling dimensions $x_\sigma=1/8$ for the $I\leftrightarrow S$-odd sector and $x_\epsilon=1$ for the leading even excitation). For $p<p_c^{(2)}$ the model is ordered and the annealed quasi-entropy $\tilde S_2(A)=-\log[\bar P_{2,A}/\bar Z_2]$ of Eq. (28), which is the free energy of a domain wall terminating on A [8], grows linearly in $|A|$ for $1\ll|A|\ll L$; for $p>p_c^{(2)}$ it saturates.

Proof. (a) Let $m=L/2$. By Eq. (40), $\mathrm{Tr}\,\mathsf{C}_{\rm comp}^T=\mathcal{W}_0^{LT}\,Z_{m,2T}$, where $Z_{m,2T}=\mathrm{Tr}(ED_hE^{\mathsf T}D_h)^T$ is the partition function of the Ising model on a periodic triangular lattice of $2T$ rows of m spins: E carries the $2m$ diagonal couplings K_d between consecutive rows, D_h the m horizontal couplings K_h within a row, so one period contains L spins, $2L$ diagonal bonds and L horizontal bonds, three bonds per spin; the factor $\mathcal{W}_0^L$ collects one $\mathcal{W}_0$ per bond factor $\mathsf w$ of the period. By Theorem 4(iii), $\lambda_1(L)$ is simple and strictly dominant, so $\lim_{T\to\infty}T^{-1}\log\mathrm{Tr}\,\mathsf{C}_{\rm comp}^T=\log\lambda_1(L)$, i.e. $\log\lambda_1(L)=L\log\mathcal{W}_0+2m\,f_m$ with $f_m=\lim_{T\to\infty}(2mT)^{-1}\log Z_{m,2T}$ the free energy per site of the periodic strip of width m. For a finite-range interaction the free energy per site has a thermodynamic limit that does not depend on the shape of the domain or the order of the limits (Ref. [26], Chap. 2), so $f_m\to f_{\rm H}(K_d,K_d,K_h)$ as $m\to\infty$, where $f_{\rm H}$ is Houtappel's closed form [9] for the anisotropic triangular lattice, Eq. (46) (Ref. [25], Chap. 6, reaches the same lattice by the star–triangle transformation). Hence $\lambda_1(L)^{1/L}\to\mathcal{W}_0e^{f_{\rm H}}$.

(b) Write $\mathsf s_d=\sinh 2K_d>0$ and $\mathsf s_h=\sinh 2K_h<0$. From Eq. (38), $e^{2K_d}=\sqrt{a/c}$ and $e^{-2K_h}=b/\sqrt{ac}$, so

$$\mathsf s_d=\frac{a-c}{2\sqrt{ac}},\qquad |\mathsf s_h|=\frac{b^2-ac}{2b\sqrt{ac}},\qquad \mathsf s_d e^{2K_h}=\frac{a-c}{2b}.$$

With $\cosh^2 2K_d=1+\mathsf s_d^2$, the bracket in Eq. (46) at $\theta_1=\theta_2=0$ equals $e^{-2K_h}+\mathsf s_d^2e^{2K_h}-2\mathsf s_d=e^{-2K_h}(1-\mathsf s_de^{2K_h})^2=:\epsilon\ge0$, and the bracket minus its value at the origin is

$$\Delta(\theta_1,\theta_2)=\mathsf s_d(2-\cos\theta_1-\cos\theta_2)-|\mathsf s_h|\big(1-\cos(\theta_1+\theta_2)\big).$$

Since $|\sin(x+y)|\le|\sin x|+|\sin y|$ and $(u+v)^2\le2u^2+2v^2$, one has $1-\cos(\theta_1+\theta_2)=2\sin^2\frac{\theta_1+\theta_2}{2}\le2(1-\cos\theta_1)+2(1-\cos\theta_2)$, so $\Delta\ge0$ on the torus whenever $\mathsf s_d\ge2|\mathsf s_h|$. By the displayed expressions this condition reads $b(a-c)\ge2(b^2-ac)$, and explicitly

$$b(a-c)-2(b^2-ac)=\frac{(\vartheta^2-1)\big[(d^4-1)\vartheta-2d^2\vartheta^2+2d^2\big]}{(d^2-1)^2(d^2+1)^2},$$

so for $\vartheta>1$ it is equivalent to $q(\vartheta)=2d^2\vartheta^2-(d^4-1)\vartheta-2d^2\le0$. The quadratic q is convex with $q(1)=-(d^4-1)<0$ and $q(d)=-d(d-1)^3(d+1)<0$, hence $q<0$ on all of $[1,d]$: $\mathsf s_d>2|\mathsf s_h|$ for every $0<p<1$. Consequently the bracket is strictly positive on the whole torus except possibly at the origin, where it vanishes iff $\mathsf s_d=e^{-2K_h}$. This is Houtappel's criticality condition $\mathsf s_d^2+2\mathsf s_d\mathsf s_h=1$ for this anisotropy pattern [8, 9] (Ref. [25], Chap. 6), since $\mathsf s_d^2+2\mathsf s_d\mathsf s_h-1=(\mathsf s_d-e^{-2K_h})(\mathsf s_d+e^{2K_h})$ and the second factor is positive. Away from such a point the integrand of Eq. (46) is real-analytic in $(\vartheta,\theta_1,\theta_2)$ on the compact torus, so $f_{\rm H}$, and with it $\lambda_\infty=\mathcal{W}_0e^{f_{\rm H}}$, is real-analytic in p. At such a point $f_{\rm H}$ is not analytic. Indeed, $\mathsf s_de^{2K_h}=(a-c)/2b=(\vartheta-\vartheta^{-1})/2\beta_d$ increases strictly through 1 at ϑ_c, so $\epsilon(\vartheta)$ has a double zero there with $\epsilon''(\vartheta_c)>0$; near the origin $c_1|\theta|^2\le\Delta_\vartheta(\theta)\le c_2|\theta|^2$ with $c_1>0$, because the quadratic part $\frac12(\mathsf s_d-|\mathsf s_h|)(\theta_1^2+\theta_2^2)-|\mathsf s_h|\theta_1\theta_2$ of Δ is positive definite exactly when $\mathsf s_d>2|\mathsf s_h|$; and $\partial_\vartheta\Delta_\vartheta$, $\partial_\vartheta^2\Delta_\vartheta$ are $O(|\theta|^2)$. Differentiating $\iint\log(\epsilon+\Delta_\vartheta)$ twice under the integral, every term stays bounded as $\vartheta\to\vartheta_c$ except $\epsilon''\iint d^2\theta/(\epsilon+\Delta_\vartheta)\ge\epsilon''(\pi/c_2)\log(1+c_2r^2/\epsilon)$, which diverges logarithmically: $\partial_\vartheta^2f_{\rm H}\to+\infty$ at ϑ_c, Onsager's logarithmic specific heat in Houtappel's form [9]. Finally, $\mathsf s_de^{2K_h}=1$ is $a-c=2b$; with Eq. (19) it reads $\vartheta^2-2\beta_d\vartheta-1=0$, whose positive root is $\vartheta_c=\beta_d+\sqrt{\beta_d^2+1}$, and $\vartheta_c\in(1,d)$ because $(a-c)/2b$ increases strictly from 0 at $\vartheta=1$ to $(d^2+1)/2d>1$ at $\vartheta=d$.

(c) Flipping the spins of every other row reverses the sign of both diagonal couplings and leaves K_h unchanged, so the model is gauge equivalent to the fully antiferromagnetic anisotropic triangular lattice whose two strongest couplings lie on the diagonals ($K_d>|K_h|$ follows from $\mathsf s_d>2|\mathsf s_h|$). The triangular-lattice Ising model is a free-fermion (Pfaffian) model for every sign pattern of

its couplings, and its exact pair correlations are Stephenson's [30, 31]: for the anisotropic antiferromagnetic lattice there is a single transition, at Houtappel's point, below which the two strong axes carry antiferromagnetic and the weak axis ferromagnetic long-range order—in the original gauge, uniform order of the bond labels—and the correlation length diverges as $|\vartheta - \vartheta_c|^{-1}$ ($\nu = 1$). The specific heat is logarithmic by (b) ($\alpha = 0$), and the long-range order vanishes with the exponent $\beta = 1/8$ of the anisotropic triangular lattice [31, 32]. In the language of the $c = 1/2$ Ising field theory, $x_\sigma = \beta/\nu = 1/8$ and $x_\epsilon = 2 - 1/\nu = 1$; Table I confirms the ratio x_σ/x_ϵ numerically for the present sign pattern. The $I \leftrightarrow S$ flip is the spin-reversal symmetry, so the flip-odd sector carries x_σ and the leading flip-even excitation x_ϵ. The ordered phase is the small-p side, because $K_d = \frac{1}{4}\log(a/c) \to \infty$ as $p \to 0$, consistent with the twofold degeneracy of λ_1 at $p = 0$ (Remark 1). Finally, $\langle \mathrm{tr}_A|$ pairs the final row with the label S on A and I elsewhere (Remark 1), so $\bar{P}_{2,A}/\bar{Z}_2$ is the ratio of the partition functions with and without a domain wall forced to terminate at the two ends of A; in the ordered phase the wall costs a surface tension per unit length and $\tilde{S}_2(A)$ grows linearly in $|A|$ for $1 \ll |A| \ll L$, while in the disordered phase the boundary condition is forgotten beyond a correlation length and $\tilde{S}_2(A)$ saturates [8]. □

Two remarks on interpretation. First, $p_c^{(2)}$ is a singularity of the *annealed* quantity $-\lim(tL)^{-1}\log \bar{Z}_2$, the record-collision (or annealed $q = 2$ record) free energy. It is distinct from the Born-weighted transition, which is not an $n = 2$ quantity: for Haar qubit circuits it lies near $p_c \approx 0.17$ [11], and for the Clifford realization of the present period Appendix B finds $p_c = 0.1597(8)$, against $p_c^{(2)}(2) = 0.2338$; at every size the quenched half-chain entropy lies strictly below the annealed quasi-entropy $\tilde{S}_2$, whose record-collision weight lowers the effective measurement density to $p_{\mathrm{eff}} = 0.092$ at $p = 0.16$ (Appendix B, Table IX), and at $p = 0.22 < p_c^{(2)}$ the quenched chain is already in its area-law phase while the annealed model is still ordered. Second, the exact solution gives the transfer-matrix chain of Sec. II D something rare: an end-to-end check with known answers. Table I reports such a check, produced with a matrix-free implementation of $\mathsf{C}_{\mathrm{comp}}$ (sector-projected Arnoldi on $2^{L/2}$ states). The finite-size crossings of the scaled $I \leftrightarrow S$ gap $L\log(\lambda_1/\lambda_{\mathrm{odd}})$ converge to Eq. (47) from below, the ratio of the odd and even gaps at $p_c^{(2)}$ approaches $x_\sigma/x_\epsilon = 1/8$ (the lattice anisotropy cancels in the ratio), and $\lambda_1(L)^{1/L}$ agrees with Eq. (46) to 3×10^{-7} at $L = 28$ and is certified exactly at rational points by the Collatz–Wielandt enclosures of Appendix A 5 ($d = 2$, $p = 0.40$: 0.6838449 versus 0.6838447). At $d = 2$, $L = 32$ the annealed quasi-entropy $\tilde{S}_2(A)$ evaluated in the Perron state at $|A| = 4$ and 8 is 2.13, 3.74 ($p = 0.12$), 1.77, 2.85 ($p = 0.18$), 0.89, 1.07 ($p = 0.30$) and 0.49, 0.52 ($p = 0.40$): growth with $|A|$ below and saturation above $p_c^{(2)}$, as Proposition 4(c) requires. These data validate the proof-to-code chain against an exact solution.

Because $\mathbb{T}_{\mathrm{period}}$ is not normal in the coordinate inner product, a nonsymmetric Arnoldi method, Schur method, or dense diagonalization is appropriate on the full space; on the compressed space Theorem 4(ii), or the Gram-metric form of Theorem 5(i) in Sec. II F below, allows a symmetric solver. A one-vector power iteration is not Lanczos and does not in general certify the subleading eigenvalue. The compression in Theorem 3 provides an independent full-spectrum route on $2^{L/2} = \sqrt{2^L}$ states. Any correlation length defined from eigenvalues must state the selected eigenvalue, the symmetry sector, observable coupling, and degeneracy treatment. The released small-size benchmarks ($L = 4, 6$, $d = 2$, $p \in \{0.01, 0.118, 0.5\}$) obtain global ranks from the complete dense spectrum of $\mathbb{T}_{\mathrm{period}}$, check it against the complete compressed spectrum (agreement to 4×10^{-16}, including the forced zero multiplicity $2^L - 3 \cdot 2^{L/2-2}$), and cross-check the leading three eigenpairs with two-seed full-space Arnoldi, right and left residuals below 2×10^{-15}, and biorthogonal condition numbers $\kappa_\lambda \in [1.3, 4.7]$. All eighteen released eigenvalues are real and positive, the nonzero counts are 3 and 6, and at $L = 6$ the fourth and fifth eigenvalues form an exact doublet carried by the two nonzero momenta of the three-site cyclic shift of the bond labels, all as Theorem 4 requires. The released small-size benchmarks validate the proof-to-code chain; they are not a finite-size-scaling data set. The crossings and gap ratios of Table I concern the exactly solved annealed model, whose transition point is known in closed form, and no quenched exponent or transition point is inferred from any of these data.

Proposition 5 (closed form of the nonzero spectrum)**.** *Let $L = 2m \geq 4$, $d \geq 2$, $0 < p < 1$, and let $\mathsf{C}_{\mathrm{comp}} = \mathcal{W}_0^L E D_h E^{\mathsf{T}} D_h$ be the compressed period operator of Eq.* (40)*, with $K_d, K_h, \mathcal{W}_0$ as in Eq.* (38)*. For $k \in [0, 2\pi)$ define*

$$\begin{aligned} Q(k) &= \cosh^2 2K_d \cosh 2K_h + \sinh^2 2K_d \sinh 2K_h \\ &\quad - \sinh 2K_h \cos k, \\ R(k) &= 2\sinh 2K_d \cos\tfrac{k}{2}, \end{aligned} \tag{48}$$

and $w_k^{\pm} = Q(k) \pm \sqrt{Q(k)^2 - R(k)^2}$ (real, since $0 \leq R^2 \leq Q^2$ for all k). Then every nonzero eigenvalue of $\mathsf{C}_{\mathrm{comp}}$ is of the form

$$\lambda = \left(2\mathcal{W}_0^2\right)^m \prod_{k \in \mathcal{K}_\sigma} w_k^{\pm_k}, \tag{49}$$

the product running over the momentum states of the flip sector σ: in the flip-even sector $\mathcal{K}_+ = \{(2j-1)\pi/m : j = 1, \ldots, m\}$ (antiperiodic) with an even *number of "$-$" choices; in the flip-odd sector $\mathcal{K}_- = \{2\pi j/m : j = 0, \ldots, m-1\}$ (periodic) with an even number of "$-$" choices below $p_c^{(2)}(d)$ and an odd number above (the $k = 0$ mode changes its branch ordering at the transition, exactly as in Kaufman's spinor analysis [27, 28]). Moreover:*

TABLE I. Benchmark of the compressed transfer matrix against the exact Ising solution. $p^*(L_1, L_2)$: crossing of $L\log(\lambda_1/\lambda_{\rm odd})$ for consecutive sizes. Ratio: $\log(\lambda_1/\lambda_{\rm odd})/\log(\lambda_1/\lambda_{\rm even,2})$ at $p = p_c^{(2)}(d)$, to be compared with $x_\sigma/x_\epsilon = 1/8$.

	$(16, 20)$	$(20, 24)$	$(24, 28)$	$(28, 32)$	exact	
p^*, $d = 2$	0.23319	0.23348	0.23361	0.23368	0.233810	
p^*, $d = 3$	0.45934	0.45950	0.45957	0.45961	0.459688	
	$L = 16$	$L = 20$	$L = 24$	$L = 28$	$L = 32$	exact
ratio, $d = 2$	0.1220	0.1231	0.1237	0.1240	0.1243	0.125
ratio, $d = 3$	0.1232	0.1238	0.1242	0.1244	0.1245	0.125

(i) $R(\pi) = 0$, *so* $w_\pi^- = Q(\pi) - |Q(\pi)| = 0$*: every state occupying the "−" branch of the* $k = \pi$ *mode lies in the kernel, and counting the nonzero states (half of the sector containing* $k = \pi$*, all of the other) recovers* $\operatorname{rank} \mathsf{C}_{\rm comp} = 3 \cdot 2^{m-2}$*, Theorem 4(iv), by mode counting.*

(ii) At $p = p_c^{(2)}(d)$ *the* $k = 0$ *mode is gapless:* $Q(0) - R(0)$ *has a double zero at* $a - c = 2b$*, so* $w_0^+ = w_0^-$ *there, and the scaled gaps have the exact limits* $L\ln(\lambda_1/\lambda_{\rm odd}) \to 2\pi\,\beta_d/4$ *and* $L\ln(\lambda_1/\lambda_\varepsilon) \to 2\pi \cdot 2\beta_d$*, ratio* $1/8 = x_\sigma/x_\varepsilon$ *for every* d*.*

(iii) $a - c = 2b \iff Q(0) = R(0)$*; the critical line of Proposition 4 is the gapless condition of the* $k = 0$ *mode.*

Proof. (a) In the spinor (Jordan–Wigner) representation with open chain ends the diagonal transfer is a product of Ising-Clifford exponentials: with $x_k = \sigma_k^x$ and $z_k = \sigma_k^z$ acting on sites k and 1 of the m-site row (the Schultz–Mattis–Lieb algebra of the row transfer [29]), $K_1 = \frac{1}{2}\operatorname{arsinh}(1/\sinh 2K_d)$ the Kramers–Wannier dual of K_d (so $\sinh 2K_1 \sinh 2K_d = 1$), $K_2 = K_d$ and $c = \sqrt{2\sinh 2K_1}$,

$$E = \sinh^m(2K_d)\left[\cosh K_d\,\tilde{Y} + \sinh K_d\, z_m \tilde{Y} z_1\right],$$
$$\tilde{Y} = c\, e^{K_1 x_m} U_{m-1} \cdots U_1, \qquad U_k = e^{K_d z_k z_{k+1}}\, c\, e^{K_1 x_k}, \tag{50}$$

the periodic boundary contributing only through the end term $z_m(\cdot)z_1$ (the analogous identity holds for D_h with the wrap coupling K_h, and $D_{h,\rm OBC} = \prod_k e^{K_h z_k z_{k+1}}$ exactly). Every factor of Eq. (50) has unit determinant, so $\det \tilde{Y} = c^{m2^m}$, one flip sector of E is exactly singular (the $k = \pi$ kernel of item (i) at the transfer level), and the two flip sectors of the open-boundary transfer have equal determinants $|\det E_{{\rm OBC},\varepsilon}| = (2\sinh 2K_d)^{m2^{m-2}}$. (b) On each flip-parity sector $\varepsilon = \pm 1$, E_ε commutes with the Majorana translation T_ε carrying the boundary sign $-\varepsilon$, hence is block-diagonal in the momentum basis $\mathcal{K}_\sigma$ fixed by that sign (antiperiodic for the even sector, periodic for the odd sector). (c) Each momentum block of $\mathsf{C}_{\rm comp} = \mathcal{W}_0^L (ED_h)(ED_h)^{\mathsf{T}}\tau$, with τ the bond-ring shift implementing $E^{\mathsf{T}} = E\tau$, is a 2×2 Majorana quadratic form; diagonalising it gives the two eigenvalues $w_k^\pm$ of Eq. (48), the half-shift $\cos(k/2)$ in $R(k)$ tracing to that shift. (d) Multiplying the per-momentum block eigenvalues over the sector and fixing the parity of "−" choices from the fermion parity of the sector (the $k = 0$ block changes its parity character at $p_c^{(2)}$, the Kaufman boundary-condition switch [27]) yields Eq. (49). (e) The kernel count (i) is immediate from $R(\pi) = 0$; the amplitude limits (ii) follow from the expansion of $\ln w_k^+ - \ln w_k^-$ near $k = 0$ at $\vartheta = \vartheta_c$, where $ac/b^2 = \beta_d^2$; and (iii) is a direct substitution using $a = \mathcal{W}_0 e^{2K_d + K_h}$, $b = \mathcal{W}_0 e^{-K_h}$, $c = \mathcal{W}_0 e^{-2K_d + K_h}$ from Eq. (38). □

Remark 3 (numerical verification). Every algebraic step has been checked numerically, and the checks are deposited with the data release: the spinor factorisation (50) holds with the stated closed-form parameters and normalisation $\sinh^m(2K_d)$ to a relative residual of 6×10^{-16} ($m = 3$–8, $d \in \{2, 3, 5\}$, both branches of the weight trajectory); the nonzero spectrum of the dense $\mathsf{C}_{\rm comp}$ agrees with Eq. (49) to $\le 10^{-10}$ in $\ln\lambda$ for $m = 4$–9, $d \in \{2, 3, 5\}$, twelve values of p including $p_c^{(2)} \pm 2\times 10^{-4}$; the commutators of $\mathsf{C}_{\rm comp}$ with the flip and the bond-ring shift vanish to 10^{-15}; the closed form reproduces the crossings of Table I to $\le 5 \times 10^{-6}$ and their ratio row to all four decimals; $\lambda_1(28)^{1/28} = 0.683844946$ at $d = 2$, $p = 0.40$, against the transfer-matrix value 0.6838449 quoted above, and the bulk 0.683844659 against Houtappel's $f_{\rm H}$; and the closed form lies inside every exact-rational Collatz–Wielandt enclosure of Appendix A 5 (verified at 60-digit precision). The closed form upgrades the annealed benchmark of Table I from agreement with Houtappel's bulk free energy to agreement with an exactly known finite-L spectrum, and gives the finite-size amplitudes at $p_c^{(2)}$ exactly.

F. Spectral structure for every replica number

Theorem 4(ii) has an operator-level origin that does not depend on the Ising structure and therefore extends to every n. Fix $n \ge 2$, $d \ge 2$, even L, and $p \in [0, 1]$. Let $\mathcal{M} = \mathcal{M}_p^{(n)\otimes L}$ with $\mathcal{M}_p^{(n)} = (1-p)\,\mathrm{id} + p\mathcal{D}_n$, let $\mathcal{E}_{\rm o/e}$ be the n-fold Haar twirl of the odd/even gate layer, and let

$$\Phi_n(p) = \mathcal{E}_{\rm e}\mathcal{M}\mathcal{E}_{\rm o}\mathcal{M} \tag{51}$$

be the post-even-gate period map on $\mathrm{End}(H^{\otimes n})^{\otimes L}$; by Proposition 1(iii) it preserves $\mathcal{G}_n^{\otimes L}$, and for $n = 2$ its restriction is $\mathbb{T}_{\mathrm{period}}$. Let $\mathcal{V}_n^{\mathrm{b}} = \mathrm{span}\{U_\sigma \otimes U_\sigma\}$ be the bond permutation span, so that $\operatorname{Ran}\mathcal{E}_{\mathrm{e/o}} = (\mathcal{V}_n^{\mathrm{b}})^{\otimes L/2}$ over the even/odd bonds, and let

$$\Psi_{\mathrm{e}} : \; y \mapsto \sum_{\tau \in S_n^{L/2}} y_\tau \bigotimes_{b \in \mathcal{G}_{\mathrm{even}}} U_{\tau_b} \otimes U_{\tau_b} \tag{52}$$

be the map from bond labels to operators, with Gram matrix $\mathsf{G} = \Psi_{\mathrm{e}}^\dagger \Psi_{\mathrm{e}} = (d^{2c(\sigma^{-1}\tau)})^{\otimes L/2}$ (invertible iff $d^2 \geq n$; otherwise Ψ_{e}^+ below is the Moore–Penrose inverse, matching the pseudo-inverse convention of Proposition 1); Ψ_{o} is defined likewise on the odd bonds. Finally let $J_{n,\mathrm{o/e}}$, $K_{n,\mathrm{o/e}}$ be the general-n analogues of Eq. (33) built from Eq. (14), acting on the $n!^L$ site labels and $n!^{L/2}$ bond labels, and $\mathsf{C}_{\mathrm{comp}}^{(n)} = K_{n,\mathrm{e}} J_{n,\mathrm{o}} K_{n,\mathrm{o}} J_{n,\mathrm{e}}$ the compressed operator on bond labels.

Theorem 5 (Spectral structure for general n)**.** *Let $n \geq 2$, $d \geq 2$, L be even, and $p \in [0,1]$.*

(i) (Positivity.) $\mathcal{D}_n$, $\mathcal{M}$, $\mathcal{E}_{\mathrm{o}}$, $\mathcal{E}_{\mathrm{e}}$ are Hilbert–Schmidt self-adjoint with spectrum in $[0,1]$, and on $\operatorname{Ran}\mathcal{E}_{\mathrm{e}}$, which contains every eigenvector of $\Phi_n(p)$ with nonzero eigenvalue,

$$\Phi_n(p)\big|_{\operatorname{Ran}\mathcal{E}_{\mathrm{e}}} = \mathsf{A}\mathsf{A}^\dagger, \qquad \mathsf{A} = \mathcal{E}_{\mathrm{e}}\mathcal{M}\mathcal{E}_{\mathrm{o}}. \tag{53}$$

Hence the nonzero spectrum of $\Phi_n(p)$, and of its restriction to $\mathcal{G}_n^{\otimes L}$, is real, strictly positive and semisimple, and $\Phi_n(p)$ is diagonalizable. In bond coordinates, $\mathsf{G}\,\mathsf{C}_{\mathrm{comp}}^{(n)} = (\mathsf{C}_{\mathrm{comp}}^{(n)})^{\mathsf{T}}\mathsf{G} \succeq 0$: the compressed operator is self-adjoint and positive semidefinite in the Gram inner product $\langle y, y' \rangle_{\mathsf{G}} = y^{\mathsf{T}}\mathsf{G}y'$, and $K_{n,\mathrm{o}}J_{n,\mathrm{e}}$ is the G-adjoint of $K_{n,\mathrm{e}}J_{n,\mathrm{o}}$.

(ii) (Compression.) $\mathsf{C}_{\mathrm{comp}}^{(n)} = \Psi_{\mathrm{e}}^+ \Phi_n(p) \Psi_{\mathrm{e}}$ and $\Psi_{\mathrm{e}}\Psi_{\mathrm{e}}^+ = \mathcal{E}_{\mathrm{e}}$. Hence the nonzero eigenvalues of $\mathsf{C}_{\mathrm{comp}}^{(n)}$, of the $n!^L$-dimensional coordinate transfer matrix $J_{n,\mathrm{e}}K_{n,\mathrm{e}}J_{n,\mathrm{o}}K_{n,\mathrm{o}}$, and of $\Phi_n(p)$ coincide with multiplicities and Jordan data, the identity (35) *holds with $N = n!^L$ and $N_{\mathrm{c}} = n!^{L/2}$, and eigenvectors lift by Ψ_{e}.*

(iii) (Strict contraction and endpoints.) $\rho(\Phi_n(p)) < 1$ for every $p \in (0,1]$. At $p = 0$ the eigenvalue 1 has multiplicity $\dim \mathrm{span}\{U_\sigma^{\otimes L} : \sigma \in S_n\}$, which is $n!$ whenever $d^L \geq n$. At $p = 1$ the rank is one and the unique nonzero eigenvalue is

$$\lambda(p{=}1) = \Big[\frac{n!}{(d^2+1)(d^2+2)\cdots(d^2+n-1)}\Big]^L, \tag{54}$$

the Haar average of $\sum_R p_R^n$ over one fully monitored period.

(iv) (Symmetry.) $\Phi_n(p)$ commutes with the global replica relabelings $X \mapsto U_\alpha^{\otimes L} X U_\beta^{\otimes L}$, $(\alpha, \beta) \in S_n \times S_n$, with the computational-basis transposition $X \mapsto X^{\mathsf{T}}$, which acts as $U_\sigma \mapsto U_{\sigma^{-1}}$ on every site and exchanges the two S_n factors, and with the translations and reflections of the bond ring. Consequently the bond weights obey $W_{p,n}(\alpha\pi\beta \mid \alpha\mu\beta, \alpha\nu\beta) = W_{p,n}(\pi \mid \mu, \nu)$, $W_{p,n}(\pi \mid \mu, \nu) = W_{p,n}(\pi \mid \nu, \mu)$, and $\mathsf{C}_{\mathrm{comp}}^{(n)}$ is block diagonal in the isotypic decomposition of $(S_n \times S_n) \rtimes \mathbb{Z}_2 \times D_{L/2}$. The relabelings applied to a single bond do not *commute with $\mathsf{C}_{\mathrm{comp}}^{(n)}$ in general.*

Proof. (i) $\mathcal{D}_n(X) = \sum_m P_m^{\otimes n} X P_m^{\otimes n}$ is a sum of Hilbert–Schmidt-orthogonal projectors, hence a self-adjoint idempotent; $\mathcal{M}$ is a tensor product of convex combinations $(1-p)\,\mathrm{id} + p\mathcal{D}_n$, hence self-adjoint with spectrum in $\{(1-p)^j : 0 \leq j \leq L\} \subset [0,1]$; the twirls are the orthogonal projectors onto the commutants of $U^{\otimes n}$ on the bonds (Schur–Weyl), and $\mathcal{E}_{\mathrm{e}}\mathcal{E}_{\mathrm{e}} = \mathcal{E}_{\mathrm{e}}$. Write $\Phi_n = \mathsf{A}\mathcal{M}$; then $\operatorname{Ran}\Phi_n \subseteq \operatorname{Ran}\mathsf{A} \subseteq \operatorname{Ran}\mathcal{E}_{\mathrm{e}}$, and on $\operatorname{Ran}\mathcal{E}_{\mathrm{e}}$ one has $\Phi_n = \mathcal{E}_{\mathrm{e}}\mathcal{M}\mathcal{E}_{\mathrm{o}}\mathcal{M}\mathcal{E}_{\mathrm{e}} = \mathsf{A}\mathsf{A}^\dagger$ because $\mathcal{E}_{\mathrm{o}} = \mathcal{E}_{\mathrm{o}}^2$ and all factors are self-adjoint. If $(\Phi_n - \lambda)^2 X = 0$ with $\lambda \neq 0$, then $Y = (\Phi_n - \lambda)X$ satisfies $\Phi_n Y = \lambda Y$, so $Y \in \operatorname{Ran}\mathcal{E}_{\mathrm{e}}$, and $X = \lambda^{-1}(\Phi_n X - Y) \in \operatorname{Ran}\mathcal{E}_{\mathrm{e}}$ as well; there Φ_n is self-adjoint and positive semidefinite, so $Y = 0$: nonzero eigenvalues are real, positive and semisimple. If $\Phi_n^2 X = 0$, then $\Phi_n X \in \operatorname{Ran}\mathsf{A} \cap \ker\Phi_n$, and on $\operatorname{Ran}\mathcal{E}_{\mathrm{e}}$ the kernel of $\Phi_n = \mathsf{A}\mathsf{A}^\dagger$ is $\ker\mathsf{A}^\dagger = (\operatorname{Ran}\mathsf{A})^\perp$; hence $\Phi_n X = 0$, the zero eigenvalue is semisimple too, and Φ_n is diagonalizable. The restriction to the invariant subspace $\mathcal{G}_n^{\otimes L}$ inherits all of this. For the coordinate statements use (ii): $\mathsf{G}\mathsf{C}_{\mathrm{comp}}^{(n)} = \Psi_{\mathrm{e}}^\dagger\Psi_{\mathrm{e}}\Psi_{\mathrm{e}}^+\Phi_n\Psi_{\mathrm{e}} = \Psi_{\mathrm{e}}^\dagger\mathcal{E}_{\mathrm{e}}\Phi_n\Psi_{\mathrm{e}} = \Psi_{\mathrm{e}}^\dagger\mathsf{A}\mathsf{A}^\dagger\Psi_{\mathrm{e}}$, which is real symmetric positive semidefinite, and likewise $\mathsf{G}\,K_{n,\mathrm{o}}J_{n,\mathrm{e}} = \Psi_{\mathrm{o}}^\dagger\mathcal{M}\Psi_{\mathrm{e}} = (K_{n,\mathrm{e}}J_{n,\mathrm{o}})^{\mathsf{T}}\mathsf{G}$. (ii) $J_{n,\mathrm{o}}$ sends the bond label $\tau \in S_n^{L/2}$ to the site labels that equal τ_b on both sites of every odd bond b, i.e. to the coordinates of the operator $\Psi_{\mathrm{o}}e_\tau$; $K_{n,\mathrm{e}}$ sends a site-label configuration $\mu \in S_n^L$, i.e. the operator $\bigotimes_i U_{\mu_i}$, to the bond coordinates of $\mathcal{E}_{\mathrm{e}}\mathcal{M}\bigotimes_i U_{\mu_i}$ on the even matching, which by Eq. (14) and the proof of Proposition 1(iii) are $\Psi_{\mathrm{e}}^+\mathcal{E}_{\mathrm{e}}\mathcal{M}\bigotimes_i U_{\mu_i}$. Hence $K_{n,\mathrm{e}}J_{n,\mathrm{o}} = \Psi_{\mathrm{e}}^+\mathcal{E}_{\mathrm{e}}\mathcal{M}\Psi_{\mathrm{o}}$ and, by the same construction on the other matching, $K_{n,\mathrm{o}}J_{n,\mathrm{e}} = \Psi_{\mathrm{o}}^+\mathcal{E}_{\mathrm{o}}\mathcal{M}\Psi_{\mathrm{e}}$. Since $\Psi_{\mathrm{o}}\Psi_{\mathrm{o}}^+$ and $\Psi_{\mathrm{e}}\Psi_{\mathrm{e}}^+$ are the orthogonal projectors onto $\operatorname{Ran}\Psi_{\mathrm{o/e}} = \operatorname{Ran}\mathcal{E}_{\mathrm{o/e}}$, i.e. equal $\mathcal{E}_{\mathrm{o/e}}$,

$$\begin{aligned}\mathsf{C}_{\mathrm{comp}}^{(n)} &= \Psi_{\mathrm{e}}^+\mathcal{E}_{\mathrm{e}}\mathcal{M}\,(\Psi_{\mathrm{o}}\Psi_{\mathrm{o}}^+)\,\mathcal{E}_{\mathrm{o}}\mathcal{M}\Psi_{\mathrm{e}} \\ &= \Psi_{\mathrm{e}}^+\mathcal{E}_{\mathrm{e}}\mathcal{M}\mathcal{E}_{\mathrm{o}}\mathcal{M}\Psi_{\mathrm{e}} = \Psi_{\mathrm{e}}^+\Phi_n\Psi_{\mathrm{e}},\end{aligned}$$

using $\mathcal{E}_{\mathrm{o}}^2 = \mathcal{E}_{\mathrm{o}}$. Now apply the argument in the proof of Theorem 3, which is valid for arbitrary rectangular X, Y, twice: for $\mathsf{X} = \Psi_{\mathrm{e}}$, $\mathsf{Y} = \Psi_{\mathrm{e}}^+\Phi_n$ one has $\mathsf{X}\mathsf{Y} = \mathcal{E}_{\mathrm{e}}\Phi_n = \Phi_n$ (because $\operatorname{Ran}\Phi_n \subseteq \operatorname{Ran}\mathcal{E}_{\mathrm{e}}$) and $\mathsf{Y}\mathsf{X} = \mathsf{C}_{\mathrm{comp}}^{(n)}$; for $\mathsf{X} = J_{n,\mathrm{e}}$, $\mathsf{Y} = K_{n,\mathrm{e}}J_{n,\mathrm{o}}K_{n,\mathrm{o}}$ one has $\mathsf{X}\mathsf{Y} = J_{n,\mathrm{e}}K_{n,\mathrm{e}}J_{n,\mathrm{o}}K_{n,\mathrm{o}}$ and again $\mathsf{Y}\mathsf{X} = \mathsf{C}_{\mathrm{comp}}^{(n)}$. This identifies the three nonzero spectra with multiplicities and Jordan data, gives Eq. (35) with $N = n!^L$, $N_{\mathrm{c}} = n!^{L/2}$ from the same block determinant, and lifts eigenvectors by $\mathsf{X} = \Psi_{\mathrm{e}}$. For $d < n$ the site labels

are overcomplete, but $K_{n,\mathrm{o/e}}$ annihilate the site-level linear relations among the U_π, being defined through the operator-level twirl, so nothing spurious enters. (iii) If $\Phi_n X = X$ with $X \in \operatorname{Ran}\mathcal{E}_\mathrm{e}$, then $\|X\| = \|\mathsf{A}\mathsf{A}^\dagger X\| \le \|\mathsf{A}^\dagger X\| \le \|\mathcal{M}\mathcal{E}_\mathrm{e} X\| = \|\mathcal{M}X\| \le \|X\|$ forces $\mathcal{M}X = X$, i.e. $X \in \ker(1-\mathcal{M}) = \big(\operatorname{span}\{P_m^{\otimes n}\}\big)^{\otimes L}$ for $p > 0$, since every site factor has eigenvalues 1 and $1-p < 1$. But $X \in \mathcal{G}_n^{\otimes L}$, and $\mathcal{G}_n \cap \operatorname{span}\{P_m^{\otimes n}\} = \{0\}$: an element of the intersection is $\mathcal{D}_n$-invariant, so by Proposition 1(i) it equals $\big(\sum_\pi x_\pi\big)C_n$, and $C_n \notin \mathcal{G}_n$ by Eq. (13). The intersection of tensor powers is the tensor power of the intersection, so $X = 0$; as $\rho(\Phi_n) \le 1$ by (i), the spectral radius is strictly below one. At $p = 0$, $\mathcal{M} = \mathrm{id}$ and the fixed points of $\mathcal{E}_\mathrm{e}\mathcal{E}_\mathrm{o}$ are the operators commuting with every bond unitary of both layers, i.e. with $U(d^2)^{\otimes n}$ on every bond; since the bond gates include the swaps, the two matchings generate every two-site unitary on every pair of sites and hence all of $U(d^L)$ [33], so the fixed space is the commutant of $U^{\otimes n}$, $U \in U(d^L)$, which is $\operatorname{span}\{U_\sigma^{\otimes L}\}$ by Schur–Weyl, of dimension $n!$ when $d^L \ge n$; these are eigenvalue-one eigenvectors and, by (i), the only ones. At $p = 1$, $\mathcal{D}_n(U_\pi) = C_n$ for every π (Proposition 1(i)), so every bond factor $\mathcal{E}_{ij}(\mathcal{D}_n \otimes \mathcal{D}_n)$ has rank one on $\mathcal{V}_n^\mathrm{b}$, with image spanned by $\mathcal{E}_{ij}(C_n \otimes C_n) = \Pi(C_n \otimes C_n)$; the trace of $C_n \otimes C_n$ against $U_\sigma \otimes U_\sigma$ is d^2 for every σ, so $\Pi(C_n \otimes C_n) = \sum_\sigma U_\sigma \otimes U_\sigma / [(d^2+1)\cdots(d^2+n-1)]$ by Proposition 1(ii) applied to $U(d^2)$, and applying $\mathcal{D}_n \otimes \mathcal{D}_n$ again returns $C_n \otimes C_n$ times $n!/[(d^2+1)\cdots(d^2+n-1)]$. Thus $\mathcal{M}\operatorname{Ran}\mathcal{E}_\mathrm{o} = \mathbb{C}\, C_n^{\otimes L}$, so $\Phi_n(1)$ has rank one, and one period contains L bond factors, each contributing the displayed factor to its unique nonzero eigenvalue. (iv) $P_m^{\otimes n}$ commutes with every U_α, so $\mathcal{D}_n$ commutes with $X \mapsto U_\alpha X U_\beta$; the twirls commute with it by Schur–Weyl; the transposition commutes with $\mathcal{D}_n$ because the $P_m^{\otimes n}$ are symmetric, and with the twirls because Haar measure is invariant under $U \mapsto \bar{U}$, and $U_\sigma^\mathsf{T} = U_{\sigma^{-1}}$; translations and reflections of the ring are symmetries of the brickwork. Since $U_\alpha(U_\pi \otimes U_\pi)U_\beta$ has coordinates $\alpha\pi\beta$ and the twirl is an intertwiner, the bond coordinates inherit bi-invariance; wire-swap symmetry is Eq. (14) itself. Relabelings on one bond only do not commute with the neighbouring bond factors because J_n embeds a bond label into *equal* labels on two sites that belong to different bonds of the other matching; numerically the commutator is nonzero already at $n = 3$, $d = 2$, $L = 6$. □

Three consequences deserve emphasis. First, the Ising-specific similarity of Theorem 4(ii) is not needed for reality and positivity of the spectrum: a symmetric solver on $\mathsf{S}_n\mathsf{S}_n^\mathsf{T}$ with $\mathsf{S}_n = \mathsf{G}^{1/2}K_{n,\mathrm{e}}J_{n,\mathrm{o}}\mathsf{G}^{-1/2}$, the general-$n$ counterpart of S in Theorem 4(ii), is available for every n (for $d^2 \ge n$; otherwise on a basis of the bond span), and biorthogonal condition numbers are unnecessary. Second, positivity in the entrywise sense fails for $n \ge 3$: Eq. (14) has negative coordinates for all $p \lesssim 0.63$ at $n = 3$, $d = 2$ (twelve of the 216 entries, e.g. -0.10 at $p = 0$), and $\mathsf{C}_\mathrm{comp}^{(3)}$ becomes entrywise positive only above an L-dependent rate ($p \gtrsim 0.31$ at $L = 4$, 0.43 at $L = 6$, $d = 2$), so Perron–Frobenius arguments for the simplicity of the leading eigenvalue are available only in that regime; elsewhere simplicity is a numerical observation. Third, the symmetry in (iv) is the $S_n \times S_n$ replica symmetry of the finite-d replica models [7], and for $n = 2$ it is the $I \leftrightarrow S$ flip. It organizes the spectrum: as $p \to 0$ the $n!$ eigenvalue-one states split into the regular bimodule $\bigoplus_\lambda V_\lambda \otimes V_\lambda$ of $S_n \times S_n$, with multiplicities $(\dim V_\lambda)^2$. At $n = 3$, $d = 2$, $L = 6$, $p = 0.02$ the top levels are 0.836807 (trivial), 0.834268 (fourfold, standard ⊗ standard) and 0.831750 (sign), and this $1+4+1$ pattern persists at all p tested; a general-n implementation must expect these exact degeneracies.

The rank of $\mathsf{C}_\mathrm{comp}^{(3)}$ is no longer p-independent, and its generic value is bounded by symmetry. Since $K_{n,\mathrm{o}}J_{n,\mathrm{e}}$ is the G-adjoint of $K_{n,\mathrm{e}}J_{n,\mathrm{o}}$ [item (i)] and the bond Gram matrix is invertible for $d^2 \ge n$, $\operatorname{rank}\mathsf{C}_\mathrm{comp}^{(n)} = \operatorname{rank}K_{n,\mathrm{e}}J_{n,\mathrm{o}} = \operatorname{rank}K_{n,\mathrm{o}}J_{n,\mathrm{e}} = \operatorname{rank}\mathsf{B}_p$, where $\mathsf{B}_p[\sigma;\tau] = \prod_k T_p(\sigma_k, \tau_{k-1})T_p(\sigma_k, \tau_k)$ is the Weingarten-free overlap matrix built from the site map $T_p(\sigma,\tau) = (1-p)d^{c(\sigma^{-1}\tau)} + pd$ of Eq. (14) (Appendix A); all rank statements below are statements about the integer-polynomial matrix B_p. At $n = 3$ the S_3-isotypic coefficients of T_p are $a_\mathrm{triv} = (1-p)d(d+1)(d+2) + 6pd$, $a_\mathrm{std} = (1-p)d(d^2-1)$, $a_\mathrm{sgn} = (1-p)d(d-1)(d-2)$; at $d = 2$ the sign coefficient vanishes identically—T_p annihilates $\sum_\pi \operatorname{sgn}(\pi)U_\pi$, the operator form of the linear dependence of the U_π for $d < n$—but the singular site Gram matrix is never inverted, and the generic rank is the same for $d = 2, \ldots, 7$ at $L \le 10$ and for $d = 2, 3$ at $L = 12$ (Appendix A). The reflection of the bond ring acts as $k \mapsto -k$ on even labels and $k \mapsto -1-k$ on odd labels, so $K_{3,\mathrm{e}}J_{3,\mathrm{o}}$ is an intertwiner between two *different* representations of $\Gamma = [(S_3 \times S_3) \rtimes \mathbb{Z}_2] \times D_{L/2}$, and Burnside counting of the multiplicities bounds the generic rank by

$$r_3(L) = \begin{cases} 6^{L/2}, & L \equiv 2\ (4), \\ 6^{L/2} - \dfrac{49\cdot 6^{L/4-1} - 4\cdot 3^{L/4-1}}{3}, & L \equiv 0\ (4), \end{cases} \tag{55}$$

i.e. $21, 216, 1202, 7776, 46080$ for $L = 4, \ldots, 12$; the bound is independent of d.

What is certified is the following. Since the entries are polynomials in p, the pointwise rank equals the generic rank r over $\mathbb{Q}(p)$ except on the finite algebraic set E_L where an invariant factor of an isotypic block vanishes, and the drop at such a point is the number of vanishing invariant factors times the irrep dimension. Exact certificates over $\mathbb{Z}[p]$ (block determinants and sums of squared minors, with exact root isolation; Appendix A, Table VIII) give, for $d = 2$: $r = r_3(L)$ for $L = 4, 6, 8, 10$, but *not* for $L = 12$, where the generic rank is $46010 = r_3(12) - 70$: in the momentum-π sector the map is antisymmetric and exchanges the two reflection-parity subspaces, and ten explicit kernel directions (one constant, nine polynomial of degree ≤ 6 in p; all verified exactly over $\mathbb{Z}$) exceed what the multiplicity count forces, in five

of the nine isotypic blocks; $E_4 = \emptyset$; $E_6 = \{p^*\}$ with $p^* = (9 - \sqrt{41})/4 = 0.6492\ldots$, where the rank is 208; $|E_8| = 11$, with ranks 1194–1201 (Table S1 of the Supplemental Material [34]); E_{12} has not been computed. The degrees of the minimal polynomials grow with the block sizes (2 at $L = 6$; up to 22 at $L = 8$, in momentum-zero blocks), and every interior drop is of first order—the vanishing invariant factors have simple zeros—so the affected singular values vanish only linearly, $\sigma \simeq c\,|p - p^*|$, which is why a floating-point rank scan on a grid does not see them. The endpoints behave differently: at $p = 0$ the ranks are $21, 198, 1184, 7663, 45822$ for $L = 4, \ldots, 12$ (at $L \le 8$ only the $k = \pm 1$ blocks lose rank; from $L = 10$ on every momentum sector does), and at $p = 1$ the rank is one, so that on every compact $K \subset (0, 1) \setminus E_L$ the rank is $r_3(L)$ and the nonzero singular values are bounded away from zero, while the numerical rank loss observed as $p \to 1$ is conditioning, not an interior drop. The points of E_L are irrational, hence invisible to sampling at rational p, and they affect only the dimension of the kernel: the nonzero spectrum, and every statement of Theorem 5, is unaffected.

For $d \ge 3$ the exceptional set changes: a rational point $p_\times(d) = (d+1)/[3(d-1)]$ appears, at which the three S_3-isotypic coefficients of the site overlap $(1-p)d^{c(\sigma^{-1}\tau)} + pd$ form a geometric progression and several isotypic blocks lose rank simultaneously (rank 21 at $L = 4$, i.e. no drop; 176 at $L = 6$, 1142 at $L = 8$, 7276 at $L = 10$, identically for $d = 3, \ldots, 7$; at $L = 12$, $d = 3$, the rank at $p_\times$ is 44898), and the irrational points move with d: at $d = 3$ there is already one at $L = 4$ (rank 20), six points at $L = 6$ (ranks 176–215) and 55 at $L = 8$ (ranks 1142–1201, minimal-polynomial degrees up to 50; Table S2 of Ref. [34]); $p_\times$ is the only rational one. The $L = 12$ deficit is d-independent as well: the same 70 directions are missing at $d = 3$ (the 27 constant kernel vectors are literally the same; the polynomial ones are recomputed and reverified), and for a generic central element of $\mathbb{Q}[S_3]$ in place of T_p.

Two remarks close the comparison. For $n = 2$ the analogous symmetry bound ($3, 8, 14, 32$ for $L = 4, 6, 8, 10$) is *not* attained—the Ising structure of Lemma 1 forces the smaller rank $3 \cdot 2^{L/2-2}$—so the $n = 2$ formula does not generalize. Thus Eq. (55) is an upper bound that is attained for $L \le 10$ and fails at $L = 12$; the generic rank for general L is open (the extra kernel is not fixed by the symmetry, and $p = 0$ cannot serve as a witness since the rank drops there), as are the structure of E_L and whether $p_\times(d)$ is exceptional for every $L \ge 6$. A closed-form spectrum, and a proof of the existence and regularity of the $L \to \infty$ growth rate, remain open for $n \ge 3$ (the symmetry bound on the generic rank, Eq. (55), is known); an end-to-end check of the $n = 3$ chain against brute-force record enumeration in a four-qubit Haar brickwork ($\overline{Z}_3(t) = \mathbb{E}_U \sum_R p_R^3$) agrees to within Monte-Carlo error. Sec. II G below supplies what the exact route does not: sector-resolved scaling spectra of $\mathsf{C}^{(3)}_{\mathrm{comp}}$ itself, which locate the annealed three-replica critical point numerically and find it continuous, with three-state-Potts scaling amplitudes.

The exact closure has a precise scope: it applies to the linear outcome-correlated replica channel and independently Haar-averaged gates (or exact 2-designs). It does not apply to a fixed gate realization, a terminal post-measurement strobe, or normalized conditioned trajectories.

G. The annealed three-replica critical point

For $n = 3$ there is no Ising equivalence and no closed form, and the open-problem list has since its first version carried the item that scaling-quality spectral data for the $n \ge 3$ operators do not exist. This subsection supplies them for $n = 3$, $d = 2$: exact, sector-resolved spectra of $\mathsf{C}^{(3)}_{\mathrm{comp}} = M_1 M_2$ on the same lattice, sizes $L = 4$–12, every number a finite transfer-matrix eigenvalue with no statistical error.

Method. The bond channel is Eq. (14) at $D = d^2 = 4$; the Gram similarity $\mathsf{C}_{\mathrm{comp}} \sim \mathsf{S}\mathsf{S}^{\mathsf{T}}$, $\mathsf{S} = \mathsf{G}^{1/2} M_1 \mathsf{G}^{-1/2}$ (Theorem 5(i)), makes every restricted spectrum a symmetric problem. The spectra are resolved by the symmetry of item (iv): at every size and rate tested, the three gaps that scale are carried by the momentum-zero sector—λ_1 (colour-trivial, the Perron value), λ_σ (the leading colour-nontrivial value, in the std⊗std, inversion-even block, fourfold degenerate as $(\dim \mathrm{std})^2 = 4$), and λ_ε (the second colour-trivial value). Sizes $L \le 10$ are dense ($6^{L/2} \le 7776$); at $L = 12$ the momentum-zero block (dimension 7826) is assembled exactly from product rows of M_1 and M_2 over shift orbits and diagonalized by Lanczos with colour-restricted iterations. The validation chain is the same as for the $n = 2$ benchmark: the deposited small-size spectra of Sec. II F reproduce to the last digit, and the identical code at $n = 2$ reproduces the Kaufman closed form to 10^{-11} and, run at $L = 16$–32 through the same momentum-block route, the crossings of Table I themselves: 0.23319, 0.23347, 0.23361 against the deposited 0.23319, 0.23348, 0.23361 (and 0.23368 for $(28, 32)$), i.e. agreement to 10^{-5} at sizes beyond the $n = 3$ chain.

Results (Table II). The crossings of $X_L = L \log(\lambda_1/\lambda_\sigma)$ converge from below; power-law, geometric-deceleration and Aitken extrapolations of the four crossings bracket $p_c^{(3)} = 0.305(3)$ at $d = 2$. Three independent diagnostics, each the exact analogue of a column of the $n = 2$ benchmark, fix the order and the universality class. (i) *Continuous:* the raw gap at the crossings decays as $0.126 \to 0.118 \to 0.100 \to 0.085$ over $L = 6 \to 12$, with $\xi/L = 1/1.32,\ 1/1.06,\ 1/1.00,\ 1/0.98$—a first-order point would saturate the gap and freeze ξ/L, and the two-phase degeneracy signature (an avoided crossing inside the colour-trivial sector) is absent at every size. (ii) *Amplitude ratio:* $\log(\lambda_1/\lambda_\sigma)/\log(\lambda_1/\lambda_\varepsilon)$ at the crossings is $0.107 \to 0.144 \to 0.161 \to 0.171$, approaching the three-state Potts value $x_\sigma/x_\varepsilon = (2/15)/(4/5) = 1/6$ [13]

TABLE II. Annealed three-replica critical point ($n = 3$, $d = 2$), from exact spectra of $\mathsf{C}^{(3)}_{\rm comp}$. Top: crossings of $X_L = L\log(\lambda_1/\lambda_\sigma)$ for consecutive sizes. Middle: the amplitude ratio $\log(\lambda_1/\lambda_\sigma)/\log(\lambda_1/\lambda_\varepsilon)$ at the crossing (value of the larger size). Bottom: the raw gap $\log(\lambda_1/\lambda_\sigma)$ at the crossing (larger size). The three-state Potts values are $x_\sigma/x_\varepsilon = 1/6$ and $\nu = 5/6$ [13]; the Ising values of the $n = 2$ row of Table I are $1/8$ and $\nu = 1$.

	$(4,6)$	$(6,8)$	$(8,10)$	$(10,12)$	limit
p^*	0.27114	0.29678	0.30244	0.30403	0.305(3)
ratio	0.107	0.144	0.161	0.171	$\to 1/6 = 0.167$
gap	0.126	0.118	0.100	0.085	$\propto 1/L$

and excluding the Ising $1/8$ already at $L = 10$; the residual overshoot at $L = 12$ (+2.5%) is of the same magnitude as the $n = 2$ control's deviation from $1/8$ at comparable sizes (0.118 at $L = 10$, 0.1243 at $L = 32$). The identification of the ε-sector is checked absolutely: at $L = 12$ and the crossing, $L\log(\lambda_1/\lambda_\varepsilon) = 6.0$, which tracks $2\pi x_\varepsilon = 5.03$ under the same lattice-amplitude factor that rescales the σ gap $2\pi x_\sigma = 0.838 \to 1.03$; a two-$\sigma$ composite ($2x_\sigma$, ratio $1/2$) or a subleading trivial operator would miss both numbers. (iii) *Slope exponent:* $\mathrm{d}X_L/\mathrm{d}p$ at p_c grows as $L^{1.15}$, i.e. $1/\nu_{\rm eff} = 1.15(10)$, against the Potts $6/5 = 1.2$ and the Ising 1. The operator content matches the assignment: the spin excitation lives in the std $\otimes$ std block (the two-dimensional standard representation of S_3 is the Potts spin), the leading symmetric excitation in the trivial block (the Potts energy), and the σ multiplet is fourfold, $(\dim \mathrm{std})^2$, as the $S_3 \times S_3$ symmetry requires. Two further records: the growth-rate chain $\lambda_1(L,p)^{1/L}$ is monotone in L at every p and its Richardson extrapolation gives a smooth $\lambda^{(3)}_\infty(p)$ (0.926 at $p = 0.05$ falling to 0.313 at $p = 0.75$), consistent with a single nonanalytic point at $p^{(3)}_c$; and λ_1 remained simple, with the $p \to 0$ multiplet splitting triv > std > sgn, at every (L,p) tested (165 of 165), extending the simplicity observation of Sec. II F below the entrywise-positivity threshold.

Interpretation and scope. All three diagnostics land on the three-state-Potts values rather than the Ising ones. Together with the exact $n = 2$ identification this suggests the annealed critical points of this circuit family follow the $q = n$-state Potts universality classes at $n = 2, 3$; the criterion $q > 4$ would then predict the annealed $n \geq 5$ points to be first order, a sharply falsifiable target for the same computation at $n \geq 4$ (the $n = 4$ bond space, $24^{L/2}$, is reachable to $L = 10$ with the present machinery). Sec. II H completes the $n = 4$ computation with full S_4 colour resolution at four sizes ($L = 4, 6, 8, 10$): $p^{(4)}_c \approx 0.383$, on the continuing trend $p^{(2)}_c = 0.2338 < p^{(3)}_c = 0.305 < p^{(4)}_c \approx 0.383$, and it adds the $n = 5$ diagnostics together with the size-ladder test ($L = 4, 6, 8$) that supports the first-order prediction at the three-size level. The replica trend also quantifies the annealed–quenched distinction of Sec. II E at the next replica number: $p^{(2)}_c = 0.233810 < p^{(3)}_c = 0.305(3)$ against the quenched $p_c = 0.1597(8)$, i.e. the annealed points move *away* from the quenched transition as n grows, so no sequence of annealed n-replica points converges to the Born-weighted transition from above. Scope: exact finite-size data with a systematic (not statistical) extrapolation, sizes $L \leq 12$, $d = 2$; the universality assignment rests on the ratio, the slope exponent and the gap law, not on a proof; and the $\lambda^{(3)}_\infty$ curve is an extrapolated record, not a theorem of existence.

H. The annealed four-replica point, and the first-order test at $n \geq 5$

The $n = 4$ bond space is $24^{L/2}$, beyond the dense route at $L \geq 8$ ($24^4 = 331776$), so this subsection uses a general-n iterative machinery, validated to the same standard as the $n = 3$ chain: central (isotypic) idempotents of S_n from the character tables give the full $S_4 \times S_4$ colour resolution—twenty-five blocks (λ, μ), $\lambda, \mu \in \{$triv, sgn, std, std $\otimes$ sgn, two$\}$—applied in $O(Nn!^2)$ through the (key, member) bijections of the free left/right regular orbits; the ring transfer $M_1 M_2$ is applied as a boundary-carry contraction with a π_0 block loop; all leading eigenvalues live at ring momentum zero, which is imposed by shift averaging; and the spectrum is extracted by Arnoldi iteration with colour classification of the eigenvectors (the deposited block-12 pattern, generalized). The validation chain: the identical code at $n = 2$ reproduces the Kaufman closed form to 3×10^{-15} at $L = 8$; at $n = 3$ it reproduces the deposited $L = 6$ scan values of Sec. II G to 10^{-15}; at $n = 4$ it reproduces the dense spectrum at $L = 4$; the twenty-five S_4 projectors are complete and idempotent to 10^{-15}; and at $p = 1$ the rank-one eigenvalue of one period equals $\left[d^2\Gamma(d^2)\Gamma(n{+}1)/\Gamma(d^2{+}n)\right]^{2\mathrm{nb}}$ exactly for $n = 2, 3, 4, 5$—for $n = 5$ this checks the pseudo-inverse Weingarten channel itself ($n = 5 > D^{1/2}$ makes the Gram singular, rank 119 of 120), whose per-bond value is $1/14$.

Results at $n = 4$ (Table III). With the true σ sector (std $\otimes$ std, ninefold as $(\dim \mathrm{std})^2 = 9$) the crossings move to 0.358 and 0.379 and, with the $L = 10$ third crossing computed ($N = 24^5 = 7{,}962{,}624$ bond labels, a dense matrix would need 6.3×10^{13} entries; the iterative ring route runs it at < 1.8 GB peak RSS), to 0.3823: the drift collapses ($0.0208 \to 0.0033$ between successive pairs), so the crossing sequence has essentially converged and the annealed four-replica point is $p^{(4)}_c \approx 0.383$, with the replica trend $0.233810 < 0.305(3) < 0.383 < 0.47$ continuing away from the quenched 0.1597(8). The amplitude ratio $R = \log(\lambda_1/\lambda_\sigma)/\log(\lambda_1/\lambda_\varepsilon)$, evaluated at the common reference $p^* = 0.383$ (the collapsed crossing) with the $L = 8$ ε values (extended scan, Supplemental Material [34], Sec. S7), reads $0.1315 \to 0.1829 \to 0.2142 \to 0.2378$ over $L = 4, 6, 8, 10$: the corrected $q = 4$

TABLE III. Annealed four-replica point ($n = 4$, $d = 2$), from exact momentum-zero, colour-resolved spectra of $\mathsf{C}^{(4)}_{\rm comp} = M_1 M_2$ with full S_4 colour resolution at $L = 4, 6, 8, 10$. Rows as in Table II: consecutive-size crossings of $X_L = L\log(\lambda_1/\lambda_\sigma)$; the amplitude ratio $\log(\lambda_1/\lambda_\sigma)/\log(\lambda_1/\lambda_\varepsilon)$ at the crossing; the raw gap $\log(\lambda_1/\lambda_\sigma)$ (larger size). The $q = 4$ Potts class has $x_\sigma/x_\varepsilon = (1/8)/(1/2) = 1/4$ ($x_\varepsilon = d - 1/\nu = 1/2$ at $1/\nu = 3/2$; the Ising value $1/8$ requires $x_\varepsilon = 1$), and $1/\nu = 3/2$, with multiplicative logarithmic corrections at the marginal point [12].

	$(4,6)$	$(6,8)$	$(8,10)$	limit
p^*	0.35820	0.37899	0.3823	≈ 0.383
ratio	0.128	0.176	—	drifts (marginal)
gap	0.146	0.131	—	decaying

target is $x_\sigma/x_\varepsilon = 1/4$, and the drift is a *logarithmically slow convergence to it*—the two-term marginal form $R = 1/4 - 0.90/\ln L + 0.46/\ln^2 L$ fits the four points with residual $\leq 8\times10^{-4}$—so the diagnostic does not degrade at the borderline class; it converges with exactly the multiplicative-logarithmic slowliness [12] that the marginal point predicts (the mixing with the marginal operator still makes the second trivial eigenvalue a noisy ε proxy at $L \leq 8$: the absolute check $L\log(\lambda_1/\lambda_\varepsilon) = 5.2$ at the $L = 8$ crossing against $2\pi x_\varepsilon = \pi = 3.14$ under the σ-side lattice-amplitude 1.3 remains inconclusive, and the ratio diagnostic carries the assignment). The same machinery at the non-marginal $n = 3$ control converges cleanly: $R_L(0.305) = 0.108 \to 0.141 \to 0.157 \to 0.167$, on the $q = 3$ target $1/6 = 0.1\overline{6}$ at $L = 10$. The slope exponent rises through the pairs as $1.00 \to 1.38 \to 1.57$ (consecutive-size crossings), through the $q = 4$ value $3/2$ with the expected $1/\ln L$ correction (at the common $p^* = 0.383$: $\mathrm{d}X_L/\mathrm{d}p = C\,L^{3/2}(1+0.45/\ln L)$ fits all four sizes with residual $\leq 3\times10^{-2}$ in ln slope, the free-exponent fit giving 1.37 and rising); the gap at the crossings decays and ξ/L stays of order unity (1.14, 0.95), so the transition is continuous; and λ_1 remains simple with the multiplet splitting triv > std > two > std⊗sgn > sgn at every tested (L, p). The colour content is exactly as the $S_4 \times S_4$ symmetry requires: at $L = 8$ and $p = 0.36$ the leading eigenvalues read λ_1 (triv ⊗ triv, once), λ_σ (std ⊗ std, ninefold), then two ⊗ two (fourfold), std ⊗ sgn (ninefold), sgn⊗sgn (once), with the second trivial eigenvalue λ_ε below all spin multiplets—the $q = 4$ analogue of the deposited $1 + 4 + 1$ pattern at $n = 3$.

The $n = 5$ first-order diagnostics. The $q = n$-Potts conjecture predicts a first-order annealed transition for $n \geq 5$. At $n = 5$ the bond space is $120^{L/2}$; the $L = 4$ space (14400) is reached by the same iterative machinery with the pseudo-inverse Weingarten channel, and a 21-point scan with full colour classification and a colour-trivial projected solve yields three $L = 4$ diagnostics, each evaluated at the corresponding estimated transition: the two-phase degeneracy measure $\log(\lambda_1/\lambda_2)$ tightens across $n = 3 \to 4 \to 5$ as $2.15 \to 1.92 \to 1.71$ (at a first-order point the two leading colour-trivial eigenvalues—the free energies of the two coexisting phases—must become degenerate; the trend is the expected one, though at $L = 4$ the gap is far from closed); the crossing sharpness $|\mathrm{d}X_4/\mathrm{d}p|$ steepens as $5.4 \to 6.2 \to 7.2$ ($n = 3 \to 4 \to 5$); and a single-size locator: the $\log(\lambda_1/\lambda_2)$ curve has an interior minimum at $p \approx 0.48$ at $n = 5$, while at $n = 4$ the same minimum sits at the crossing region (0.38 against the extrapolated 0.383), which calibrates the $n = 5$ annealed point to $p_c^{(5)} \approx 0.48$–0.50 from the single size—on the continuing trend $0.2338 < 0.305 < 0.383 < 0.47$ away from the quenched 0.1597(8), refined by the two-size test below to $p_c^{(5)} \approx 0.47$–0.48. Every sharpness measure moves in the direction the first-order prediction requires, and the colour content is the S_5 analogue of the $n \leq 4$ pattern (σ in std ⊗ std, 16-fold as $(\dim \mathrm{std})^2$). The decisive test rests on a restructured ring contraction—the same mathematical operation as the iterative kernel, validated to 4×10^{-16} against it at $n = 2$–4, nb = 2–5, and reproducing every $L = 4$ eigenvalue to 5.5×10^{-15}—which makes the full 21-point, $L = 6$, colour-restricted sweep on the same p-grid as $L = 4$ routine, with float64 spot-checks at the decisive points (implementation, timings and memory profile in the Supplemental Material [34], Sec. S7).

The size-ladder test at $n = 5$ (Table IV). Every measured quantity moves in the first-order direction, and the closing of the two-phase gap accelerates monotonically across replica number at matched sizes. At the coexistence locator $p \approx 0.47$–0.48 (where the $\log(\lambda_1/\lambda_2)$ curve has its interior minimum at both sizes) the gap closes from 1.711 at $L = 4$ to 0.806 at $L = 6$—a factor 2.12, against 1.94 at $n = 4$ and 1.81 at $n = 3$ evaluated at the same two sizes at the corresponding locators: the continuous ($n = 3$) baseline closes slowly, the marginal ($n = 4$) faster, the first-order prediction ($n = 5$) fastest. The scaled gap $L\log(\lambda_1/\lambda_2)$ makes the contrast structural: at $n = 3$ it saturates (8.59, 7.11, 6.53, 6.21 at $L = 4, 6, 8, 10$—the energy-operator amplitude $2\pi x_\varepsilon$ of a continuous transition), whereas at $n = 5$ it falls below that envelope ($6.85 \to 4.84$ over $L = 4 \to 6$), the direction of an exponentially closing tunnelling splitting. The coexistence plateau narrows by the full size ratio ($0.060 \to 0.040$, i.e. 4/6: fixed ξ, against the ≈ 1.1 narrowing that a continuous $1/\nu = 1.15$ scaling gives); the X-curves $X_L = L\log(\lambda_1/\lambda_\sigma)$ cross at $p \simeq 0.449$ with slope ratio 2.18 between the two sizes (the $n = 3$ continuous baseline measures $1.69 \approx (6/4)^{1.15}$); and the gap-minimum locator is stable across the ladder (0.48, 0.47, 0.465 at $L = 4, 6, 8$, the last on the finer grid); the sequence is monotone decreasing, no size extrapolation is applied at $n = 5$ (the $n = 3$ and $n = 4$ points state theirs), and the quoted $p_c^{(5)} \approx 0.47$–0.48 is accordingly a plausibility range over the locator trend rather than a derived interval—an extrapolated value would sit at or below the last locator 0.465, the firm constraint being $p_c^{(5)} > p_c^{(4)} \approx 0.383$. The $q = n$-Potts first-order pre-

diction is therefore *supported* at the two-size level: the claim is not a prediction supported by single-size diagnostics alone but a measured finite-size trend whose every element (gap closing and its acceleration across n, subcontinuous $L\,\mathrm{gap}_{12}$, plateau narrowing at fixed ξ, slope-ratio excess) points the same way; two sizes alone cannot separate an exponential from a power-law closing, and the $L = 8$ rung ($120^4 = 2.1 \times 10^8$ bond labels, carried by the symmetry-reduced block route of the Supplemental Material [34]) supplies the third size, with the numbers below.

The $L = 8$ rung—the third rung of the size ladder $L = 4, 6, 8$ and the discriminator between exponential and power-law closing: the mom0 triv.triv sector at $n = 5$, $L = 8$ has dimension 3865 (the orbit count of $(S_5 \times S_5) \rtimes \mathbb{Z}_4$ on $(S_5)^4$), assembled as an exact dense block (Supplemental Material [34], Sec. S7; the assembler validated against the deposited $L = 4$ spectrum to 1.6×10^{-14} and the $L = 6$ blocks against unrestricted colour-restricted Arnoldi solves), with $\mathrm{gap}_{12} = \log(\lambda_1/\lambda_2)$ equal to $p = 0.44$: 0.505, $p = 0.46$: 0.481, $p = 0.47$: 0.481, $p = 0.48$: 0.492, $p = 0.50$: 0.545 over the locator grid, refined by the finer locator grid ($p = 0.455$: 0.484, $p = 0.465$: 0.480, $p = 0.475$: 0.485, $p = 0.485$: 0.501); at the refined $p = 0.465$ locator the closing factor extends to $\times 1.68$ from $L = 6$ to $L = 8$ (against 2.12 from $L = 4$ to $L = 6$), and $L\,\mathrm{gap}_{12} = 3.84$ falls further below the continuous envelope (6.53 at $L = 8$ for the $n = 3$ baseline)—the first-order reading of the $q = 5$ prediction at the three-size level. The same orbit basis serves the local-dimension extension at $d = 3$ (only the per-bond channel changes, through the Gram $D = 9$): the $d = 3$ ladder reads 1.482 ($L = 4$, $p \approx 0.85$), 0.621 ($L = 6$, $p = 0.85$), 0.453 ($L = 8$, the interior minimum at $p = 0.84$, bracketed by the rung values 0.948 at $p = 0.82$ and 0.519 at $p = 0.86$)—the gap level sits below the $d = 2$ value at every size and $8\,\mathrm{gap}_{12} = 3.62$ below the $d = 2$ value 3.84, while the interval closing factors read $\times 2.39$ then $\times 1.37$ (against $\times 2.12$, $\times 1.68$ at $d = 2$): the level ordering carries the first-order direction with local dimension, and the interval rates sit in the crossover region at both dimensions (deposits: Supplemental Material [34], Sec. S7).

The same three sizes admit a dual reading as effective rates (Table V). If the closing were asymptotically exponential—the tunnelling picture of Remark 4—then σ_{eff} would converge to the interface tension in units of the ring length; if it were a power law, α_{eff} would converge to its exponent. At matched sizes both readings rise monotonically across replica number, and the $n = 3$ control behaves as a continuous transition must: σ_{eff} decays toward zero with size while $L\,\mathrm{gap}_{12}$ saturates, so α_{eff} drifts down toward the value a saturating envelope implies. At $n = 4$ the closing sits at the marginal level over $L = 6 \to 8$ ($\alpha_{\mathrm{eff}} = 1.489$), and at $n = 5$ it is above it at every measured interval.

Three sizes do not yet separate the two asymptotics at $n = 5$, and the honest statement is directional: every matched-size rate is super-continuous and n-monotone, the level ordering continuous < marginal < first-order persists in both readings, and the accessible sizes sit in the crossover region that $q > 4$ Potts systems also display before their asymptotic exponential regime. The rates add to the size-ladder test a second, model-free summary of the same deposits, and they are the quantities a finite-size extraction of the interface tension would converge to if the asymptotic regime were reached. The test is therefore registered with its completion rather than declared finished: if the $n = 5$ closing is genuinely exponential, the fourth size ($L = 10$) will show σ_{eff} stabilizing at a positive interface tension, and if the asymptotic is a slowly decaying power law, σ_{eff} will continue to decay; the current data cannot distinguish these outcomes, and the registered conditional—not a performed discrimination—is the claim.

Scope. Exact finite-size data with systematic extrapolation, $d = 2$, $L \le 10$ at $n = 4$ and $L \le 8$ at $n = 5$ (the local-dimension extension $d = 3$ at $n = 5$ through $L = 8$); the universality assignments rest on the ratio, slope and gap law, not on a proof; the $n = 5$ first-order test is three-size, its every measured quantity in the first-order direction.

Remark 4 (Why $q = n$, and why the annealed points recede from the quenched one). Two structural facts organize the whole chain. (i) *Why the Potts classes appear.* The site channel $T_p(\sigma, \tau) = (1-p)d^{c(\sigma^{-1}\tau)} + pd$ is central in $\mathbb{Q}[S_n]$, because the cycle count c is a class function; the transfer problem therefore decomposes over the irreducible representations of S_n, in the same way as the q-state Potts transfer matrix decomposes over its representation sectors, and at $n = 2, 3$ the measured amplitude ratios and slope exponents land on the $q = n$ values. The replica permutations play the role of the Potts spin algebra: the standard representation of S_n has dimension $n - 1 = q - 1$, the number of Potts spin components, and the σ excitation lives in std $\otimes$ std with multiplicity $(n-1)^2$, while the leading symmetric excitation sits in the trivial sector—the energy–spin hierarchy of the Potts spectrum. The identification remains empirical (no Temperley–Lieb or free-fermion equivalence is exhibited for $n \ge 3$), but every measured number supports it, and the $q > 4$ first-order criterion then predicts, correctly, what the $n = 5$ diagnostics and the size-ladder test measure. (ii) *Why the annealed points move away from the quenched transition as n grows.* The annealed average $\mathbb{E}_\omega Z^n_{q,\omega}$ is the disorder law tilted toward high-Z realizations, and the tilt sharpens with n; Proposition 8 makes this exact through the monotone ladder $g(r)/r$ and the tilted variance. For this circuit family, high-Z disorder means measurement-poor records—the p_R^2 record-collision weight of Sec. II E is the $n = 2$ instance, and its effective density shift $p \to p_{\mathrm{eff}} < p$ (Appendix B) is the mechanism in microcosm—so the annealed singularity sits at a higher monitoring rate than the quenched one and recedes further as the tilt strengthens: the chain $0.1597(8) < 0.2338 < 0.305 < 0.383 < 0.47$ is the n-axis

TABLE IV. The $n = 5$ size-ladder test ($d = 2$; colour-trivial sector; float32 sweep validated to 4.3×10^{-6} against the deposited $L = 4$ values with float64 spot-checks at the decisive points). Closing factors at matched sizes and locators: $n = 3$ (continuous) closes slowest, $n = 5$ fastest, and the ordering persists at the third size; $L\,\mathrm{gap}_{12}$ saturates at $n = 3$ and falls below the continuous envelope at $n = 5$. The $L = 8$ column is available at $n = 3$ from the deposited $n = 3$ chain, at $n = 4$ from the deposited λ_ε top-up (Supplemental Material [34], Sec. S7, hashed in `certificate_sha256_v9.txt`), and at $n = 5$ from the symmetry-reduced block route, the $L = 8$ value being the minimum of the finer locator grid ($p = 0.455$–0.485, minimum at $p = 0.465$; Supplemental Material [34], Sec. S7).

	$n = 3$ ($p = 0.30$)	$n = 4$ ($p = 0.38$)	$n = 5$ ($p = 0.47$)
$\mathrm{gap}_{12}(L{=}4)$	2.147	1.919	1.711
$\mathrm{gap}_{12}(L{=}6)$	1.185	0.990	0.806
$\mathrm{gap}_{12}(L{=}8)$	0.816	0.645	0.480
closing factor $L{=}4 \to 6$	1.81	1.94	2.12
closing factor $L{=}6 \to 8$	1.45	1.54	1.68
$4\,\mathrm{gap}_{12} \to 6\,\mathrm{gap}_{12}$	$8.59 \to 7.11$	$7.67 \to 5.94$	$6.85 \to 4.84$
$8\,\mathrm{gap}_{12}$	6.53	5.16	3.84

TABLE V. Effective closing rates of the two-phase gap at the coexistence locators, computed from the size-ladder values of Table IV (the$n = 5$ $L{=}8$ value being the finer-grid minimum; the $n = 3$ $L = 10$ rate from the deposited envelope value $10\,\mathrm{gap}_{12} = 6.21$). $\sigma_{\mathrm{eff}} = \frac{1}{\Delta L}\log[\mathrm{gap}_{12}(L)/\mathrm{gap}_{12}(L{+}\Delta L)]$ is the exponential (interface-tension) reading; $\alpha_{\mathrm{eff}} = \log[\mathrm{gap}_{12}(L)/\mathrm{gap}_{12}(L{+}\Delta L)]/\log[(L{+}\Delta L)/L]$ the power-law reading, per consecutive-size pair ($\Delta L = 2$). At matched intervals both rates rise monotonically with n; at $n = 3$ they decay with size as a continuous transition requires, at $n = 4$ they sit at the marginal level, and at $n = 5$ they stay above it. Extraction script and cross-checks: Supplemental Material [34], Sec. S7.

	$n = 3$ ($p = 0.30$)	$n = 4$ ($p = 0.38$)	$n = 5$ ($p = 0.47$)
σ_{eff}, $L{=}4 \to 6$	0.297	0.331	0.376
σ_{eff}, $L{=}6 \to 8$	0.187	0.214	0.259
σ_{eff}, $L{=}8 \to 10$	0.137	—	—
α_{eff}, $L{=}4 \to 6$	1.466	1.632	1.857
α_{eff}, $L{=}6 \to 8$	1.297	1.489	1.802
α_{eff}, $L{=}8 \to 10$	1.224	—	—

trace of that single mechanism. At a first-order point the same picture identifies the gap $\log(\lambda_1/\lambda_2)$: the two leading colour-trivial eigenvalues are the free energies of the two coexisting phases, and the gap is their tunnelling splitting—exponentially small in L through an interface whose tension the latent heat sets, against the power-law gap of a continuous transition. That is why the closing-factor and $L\,\mathrm{gap}_{12}$ diagnostics of Table IV separate the two behaviours, and why the coexistence plateau narrows at fixed ξ.

III. BORN NORMALIZATION, REPLICA DATA, AND COMPLEXITY

A. Correct Mellin derivative and a scalar-moment no-go

For a finite circuit with Kraus operator K_R for the record R, let $\tilde\rho_R = \mathsf{K}_R\rho_0\mathsf{K}_R^\dagger$ be the unnormalized conditional state, $p_R = \mathrm{Tr}\,\tilde\rho_R$, $\rho_R = \tilde\rho_R/p_R$ for $p_R > 0$, and $\tilde\rho_{A,R} = \mathrm{Tr}_{\bar A}\,\tilde\rho_R = p_R\rho_{A,R}$. Let records with positive Born probability form $\mathcal{R}_+$, so $\sum_R p_R = 1$, and let $x_R = \mathrm{Tr}(\rho_{A,R}^n) \in (0, 1]$. Define

$$f_n(z) = \sum_{R\in\mathcal{R}_+} p_R x_R^z = \sum_R p_R^{1-nz}\big[\mathrm{Tr}(\tilde\rho_{A,R}^{\,n})\big]^z. \tag{56}$$

Proposition 6 (Born-weighted replica derivative)**.** *For finite $\mathcal{R}_+$,*

$$\overline{S}_n(A) = \frac{1}{1-n}f_n'(0) = \frac{1}{1-n}\,\frac{d}{dz}\log f_n(z)\Big|_{z=0}. \tag{57}$$

Moreover f_n is entire and $|f_n(z)| \le 1$ for $\mathrm{Re}\,z \ge 0$.

Proof. The sum is finite, so differentiation term by term gives $f_n'(0) = \sum_R p_R \log x_R$. Since $f_n(0) = \sum_R p_R = 1$, the logarithmic derivative is equal to the ordinary derivative at zero. The Rényi definition supplies $(1-n)^{-1}$. Finally, f_n is a finite exponential sum and hence entire, while $|x_R^z| = x_R^{\mathrm{Re}\,z} \le 1$ in the right half-plane. □

Thus the undifferentiated $z \to 0$ limit is 1, not an entropy. The complete sequence $f_n(k)$, $k = 0, 1, \ldots$, is the

Hausdorff moment sequence of the finite probability measure $\sum_R p_R \delta_{x_R}$ on $[0,1]$ and therefore determines that measure uniquely. This uniqueness follows from density of polynomials in $C[0,1]$; it does not license an arbitrary analytic interpolant from finitely many replica values. More importantly, the symmetric scalar moments below encode a different measure.

Theorem 6 (Symmetric scalar moments do not determine quenched entropy)**.** *Fix $n > 1$. There is no rule, linear or nonlinear, that takes only*

$$\left\{ \widetilde{F}_n(k) = \sum_R (p_R^n x_R)^k : k = 1, 2, \dots \right\} \tag{58}$$

and returns $\sum_R p_R \log x_R$ correctly for every finite Born ensemble. The conclusion remains true if the canonical continuation $\widetilde{F}_n(z) = \sum_R (p_R^n x_R)^z$ is also supplied.

Proof. Choose $u \in (0,1)$, $u \neq 1/2$, and $0 < \varrho < \min\{u, 1-u, 1/2\}^n$. Ensemble I has probabilities $(1/2, 1/2)$ and $x_i = \varrho / p_i^n$; ensemble II has probabilities $(u, 1-u)$ and the same rule $x_i = \varrho/p_i^n$. All x_i lie in $(0,1]$, and both ensembles give $\widetilde{F}_n(k) = 2\varrho^k$, indeed $\widetilde{F}_n(z) = 2\varrho^z$. Their desired averages are

$$\sum_i p_i \log x_i = \log \varrho - n \sum_i p_i \log p_i = \log \varrho + n H(p_1, p_2), \tag{59}$$

which differ because $H(u, 1-u) < \log 2$. Identical input data therefore require two outputs. The values are physically realizable: for any $x \in (0,1]$, in sufficiently large finite dimension one can continuously interpolate a density matrix between a pure state and the maximally mixed state until $\operatorname{Tr} \rho^n = x$, purify it, and prepare the purification controlled by the classical record. □

The theorem concerns scalar symmetric data; record-resolved methods, an asymmetric generating functional computing Eq. (56), and special ensembles with additional constraints lie outside its hypotheses. What exactly is missing from the symmetric data is made precise by the following statement.

Proposition 7 (Bivariate moments suffice)**.** *The finite measure $\mu = \sum_R \delta_{(p_R, x_R)}$ on $[0,1]^2$ is determined by the bivariate moment family $M(j,k) = \sum_R p_R^j x_R^k$, $j, k \in \{0, 1, 2, \dots\}$; consequently that family determines $f_n(z)$ for all z and $\overline{S}_n(A)$. The symmetric data* (58) *are the diagonal slice $\widetilde{F}_n(k) = M(nk, k)$, and Theorem 6 shows that this slice does not suffice. Whether a one-parameter family of the form $M(nk + j_0, k)$ or a parametric constraint $p_R = g(x_R)$ suffices is not asserted.*

Proof. Polynomials in two variables are dense in $C([0,1]^2)$ (Stone–Weierstrass), so two finite measures with equal bivariate moments agree on all continuous functions and hence coincide. Then $f_n(z) = \int p\, x^z \, d\mu(p, x)$. □

B. Worst-case trajectory complexity

The precise definitions, the completeness proof, and the decision corollaries are collected in Appendix C; we state the results. For the finite universal gate set generated by Hadamard, Toffoli, and computational-basis permutations, the promise problem Conditional-Trajectory Decision (CTD)—given a polynomial-size circuit $\mathcal{C}$, a computational-basis record R with $p_R > 0$, and an output qubit, decide whether $\Pr[\text{out} = 1 \mid R] \geq \frac{2}{3}$ (YES) or $\leq \frac{1}{3}$ (NO)—is complete for PostBQP under promise-preserving polynomial-time many-one reductions (Proposition 16); since PostBQP=PP [38], its promised instances are PP-hard and the decision rule has a PP extension, and completeness persists under compilation to one-dimensional nearest-neighbour architectures, already for terminal postselection. Two corollaries carry the hardness to estimation. Additive-1/7 estimation of a one-qubit conditional expectation $\operatorname{Tr}(|1\rangle\langle 1| \rho_R)$ decides CTD, so the estimation problem is PP-hard under randomized polynomial-time Turing reductions and, at inverse-polynomial precision given in unary, lies in $\mathrm{FP}^{\mathrm{PP}}$; and the conditional-purity decision problem CPD is PostBQP-complete, so the conditional Rényi-2 entropy inherits a PP promise pair. The boundary of efficient estimability is sharp: for fixed integers $n \geq 2$, $k \geq 0$ and $j \geq \max\{nk, 1\}$, all $O(1)$, the bivariate moments $M(j,k)$ of Proposition 7 are BQP-estimable by coherent record-collision tests, while the Born average $M(1,1)$ and every $j < nk$ sit outside this family, and relative-error estimation of record probabilities already decides CTD. All statements are worst-case, for universal circuit descriptions; the ensemble randomness of the physical circuits plays no role in them, and neither an average-case Haar hardness result nor hardness of the scalar Born-averaged entropy follows (Sec. IV F; Discussion).

IV. EXTENSIVE OBSERVABLES AND RECORD MULTIFRACTALITY

A. Why endpoint entropy increments do not define a time SCGF

For any trajectory and subsystem A,

$$\sum_{t=1}^{T} [S_2(A,t) - S_2(A,t-1)] = S_2(A,T) - S_2(A,0). \tag{60}$$

At fixed L the right-hand side is bounded in magnitude by $|A| \log d$. Therefore its logarithmic moment generating function divided by T vanishes as $T \to \infty$. Resampling cannot alter this pathwise identity. Endpoint entropy may instead be made extensive in a spatial thermodynamic limit, but it is not a time-additive observable at fixed size.

B. Born-record surprisal and its exact transforms

Fix circuit disorder ω, including gates and either a fixed measurement schedule or the random schedule itself. Let X_{t-1} be the normalized conditional state before event t, Y_t the outcome, and

$$r_{\omega,t}(y \mid X_{t-1}) = \Pr_{\omega}(Y_t = y \mid X_{t-1}) \tag{61}$$

its normalized predictive Born probability. If schedule coins are included in Y_t, their Bernoulli probabilities must be included in r; otherwise the schedule is held fixed as part of ω. All logarithms are natural, so rates are measured in nats per event; a base-b convention divides every exponent below by $\log \mathsf{b}$.

Definition 2 (Sample and disorder-averaged quenched SCGFs)**.** *The path surprisal and finite-time sample log-MGF are*

$$\mathcal{A}_T = -\sum_{t=1}^{T} \log r_{\omega,t}(Y_t \mid X_{t-1}) = -\log P_\omega(Y_{1:T}), \tag{62}$$

$$\phi_{L,T,\omega}(k) = \frac{1}{T} \log \mathbb{E}_{P_\omega} e^{k\mathcal{A}_T} = \frac{1}{T} \log \sum_{Y:P_\omega(Y)>0} P_\omega(Y)^{1-k}. \tag{63}$$

When the limits exist, set $\psi_{L,\omega}(k) = \lim_{T\to\infty} \phi_{L,T,\omega}(k)$ and $\psi_L^{\mathrm{q}}(k) = \lim_{T\to\infty} \mathbb{E}_\omega \phi_{L,T,\omega}(k)$. The latter is a disorder-averaged quenched free energy, not by itself the log-MGF of one probability law. The annealed expression with the disorder average inside the logarithm is a third object; at $k = -1$ it is computed exactly by the transfer matrix of Sec. II D (Remark 1), and its spatial thermodynamic limit per site and period, $\log \lambda_\infty(p)$ of Proposition 4, exists and is real-analytic in $p \in (0,1)$ except at $p_c^{(2)}(d)$; more generally $k = 1 - n$ is computed by the n-replica operator of Theorem 5, whose leading eigenvalue is real and positive for every n.

For every finite T and fixed ω, $\phi_{L,T,\omega}$ is convex on its domain, $\phi_{L,T,\omega}(0) = 0$, and, whenever the derivatives are finite,

$$\phi'_{L,T,\omega}(0) = \frac{1}{T}\mathbb{E}_{P_\omega}\mathcal{A}_T, \quad \phi''_{L,T,\omega}(0) = \frac{1}{T}\operatorname{Var}_{P_\omega}(\mathcal{A}_T) \geq 0. \tag{64}$$

These are finite-record identities, requiring no long-time or self-averaging assumption. For Clifford circuits with stabilizer inputs and Pauli measurements the variance vanishes identically (Remark 2).

For a fixed time-homogeneous environment with outcome kernel $K_\omega(x, dy)$, update $F_\omega(x, y)$, and additive increment $g_\omega(x, y)$, define the Feynman–Kac operator [40]

$$(\mathcal{Q}_{k,\omega} f)(x) = \int K_\omega(x, dy) e^{k g_\omega(x,y)} f(F_\omega(x, y)). \tag{65}$$

If a Perron–Frobenius theorem applies and the initial measure has nonzero overlap with the leading positive eigendirection, then $\psi_{L,\omega}(k) = \log \rho(\mathcal{Q}_{k,\omega})$, and $k \mapsto \psi_{L,\omega}(k)$ is real-analytic wherever that leading eigenvalue is simple and isolated. Temporally refreshed disorder instead requires a lifted stationary environment or a random-operator Lyapunov theorem; Eq. (65) alone does not cover it. For a finite reachable state set with i.i.d. tilted operators $\mathcal{Q}_{k,t}$, the Furstenberg–Kesten theorem [41] gives the almost-sure limit of $T^{-1} \log \|\mathcal{Q}_{k,T} \cdots \mathcal{Q}_{k,1}\|$ whenever $\mathbb{E} \log^+ \|\mathcal{Q}_{k,1}\| < \infty$; identifying that norm Lyapunov exponent with the specific matrix element $\langle \mathbf{1}|\mathcal{Q}_{k,T} \cdots \mathcal{Q}_{k,1}|\delta_{x_0}\rangle$ that defines $\phi_{L,T,\omega}$ requires a positivity (Doeblin-type) condition on the reachable set, and analyticity of the top Lyapunov exponent in k requires strictly positive, or at least allowable, random matrices in the sense of Ruelle and Peres [42, 44]; irreducibility and aperiodicity of the unperturbed chain do not suffice. Two monitored-circuit families have an exactly finite reachable set. Clifford circuits with stabilizer inputs are one, and for them the whole construction is trivial (Remark 2). The fully monitored point $p = 1$ is the other, and it is not trivial: when every site is measured in every layer, grouping each measurement layer into one event makes the record a Markov chain on the d^L computational-basis states with kernel $K(y \mid x) = \prod_{\text{bonds}} |U_b(y_b, x_b)|^2$ for arbitrary gates, and for fixed time-periodic gates with all $|U_b(y_b, x_b)| > 0$ the finite-state theory applies verbatim: $\psi_{L,\omega}(k) = \log \rho(\mathcal{Q}_{k,\omega})$ is convex and real-analytic on all of $\mathbb{R}$, $\psi'_{L,\omega}(0)$ is the record entropy rate and $\psi''_{L,\omega}(0) > 0$ for generic gates (at $L = 4$, $d = 2$, one Haar draw gives $\psi'(0) = 4.272$ nats per period and $\psi''(0) = 1.65$). For gates refreshed every layer the tilted period matrices are i.i.d., almost surely entrywise positive, and satisfy $\mathbb{E} \log^+ \|\mathcal{Q}_{k,1}\| < \infty$ for every k because $\mathbb{E}|\log |U_{yx}|^2| < \infty$ under Haar measure; the Furstenberg–Kesten limit therefore exists for every k at fixed L, is deterministic, and equals the growth rate of the matrix element, since the column ratios of a product of positive matrices are controlled by its first factor, whose entry logarithms are integrable. Its analyticity in k is no longer conditional: Theorem 9 and Corollary 1 below close it for exactly this family, since almost-surely entrywise strictly positive tilted matrices are allowable. The same point has a closed-form annealed record SCGF, $\mathbb{E}_U \sum_y P(y \mid x)^{1-k} = d^2\Gamma(d^2)\Gamma(2-k)/\Gamma(d^2+1-k)$ per bond, which is finite only for $k < 2$ although the quenched $\psi_{L,\omega}(k)$ is finite for all k; at $k = -1$ and $k = -2$ it reproduces the $p = 1$ bond eigenvalue $a+c = 2/(d^2+1)$ of Eq. (18) and the $n = 3$ case of Eq. (54). This is entanglement-trivial but record-nontrivial, and it illustrates concretely that the annealed SCGF is a third object.

Theorem 7 (Quenched record large deviations and multifractal identities)**.** *Assume that for almost every fixed disorder realization ω the sample limit $\psi_{L,\omega}(k) = \lim_{T\to\infty} \phi_{L,T,\omega}(k)$ exists in $(-\infty, +\infty]$ for every $k \in \mathbb{R}$,*

that 0 is an interior point of $\{k : \psi_{L,\omega}(k) < \infty\}$, and that $\psi_{L,\omega}$ is lower semicontinuous and essentially smooth (differentiable in the interior of its domain and steep at its boundary). Then under P_ω, $h = \mathcal{A}_T/T$ obeys the quenched full LDP with good rate function

$$I_{L,\omega}(h) = \sup_k\{kh - \psi_{L,\omega}(k)\}. \tag{66}$$

If additionally $\psi_{L,\omega} = \psi_L$ almost surely, then the rate is self-averaging and deterministic.

For $Z_{q,\omega}(T) = \sum_{Y:P_\omega(Y)>0} P_\omega(Y)^q$ define, when the limit exists,

$$\tau_{L,\omega}(q) = -\lim_{T\to\infty} \frac{\log Z_{q,\omega}(T)}{T}. \tag{67}$$

Then exactly

$$\tau_{L,\omega}(q) = -\psi_{L,\omega}(1-q), \tag{68}$$

$$D_{L,\omega}(q) = \frac{\tau_{L,\omega}(q)}{q-1} \quad (q \neq 1), \qquad D_{L,\omega}(1) = \tau'_{L,\omega}(1), \tag{69}$$

$$\alpha(q) = \tau'_{L,\omega}(q), \qquad f_{L,\omega}(\alpha(q)) = q\alpha(q) - \tau_{L,\omega}(q), \tag{70}$$

whenever the indicated derivatives and exposed saddle exist. At exposed points of the sample LDP,

$$f_{L,\omega}(\alpha) = \alpha - I_{L,\omega}(\alpha). \tag{71}$$

Normalization gives $\tau_{L,\omega}(1) = 0$. If the sample limits are uniformly integrable, disorder averaging also gives $\tau^{\mathrm{q}}_L(q) = -\psi^{\mathrm{q}}_L(1-q)$; without self-averaging, however, a Legendre transform and a disorder average need not commute.

Proof. The first assertion is the Gärtner–Ellis theorem (Ref. [45], Theorem 2.3.6, with essential smoothness as in Definition 2.3.5 there) applied separately under each fixed P_ω to the variables $\mathcal{A}_T/T$ at speed T, whose scaled log-MGF is $\phi_{L,T,\omega}$; it is not applied to $\mathbb{E}_\omega\phi_{L,T,\omega}$. The hypotheses are stated on all of $\mathbb{R}$ because the increments $-\log r_t$ are unbounded, so exponential tightness is not automatic and must come from $0 \in \mathrm{int}\{\psi < \infty\}$; a limit on an interval alone would give only a weak LDP. Since $\mathcal{A}_T = -\log P_\omega(Y)$,

$$\mathbb{E}_{P_\omega} e^{(1-q)\mathcal{A}_T} = \sum_Y P_\omega(Y)P_\omega(Y)^{q-1} = Z_{q,\omega}(T), \tag{72}$$

so $\phi_{L,T,\omega}(1-q) = T^{-1}\log Z_{q,\omega}(T)$, and letting $T \to \infty$ proves Eq. (68). In Eq. (70), $D(q)$, $\alpha(q)$ and $f(\alpha(q))$ are the generalized dimension, singularity strength and singularity spectrum defined from τ; $D(1) = \tau'(1)$ is the $q \to 1$ limit of $\tau(q)/(q-1)$ because $\tau(1) = 0$. For Eq. (71) let $\alpha = \alpha(q) = \tau'(q) = \psi'_{L,\omega}(1-q)$ be an exposed point. Since $\psi_{L,\omega}$ is convex and differentiable there, the supremum in Eq. (66) is attained at $k = 1-q$, so $I_{L,\omega}(\alpha) = (1-q)\alpha - \psi_{L,\omega}(1-q)$ and therefore

$$\alpha - I_{L,\omega}(\alpha) = q\alpha + \psi_{L,\omega}(1-q) = q\alpha - \tau_{L,\omega}(q) = f_{L,\omega}(\alpha).$$

(Interpretation: records with surprisal rate α number $e^{Tf(\alpha)}$ and carry Born weight $e^{-T\alpha}$ each, so their total probability decays at the rate $\alpha - f(\alpha) = I(\alpha)$.) Finally $Z_{1,\omega} = 1$ gives $\tau_{L,\omega}(1) = 0$. For the averaged identity, $\mathbb{E}_\omega\phi_{L,T,\omega}(1-q) = \mathbb{E}_\omega T^{-1}\log Z_{q,\omega}(T)$ holds at every finite T, and uniform integrability of the sample limits justifies interchanging the disorder expectation and the long-time limit on both sides. □

The physical record q-ensemble is

$$P_{q,\omega}(Y) = \frac{P_\omega(Y)^q}{Z_{q,\omega}(T)}, \tag{73}$$

so its cloning field is $k = 1-q$, not $k = q$. A finite-q freezing claim requires a thermodynamic limiting linear branch of $\tau(q)$ and a corresponding nonanalytic onset in $\psi(k)$; the first derivative need not jump. Concentration as $q \to \infty$ is not by itself a finite-q transition; an independently biased coin has analytic $\tau(q) = -\log[p^q + (1-p)^q]$ for every finite q, and a Clifford circuit has $\tau(q)$ exactly linear (Remark 2). The annealed Ising point $p_c^{(2)}$ of Sec. II E is a singularity in p of the annealed $\tau(2)$ per site in the spatial thermodynamic limit, not a freezing transition in q and not a quenched statement.

The identities of Theorem 7 are not vacuous with respect to freezing: there is an explicit Born-realizable family of record measures whose quenched $\tau(q)$ acquires a linear branch at a finite q_c. The family is a random-energy model [46] on the records; its members are indexed by the horizon T and are *not* the marginals of one time-homogeneous process, so the statement concerns a thermodynamic sequence of finite-horizon monitored processes, in the sense made precise in Remark 5.

Theorem 8 (Freezing in a Born-record free-energy sequence)**.** *Let $a = \log 2$ and $s_c = \sqrt{2a} = 1.17741$. On one probability space let $\{E_{T,\omega}(Y) : T \geq 1,\ Y \in \{0,1\}^T\}$ be independent Gaussian variables with $E_{T,\omega}(Y) \sim \mathcal{N}(0,T)$, and for real s put*

$$\mathcal{Z}_{s,T,\omega} = \sum_{Y\in\{0,1\}^T} e^{-sE_{T,\omega}(Y)}, \qquad P_{T,\omega}(Y) = \frac{e^{-E_{T,\omega}(Y)}}{\mathcal{Z}_{1,T,\omega}}. \tag{74}$$

(i) (Born realizability.) For every T and ω, $P_{T,\omega}$ is the record law of the sequential computational-basis measurement of the T qubits of the pure state $|\Psi_{T,\omega}\rangle = \sum_Y \sqrt{P_{T,\omega}(Y)}\,|Y\rangle$, and the predictors of Definition 2 are $r_{T,\omega,t}(y_t \mid y_{<t}) = P_{T,\omega}(y_{\leq t})/P_{T,\omega}(y_{<t})$ with $P_{T,\omega}(y_{\leq t}) = \sum_{y_{>t}} P_{T,\omega}(y_{\leq t}y_{>t})$, so that $\mathcal{A}_T = -\log P_{T,\omega}(Y)$ and $\phi_{T,\omega}(k) = T^{-1}\log\sum_Y P_{T,\omega}(Y)^{1-k}$ are exactly the objects of Eq. (63).

(ii) (Extremes.) As $T \to \infty$, almost surely, $\max_Y(-E_{T,\omega}(Y))/T \to s_c$ and $\min_Y(-E_{T,\omega}(Y))/T \to -s_c$.

(iii) (Quenched free energy.) Almost surely, simultaneously for all real s,

$$\lim_{T\to\infty}\frac{1}{T}\log \mathcal{Z}_{s,T,\omega}=\varphi(s)=\begin{cases}a+s^2/2, & |s|\le s_c,\\ |s|\,s_c, & |s|\ge s_c,\end{cases} \tag{75}$$

and the convergence is locally uniform in s; φ is convex and C^1 on $\mathbb{R}$, φ'' jumps from 1 to 0 at $\pm s_c$, and $\varphi(1)=a+\frac{1}{2}$ because $1<s_c$.

(iv) (Record exponents.) Almost surely, for every real k and every $q\ge 0$,

$$\psi(k)=\lim_{T\to\infty}\phi_{T,\omega}(k)=\varphi(1-k)-(1-k)\varphi(1), \tag{76}$$

$$\begin{aligned}\tau(q)&=-\psi(1-q)=q\varphi(1)-\varphi(q)\\ &=\begin{cases}q\big(a+\frac{1}{2}\big)-a-\frac{q^2}{2}, & 0\le q\le q_c,\\ q\big(a+\frac{1}{2}-s_c\big), & q\ge q_c,\end{cases}\end{aligned} \tag{77}$$

with $q_c=s_c$. Thus τ is C^1 on $[0,\infty)$ with a linear branch of slope $a+\frac{1}{2}-s_c=0.01574$ for $q\ge q_c$, τ'' jumps from -1 to 0 at q_c, and ψ is convex, finite and C^1 on $\mathbb{R}$ but not analytic at $k_c=1-s_c=-0.17741$ (and, on the unphysical side $q<0$, at $1+s_c=2.17741$). Explicitly $\psi(k)=\frac{1}{2}k^2+(a-\frac{1}{2})k$ for $k_c\le k\le 1+s_c$ and ψ is affine outside this interval.

(v) (Large deviations.) ψ is essentially smooth, so Theorem 7 applies to the sequence $(P_{T,\omega})_T$: the surprisal rate $h=\mathcal{A}_T/T$ satisfies the full LDP with the good rate function $I(h)=\frac{1}{2}(h-h^*)^2$ for $h\in[h_{\min},h_{\max}]$ and $I(h)=+\infty$ otherwise, where $h^*=\psi'(0)=a-\frac{1}{2}=0.19315$ is the typical surprisal rate, $h_{\min}=\psi'(k_c)=a+\frac{1}{2}-s_c=0.01574$ and $h_{\max}=\psi'(1+s_c)=a+\frac{1}{2}+s_c=2.37056$. The singularity spectrum is $f(\alpha)=\alpha-I(\alpha)$ on $[h_{\min},h_{\max}]$, with $f(h_{\min})=0$: the most probable records, $P_{T,\omega}(Y)=e^{-Th_{\min}+o(T)}$, are subexponentially many, and the tilted ensembles $P_{q,\omega}$ with $q>q_c$ all concentrate on them.

Proof. The proof—the extremal estimates of item (ii), the frozen and annealed upper bounds with the Tsirelson–Ibragimov–Sudakov concentration, the truncation lower bound with its countable-intersection uniformity, and the Legendre algebra of the affine branches in item (v)—is given in the Supplemental Material [34], Sec. S5. □

Remark 5 (Scope of Theorem 8). The measures $P_{T,\omega}$ are defined separately for each horizon T from independent energies and are not projectively consistent: $P_{T+1,\omega}(y_{1:T}0)+P_{T+1,\omega}(y_{1:T}1)\neq P_{T,\omega}(y_{1:T})$ almost surely. Theorem 8 is therefore a statement about the thermodynamic limit of a *sequence* of finite-horizon monitored processes, each realized by a nonlocal, disorder-dependent state on T qubits, and not about the long-time SCGF of one fixed monitored dynamics of the kind treated by the Feynman–Kac operator (65); it says nothing about local Haar or Clifford brickwork circuits or about the measurement-induced entanglement transition. Its content is that the exact identities of Theorem 7 accommodate a genuine finite-q freezing transition and that this transition can be invisible in ψ' and in the interior of the rate function: ψ remains essentially smooth, the LDP holds, and the nonanalyticity appears only as the onset of the affine branches at the edges $h_{\min},h_{\max}$ of the support of I. Any freezing claim for a circuit ensemble must therefore be tested on τ'' or on the linearity of τ over an interval, not on τ'. A deterministic-environment analogue with feedback is the Curie–Weiss record measure $P_{\beta,T}(\sigma)\propto\exp[\beta(\sum_i\sigma_i)^2/2T]$ on $\sigma\in\{\pm1\}^T$, realized by adaptive coins whose predictor at step t is $F_{t+1}(m+\sigma_t)/[F_{t+1}(m+1)+F_{t+1}(m-1)]$ with $m=\sum_{i<t}\sigma_i$ and $F_t(m)=\sum_{\sigma_t,\dots,\sigma_T}\exp[\beta(m+\sum_{j\ge t}\sigma_j)^2/2T]$ (a logistic rule in m/T is the single-spin Glauber conditional, not this predictor); there $\tau(q)=qf_{\mathrm{CW}}(\beta)-f_{\mathrm{CW}}(q\beta)$ with f_{CW} the Curie–Weiss free energy, which for $\beta<1$ is nonanalytic at $q=1/\beta$, but with the linear branch on $[0,1/\beta]$ and the curved branch beyond it, the reverse of freezing.

C. No freezing for finite reachable sets, and the replica interpolation identity

The freezing question raised by Theorem 8 can be closed for a defined class of monitored processes, and the disorder-replica interchange can be reduced to an exact identity with a measurable criterion. Call the monitored process *finite reachable* if its tilted Feynman–Kac dynamics closes on a finite state set S, the operators $\mathcal{Q}_{k,t}$ of Eq. (65) being $|S|\times|S|$ nonnegative matrices, and *allowable* if the i.i.d. tilted matrices are almost surely entrywise strictly positive (some finite product in the support of the disorder law is strictly positive, in the sense of Ruelle and Peres [42, 44]) and strongly irreducible. Clifford circuits with stabilizer inputs are finite reachable with degenerate (zero-variance) record statistics (Remark 2); the fully monitored point $p=1$ with layer-refreshed gates is finite reachable and, under Haar refreshment, allowable, since every tilted period matrix is almost surely entrywise strictly positive.

Theorem 9 (No finite-q freezing for allowable finite-reachable processes)**.** *Let the monitored process be finite reachable with i.i.d. tilted operators $\mathcal{Q}_{k,t}$ that are allowable for every k in a compact interval and satisfy $\mathbb{E}|\log\min_{ij}(\mathcal{Q}_{k,1})_{ij}|<\infty$ and $\mathbb{E}\log^+\|\mathcal{Q}_{k,1}\|<\infty$. Then: (i) for almost every ω and all fixed positive vectors u,v, $\psi_{L,\omega}(k)=\lim_{T\to\infty}T^{-1}\log\langle u|\mathcal{Q}_{k,T}\cdots\mathcal{Q}_{k,1}|v\rangle$ exists, is deterministic, and equals the top Lyapunov exponent $\gamma(k)$ of the i.i.d. cocycle: products of almost-surely strictly positive matrices contract the positive cone in the Hilbert projective metric, so every positive functional shares the norm growth rate [41]; (ii) $k\mapsto\gamma(k)$ is real-analytic on*

all of $\mathbb{R}$ *under the stated allowability [42–44]; (iii) consequently* $\tau_L^{\mathrm{q}}(q) = -\gamma(1-q)$ *is real-analytic on* $\mathbb{R}$*, and no finite-q freezing transition of the kind exhibited by Theorem 8 occurs. For if* $\psi_L''(0) > 0$ *(nondegenerate records), an analytic function affine on an open interval would be affine everywhere by the identity theorem, contradicting strict convexity at* $k = 0$*; in the degenerate case* τ *is affine everywhere with no finite-q onset. If the reachable set decomposes into* $r \geq 2$ *invariant communicating classes with allowable restrictions, then* $\psi = \max_{1\leq i\leq r} \gamma_i$ *develops corners at exchanges—a quenched dynamical phase transition detected by jumps of* τ'*, realizable by projectively consistent local dynamics with a conserved sector—but every branch is analytic on all of* $\mathbb{R}$*, so still no affine branch arises. Freezing requires an* unbounded *reachable set, as in the random-energy sequence of Theorem 8.*

Corollary 1 (The fully monitored point, unconditional)**.** *At* $p = 1$ *with Haar-refreshed gates the tilted period matrices are almost surely entrywise strictly positive with* $\mathbb{E}|\log \min_{ij} K_{ij}| < \infty$*, so Theorem 9 applies: the quenched record SCGF of that family exists for every* k*, is deterministic and real-analytic on* $\mathbb{R}$*, and its* τ *has no affine branch. The analyticity conditionality flagged above is removed for this family.*

Proposition 8 (Replica interpolation identity)**.** *Let* $X > 0$ *be a random variable with* $\mathbb{E}|\log X| < \infty$ *and* $\mathbb{E}X^r < \infty$ *for* r *in a neighbourhood of* $[0, m]$*, and set* $g(r) = \log \mathbb{E}X^r$*, with* Var_r *the variance under the* r*-tilted law* $\Pr_r(\mathrm{d}x) \propto X^r \Pr(\mathrm{d}x)$*. Then exactly*

$$g(r) - r\,\mathbb{E}\log X = \int_0^r (r-s)\ \mathrm{Var}_s(\log X)\,\mathrm{d}s, \qquad (78)$$

$r \mapsto g(r)/r$ *is nondecreasing on* $(0,\infty)$ *with* $\lim_{r\downarrow 0} g(r)/r = \mathbb{E}\log X$ *and* $\lim_{r\to\infty} g(r)/r = \log \operatorname{ess\,sup} X$ *for bounded* X*, and* $\partial_r \mathbb{E}X^r|_{r=0} = \mathbb{E}\log X$*.*

Proof. g is C^∞ near $[0, m]$ with $g(0) = 0$, $g'(r) = \mathbb{E}_r \log X$ and $g''(r) = \mathrm{Var}_r(\log X) \geq 0$, the classical exponential-family identities; Eq. (78) is Taylor's formula with integral remainder. Monotonicity of $g(r)/r$ is Lyapunov's inequality (the power means $(\mathbb{E}X^r)^{1/r}$ are nondecreasing in r), the $r \downarrow 0$ limit is l'Hôpital's rule with $g(0) = 0$, the $r \to \infty$ limit for bounded X is classical, and the final identity is dominated convergence. □

Applied to $X = Z_{q,\omega}(T)$ at fixed (L, T), Proposition 8 organizes the disorder-replica structure: the annealed replica free energies $f_m = (1/m)\log \mathbb{E}_\omega Z_{q,\omega}^m$ form a monotone family squeezed between the quenched value $\mathbb{E}_\omega \log Z_{q,\omega}$, approached quadratically as $m \downarrow 0$ ($f_m = \mathbb{E}\log Z + \frac{m}{2}\mathrm{Var}_0(\log Z) + O(m^2)$), and the extremal value $\log \operatorname{ess\,sup}_\omega Z$ as $m \to \infty$. The replica derivative $\partial_m \mathbb{E}_\omega Z^m|_{m=0} = \mathbb{E}_\omega \log Z$ is exact at finite (L, T); what remains open is its interchange with $L, T \to \infty$, and Eq. (78) reduces that interchange to the tilted-variance profile Var_s along $s \in [0, m]$: a sufficient condition is the self-averaging bound $\sup_T \mathrm{Var}\big(T^{-1}\log Z_{q,\omega}(T)\big) < \infty$, a measurable quantity for the sequential Monte Carlo machinery of the next subsection, while the naive $m \to \infty$ replica limit overshoots the quenched object to the extremal value and is not an interchange at all.

Numerical verification (script `no_freeze_theory_v1.py`, all checks pass). (i) Deterministic, i.i.d.-random, and reducible allowable chains: ψ smooth and strictly convex on $k \in [-3, 3]$, τ with no affine segment (the longest run of $|\tau''|$ below 5% of its median is a single grid point); the reducible two-sector chain shows the corner (τ' jump 0.57 in slope units) with both branches curved. (ii) The random-energy benchmark reproduces the opposite behaviour exactly: the closed form $\varphi(s) = \min$-piecewise of Theorem 8 has a terminal affine branch ($|\varphi''| \leq 9 \times 10^{-16}$ above s_c against 10^{-4} below) with φ' continuous at s_c—the side-by-side dichotomy. (iii) The $p = 1$ Haar-refreshed family ($L = 4$, 12 sweeps, 400 exact enumerations): ψ smooth and strictly convex over $k \in [-2.5, 2.5]$ (Corollary 1 empirically). (iv) The interpolation identity (78) holds to 4×10^{-8} relative on an exact synthetic law and to 6×10^{-7} relative on 6000 exactly-enumerated $p = 1$ Haar record partition functions ($\mathbb{E}\log Z = 13.256$, $\mathrm{Var}\log Z = 0.037$); the quadratic small-$m$ rate $(f_m - \mathbb{E}\log Z)/(m\,\mathrm{Var}_0/2)$ measures 0.994–0.998 at $m \in \{0.125, 0.25, 0.5\}$; the replica derivative matches $\mathbb{E}\log Z$ to 1.2%; and $f_8 = 13.390 < \log\max Z = 13.869$ exhibits the extremal overshoot. Lyapunov monotonicity is exact on a 400-point m-grid with the two one-sided limits matching their predicted values including the $(\log p_{\max})/m$ finite-m correction.

Remark 6 (Corners versus affine branches—the physical dichotomy). Theorem 9 separates two mechanisms that a glance at a plotted $\tau(q)$ can conflate. A *corner*—a jump of τ'—is phase competition: the tilted process has (at least) two invariant sectors with distinct growth rates, and the dominant one exchanges as k varies; the quenched free energy is then a maximum of analytic branches, exactly as at a first-order thermodynamic transition, and the reducible allowable chains realize it with conserved sectors. An *affine branch*—τ linear over an interval of q—is extremal domination: a single unbounded configuration class controls the growth rate throughout the interval, as in the random-energy sequence of Theorem 8, where beyond $q_c = \sqrt{2\log 2}$ the maximum energy alone fixes the free energy while τ' stays continuous. Finite-reachable allowable processes can produce the first behaviour but never the second: within a finite set the extremal configuration is bounded and the process mixes (the Hilbert-metric contraction), so the growth mode is unique and analytic in k—there is no ladder of ever-more-extreme configurations for the tilt to climb. Freezing is thus a statement about the *reachable-set geometry*, not about the record observable itself: it requires unbounded extremes, which is precisely the point on which the theorem's hypothesis and its counterexample meet. The numerical verification above displays the dichotomy side by side: corners with curved branches in the reducible

chains, an affine terminal branch only in the random-energy benchmark.

D. Sequential Monte Carlo contract

Definition 3 (Bootstrap estimator for the record SCGF)**.** *For fixed L, k, ω, initialize N normalized conditional states. At event t, independently for each particle i, sample Y_t^i from its Born predictor, update and normalize the state, and set*

$$\ell_t^i = -k \log r_t^i. \tag{79}$$

Here an "event" may be one primitive outcome or a declared compound block. For a compound block, r_t^i is the product of its sequential conditional Born probabilities and $-\log r_t^i$ is their sum; resampling only at block boundaries leaves the Feynman–Kac normalizer unchanged but can increase variance. The released implementation uses one full brickwork period as a block, so its T counts periods. With resampling at every declared event, compute

$$m_t = \max_i \ell_t^i, \tag{80}$$

$$\log c_t = m_t + \log\left(\frac{1}{N}\sum_i e^{\ell_t^i - m_t}\right), \tag{81}$$

$$\bar{w}_t^i = \frac{e^{\ell_t^i - m_t}}{\sum_j e^{\ell_t^j - m_t}}, \tag{82}$$

then resample with a conditionally unbiased scheme and reset weights. The estimates are

$$\widehat{Z}_T(k) = \prod_{t=1}^{T} c_t, \qquad \widehat{\psi}_{L,\omega}(k) = \frac{1}{T}\sum_{t=1}^{T} \log c_t. \tag{83}$$

Proposition 9 (Finite-population normalization property)**.** *Under conditionally unbiased resampling, $\widehat{Z}_T(k)$ in Eq. (83) is unbiased for $Z_T(k) = \mathbb{E}_{P_\omega} e^{k\mathcal{A}_T}$ at every finite N, T [40]. In general $\log \widehat{Z}_T$ is biased. If outcomes are drawn from an auxiliary proposal Q_t rather than the Born kernel K_t, the incremental weight must be*

$$w_t = e^{kg(x,y)} \frac{K_t(x, dy)}{Q_t(x, dy)}. \tag{84}$$

Proof. The Del Moral induction—the per-event identity $\mathbb{E}[c_t \widehat{\eta}_t^N(f) \mid \mathcal{F}_{t-1}] = \eta_{t-1}^N(\mathcal{Q}_{k,\omega} f)$ from the independent predictor draws, combined with conditionally unbiased resampling and iterated from $\widehat{Z}_0 = 1$—together with the resulting unbiasedness $\mathbb{E}\widehat{Z}_T = Z_T$ for $f = 1$, the strictness of Jensen's inequality, and the Radon–Nikodym change-of-measure identity (84), is given in the Supplemental Material [34], Sec. S6. □

If resampling is skipped at some events, normalized weights must be carried forward and the weighted particle-filter normalizer used; the equal-input formula for c_t cannot simply be reused. All particles in one quenched run share ω but have independent physical outcomes. Independent filters and independent disorder samples are top-level replicates. Averaging sample-wise log normalizers estimates the quenched object; averaging normalizers and then taking one logarithm estimates an annealed object [53, 54]. For $k = -1$ the annealed object is $\bar{Z}_2$, so the transfer matrix of Sec. II D provides an exact reference value for the annealed estimator at every L reachable by the compressed operator.

At minimum, computations must report the one-step weight concentration

$$Y_{2,t}^{\mathrm{wt}} = \sum_i (\bar{w}_t^i)^2 = N_{\mathrm{eff}}^{-1} \tag{85}$$

and long-lag genealogical concentration

$$Y_{2,t}^{\mathrm{gen}}(s) = \sum_g [n_g(t; s)/N]^2, \tag{86}$$

where $n_g(t; s)$ counts descendants of ancestor g at lag s. Convergence requires increases $N, 2N, 4N$, longer T after verified burn-in, independent disorder and particle seeds, block or hierarchical uncertainty estimates, a refined k grid, convexity of ψ, $\psi(0) = 0$, $\tau(1) = 0$, and agreement of $\psi'(0)$ with direct sampling. Clone collapse at finite N diagnoses estimator difficulty; it is not evidence of a physical transition.

E. First numerical record SCGFs for Haar circuits

The estimator of the preceding subsection is run: $N = 384$ particles per circuit, one brickwork period per block ($T = 2L$ periods), systematic resampling, the stabilized normalizer of Eq. (83), on the deposited Haar protocol ($d = 2$, periodic brickwork, $|0 \cdots 0\rangle$, schedule coins in ω), for $L = 6$–12, $p \in \{0.10, 0.1597, 0.2338, 0.4\}$ and $k \in [-2, 2]$ in half-unit steps, with 8 circuits per point (6 at $L = 12$). This is, to our knowledge, the first numerical quenched record SCGF for a nontrivial circuit ensemble: Clifford circuits have ψ exactly linear, and the fully monitored point is record-nontrivial but entanglement-trivial; between them nothing has been measured.

Validation chain. (i) $\widehat{\psi}(0) = 0$ exactly and the particle surprisal statistics reproduce the deposited Born-sampling data at matched (L, p, t) ($L = 8$, $p = 0.2338$, $t = 16$: rate 2.295 ± 0.009 against deposited 2.287 ± 0.005, $z = 0.9$). (ii) At $L = 4$ all 2^M records of a fixed circuit are enumerated, $Z_T(k) = \sum_{Y: P_\omega(Y) > 0} P_\omega(Y)^{1-k}$ computed exactly, and $\mathbb{E}\widehat{Z}_T$ over 600 independent runs agrees within 2σ for every $k \in \{-1.5, -0.5, 0.5, 1.5\}$—the unbiasedness of Proposition 9 verified directly, across four decades of Z. (iii) At $p = 1$ the exact basis-state Markov chain ($2^L \times 2^L$ tilted kernels) is compared with $\mathbb{E}\widehat{Z}_T$ over 400 runs: agreement within 1.4σ for $k \in \{-1, 0.5, 1.5\}$,

i.e. up to $Z \sim 4 \times 10^{10}$. (iv) The finite-record identities $\psi'(0) = \mathbb{E}\mathcal{A}_T/T$ and $\psi''(0) = \mathrm{Var}(\mathcal{A}_T)/T$ hold between the k-grid and the particle statistics: at $L = 8$, $p = 0.2338$, the central-difference curvature is 0.428 against the direct variance rate 0.427.

Results (Table VI). (i) *Genuine record multifractality.* $D(q)$ is not constant: at $L = 8$, $p = 0.2338$ it spans 0.85 nats per period over $q \in [-1, 3]$, and at $L = 12$ the spread is 1.19; the strength grows with the monitoring rate (D-spread at $L = 8$: 0.20, 0.32, 0.85, 1.96 at $p = 0.10, 0.1597, 0.2338, 0.4$). For Clifford circuits D is exactly constant, so this is the first measurement of the object that Sec. IV defines. (ii) *Entropy rate and variance.* $\psi'(0)$ equals the record entropy rate per period, 2.36 nats at $L = 8$, $p = 0.2338$, rising monotonically with p (1.02, 1.50, 2.36, 3.62 at the four rates), and with L at fixed rate sub-extensively: the per-site rate $A_T/(TL)$ is L-independent to 5% over $L = 6$–10 at $p = 0.2338$ (0.287, 0.295, 0.288 nats per site-period); the variance rate $\psi''(0)$ likewise rises (0.09 to 0.99), with no nonanalyticity resolved at the quenched p_c at these sizes. (iii) *No finite-q freezing.* Over the measured range $q \leq 3$ and all (L, p), the smallest magnitude of the discrete second difference of τ is 0.09 nats per period: τ remains strictly convex, so no linear branch—the freezing signature of Theorem 8—appears; these are the first finite-size bounds on that question, though $T = 2L$ and $L \leq 12$ are far from the asymptotic regime in which the theorem's criteria would apply. (iv) *Annealed versus quenched.* At $L = 8$, $p = 0.2338$, $k = 1$ the annealed value $(1/T) \log \mathbb{E}_\omega Z_{T,\omega} = 2.75$ exceeds the quenched 2.60, the disorder average failing to commute with the logarithm—the two are distinct objects, as Sec. IV insists, here at the 6% level and growing with L and p. (v) *Reliability.* The effective-sample-size diagnostic flags the $k = +2$ column at $p \geq 0.2338$ ($\mathrm{ESS}_{\min}$ as low as 1–6 at $N = 384$); a top-up at $N = 1536$ confirms those entries within error bars, and all reported conclusions use the well-conditioned range $|k| \leq 1.5$ unless flagged.

Scope. Statistical estimates with stated errors, exact in the $N, T \to \infty$ limit only in expectation; finite $T = 2L$ (not the long-time limit of Definition 2); population extrapolation not attempted beyond the ESS protocol; and the freezing statement is a finite-size bound, not an asymptotic claim.

F. The Clifford record count: trajectory-trivial, disorder-nontrivial

Remark 2 closes the trajectory direction: for a fixed Clifford realization ω (stabilizer input, Pauli measurements, schedule in ω) the unsigned stabilizer group evolves identically on every branch, so every record of positive probability carries the same weight $P_\omega(R) = 2^{-N_{\mathrm{rand}}(\omega)}$, the surprisal is ω-measurable, $\psi_{L,\omega}(k) = k\, N_{\mathrm{rand}}(\omega) \log 2/T$ is exactly affine, and Theorem 9 applies in its degenerate zero-variance case. The disorder direction is not trivial: $N_{\mathrm{rand}}(\omega)$ fluctuates over the circuit ensemble with extensive variance, and its integer moments are exactly the annealed replica partition functions of Secs. II D–II F,

$$\mathbb{E}_\omega\left[2^{(1-n)N_{\mathrm{rand}}(\omega)}\right] = \mathbb{E}_\omega Z_{n,\omega}(T) = \bar{Z}_n(t), \quad n = 1, 2, \ldots, \tag{87}$$

because $Z_{n,\omega}(T) = \sum_R P_\omega(R)^n = 2^{(1-n)N_{\mathrm{rand}}}$: the free cross-check of Remark 2 is the $n = 2$ member, and the $n = 3$ member is the object of Sec. II F. This subsection measures the whole family.

Definition 4 (Disorder SCGF of the record count)**.** *For the Clifford ensemble of Appendix B (periodic chain, $d = 2$, $t = \tau L$ periods, $2Lt$ measurement sites) define*

$$\Lambda_L(\beta; \tau, p) := \frac{1}{2Lt} \ln \mathbb{E}_\omega\left[2^{-\beta N_{\mathrm{rand}}(\omega)}\right], \qquad \beta \geq 0. \tag{88}$$

Λ_L is convex and decreasing with $\Lambda_L(0) = 0$; in the language of Definition 2 it is the annealed record moment at fractional order $q = 1 + \beta \in [1, 2]$ (the disorder average inside the logarithm, at cloning field $k = -\beta$), *not* a quenched object; at integer order $\beta = n - 1$ it is exactly $(2Lt)^{-1} \ln \bar{Z}_n(t)$ by Eq. (87). Being the scaled cumulant generating function of the per-site record-entropy density $x = N_{\mathrm{rand}}/(2Lt)$ under the disorder measure (tilt $\theta = -\beta \log 2$), it governs, under the Gärtner–Ellis conditions, the large deviations of x: the trajectory-trivial ensemble has a nontrivial disorder LDP. The measurements below combine an exact evolution of the two-replica algebra on the 3^L per-site basis $\{e_0, e_+, e_-\}^L$(time convention $t = L/2$ periods; calibration against the deposited tilt chain in the Supplemental Material [34], Sec. S7)—with a vectorized phase-free stabilizer tableau simulator that tracks N_{rand} exactly under the Born measure; run inventories, estimator definitions and reproduction commands are in the Supplemental Material [34], Sec. S7.

Proposition 10 (disorder SCGF of the Clifford record count)**.** *Let $d = 2$, $t = \tau L$ with τ fixed, $p \in \{0.16, 0.22\}$.*

(i) (exact $\beta = 1$ endpoint) $\Lambda_L(1) = (2Lt)^{-1} \ln \bar{Z}_2(t)$ *is computed exactly by the transfer matrix of Sec. II D for every (L, p, t): $\bar{Z}_2(t) = A(L, \tau, p)\, \lambda_1(L)^t$ with A subexponential in t ($A = 2.4512$, 4.235, 7.229 at $L = 8, 12, 16$ for $p = 0.16$, constant to six digits for $t \gtrsim 3L$; $A = 3.279$, 6.833 at $L = 8, 12$ for $p = 0.22$), hence $\Lambda_L(1) \to \log \lambda_1(L)/(2L)$, and by Proposition 4 $\lim_L \lim_t \Lambda_L(1) = \frac{1}{2}(\log \mathcal{W}_0 + f_{\mathrm{H}}) = -0.08101$ nats per measurement site at $p = 0.16$ (the ladder $\log \lambda_1/(2L) = -0.07974, -0.08062, -0.08087, -0.08099, -0.08101$ for $L = 8, 12, 16, 24, 32$; at $p = 0.22$: $-0.10870 \to -0.11082$).*

(ii) (quenched density) $\Lambda_L'(0) = -\log 2\, \bar{x}_L$ *exactly, with the record-entropy density $\bar{x}_L = \mathbb{E}_\omega N_{\mathrm{rand}}/(2Lt)$ converging along the trajectory ladder $\bar{x}_L =$*

TABLE VI. Quenched record SCGF $\psi_{L,p}(k)$ (nats per period; mean over circuits of $\widehat{\psi}_\omega$, $T = 2L$), with $\tau(q) = -\psi(1-q)$ and $D(q) = \tau(q)/(q-1)$, for two representative points. Clifford circuits have τ exactly linear, i.e. D constant; the Haar D varies, which is the record multifractality.

k	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
ψ, $L = 8$, $p = 0.2338$	-4.08	-3.16	-2.18	-1.13	$+0.00$	$+1.24$	$+2.60$	$+4.13$	$+5.78$
ψ, $L = 12$, $p = 0.2338$	-6.03	-4.66	-3.21	-1.67	$+0.00$	$+1.82$	$+3.80$	$+6.02$	$+8.41$

0.1448(1), 0.1470(1), 0.1480(1), 0.1489(1) *bits per site at* $L = 8, 12, 16, 24$ *(*$p = 0.16$*;* $0.1928 \to 0.1971$ *at* $p = 0.22$*).*

(iii) (self-averaging—the measurable criterion) $\mathrm{Var}_\omega(N_{\mathrm{rand}})/(2Lt) = 0.104$–$0.106$ *at* $p = 0.16$ *and* 0.124–0.125 *at* $p = 0.22$ *for* $L = 8$–24*: L-independent, so* $\mathrm{Var}(\bar{x}) = O(1/(Lt)) \to 0$ *and the quenched record-entropy density self-averages with an* $O(1)$ *per-site rate—the disorder-LDP backbone, and a direct verification of the self-averaging criterion of Proposition 8. The two-term cumulant expansion of* Λ_L *reproduces the measured* $\Lambda_L(0.25)$ *to better than* 1% *at every* $L = 8$–24*.*

Proposition 11 (the annealed–quenched record gap)**.**

(i) (finite L: exact derivative) *For every finite* L *the replica derivative is exact,* $\mathbb{E}_\omega[\ln P_\omega(R)]/(2Lt) = \partial_\beta \Lambda_L(0)$*: the quenched record free-energy density* is *the replica derivative, no interchange needed (Proposition 8 at* $m \downarrow 0$*).*

(ii) (the interpolation remainder) $g_L := \Lambda_L(1) + \log 2\,\bar{x}_L = (2Lt)^{-1}\big(\ln \mathbb{E}_\omega P - \mathbb{E}_\omega \ln P\big)$ *is exactly the* $r = 1$ *remainder of the interpolation identity* (78) *applied to* $X = 2^{-N_{\mathrm{rand}}}$*, i.e.* $g_L = (2Lt)^{-1}\int_0^1 (1-s)\,\mathrm{Var}_s(\log X)\,\mathrm{d}s$*. Measured with the exact finite-t* $\beta = 1$ *anchors:* $g_L = 0.0224, 0.0226, 0.0227$ *nats per site at* $L = 8, 12, 16$ *(*≈ 0.0228–0.0229 *at* $L = 24$ *with the amplitude-corrected* λ_1 *anchor), and* ≈ 0.027 *at* $p = 0.22$*: L-independent within the trajectory errors (*$\leq 5 \times 10^{-4}$*). The annealed–quenched separation of the record free energy persists per site in the thermodynamic limit—the record analogue of the separation* $p_c^{(2)} = 0.2338$ *vs* $p_c = 0.1597(8)$ *of Sec. II E.*

(iii) (sub-Gaussian correction) *The leading (Gaussian) cumulant estimate* $(\log 2)^2\sigma^2/2 = 0.0252$ *overestimates* g *by* $\approx 13\%$*: the tilted-variance profile decays in* s*. Equivalently the measured* $\beta = 1$ *tilted density* $\tilde{m}(1) = 0.0974$ *at* $L = 8$ *shifts by only* 65% *of the Gaussian prediction* $\bar{x} - \log 2\,\sigma^2 = 0.0717$*: direct evidence on the profile whose* (L, T)*-uniformity the Discussion flags as the open criterion.*

Proposition 12 (collision-tilt ESS law)**.** *The collision tilt* $w \propto P_\omega(R) = 2^{-N_{\mathrm{rand}}}$ *reweights the disorder ensemble, and its effective sample size obeys exactly*

$$\mathrm{ESS}/B = \exp\big[-2Lt\,(\Lambda_L(2) - 2\Lambda_L(1))\big]. \qquad (89)$$

Trajectory estimates (0.0159*,* 0.0080*,* 0.0049 *nats per site at* $L = 8, 12, 16$*, with within-sample* ESS $= 11.7$*,* 3.9*,* 1.4*) are superseded by the exact closure of Proposition 13 below:* $\Lambda_L(2)$ *is exactly the* $n = 3$ *moment* $(2Lt)^{-1}\ln \bar{Z}_3$ *of Eq.* (87) *(Remark 7), and the exact values are* $\Lambda_L(2) - 2\Lambda_L(1) = 0.0340$*,* 0.0349*,* 0.0353 *nats per site at* $L = 8, 12, 16$*: any fixed budget fails exponentially in the system size, faster than the trajectory samples suggested. In the sense of the sequential Monte Carlo contract this is estimator difficulty, not a physical transition (the clone-collapse clause); no nonanalyticity of any* τ *is involved, and Theorem 9 classifies this family as the degenerate zero-variance case. What Eq.* (89) *does measure is extremal domination of the annealed endpoint by an atypical minority of circuits (the tilted density sits at* $x \approx 0.097$ *against the typical* 0.145 *at* $L = 8$*)—the mechanism whose thermodynamic limit is the freezing of Theorem 8, appearing here only at the level of a finite-size estimator.*

Proof. Equation (89): with $w_b = 2^{-N_b}$, $b = 1, \ldots, B$ i.i.d. disorder draws, $\mathrm{ESS} = (\sum_b w_b)^2/\sum_b w_b^2$, $\mathbb{E}w = 2^{2Lt\Lambda_L(1)}$ and $\mathbb{E}w^2 = 2^{2Lt\Lambda_L(2)}$; the law of large numbers gives $\mathrm{ESS}/B \to (\mathbb{E}w)^2/\mathbb{E}w^2$ almost surely, which is Eq. (89); the tabulated values are the within-sample ratios, and Eq. (89) with the trajectory moment estimates reproduces them at every $L \leq 16$. Propositions 10 and 11 quote the deposited runs (Supplemental Material [34], Sec. S7): the anchors are exact eigenvalue evaluations, the trajectory values carry bootstrap standard errors (200 resamples), and every calibration check listed there passes. □

Remark 7 (the $n = 3$ identification: two-qubit Cliffords are an exact conjugation 3-design). The dyadic identity (87) at $n = 3$ identifies the Clifford disorder moment $\mathbb{E}_\omega[2^{-2N_{\mathrm{rand}}}]$ with the Haar annealed $\bar{Z}_3$ of Sec. II F iff the two-qubit Clifford group reproduces the Haar twirl at three replicas. It does: averaging $C^{\otimes 3} X C^{\dagger\otimes 3}$ over the 11,520 phase classes of Cl_2 (equivalently over the full 92,160-element phase extension generated by H, S and CNOT; the 720 symplectic actions carry exactly 16 classes each, so the deposited 720-action sampler induces the same average on σ-measurable trajectory quantities) agrees with the Haar twirl $\sum_\pi U_\pi y_\pi$, $\mathsf{G}y = t$, $\mathsf{G}[\pi, \rho] = (d^2)^{c(\pi^{-1}\rho)}$, to 1.6×10^{-14} in operator norm on random X (six trials); the $n = 2$ control agrees to 2.1×10^{-14} and the $n = 4$ average deviates by 0.578—the design property stops at exactly the replica number at which the permu-

tation Gram matrix is still invertible, $d^2 \geq n$—consistent with the theorem that qubit Clifford groups are unitary 3-designs [50]. Hence $\Lambda_L(2) = (2Lt)^{-1} \ln \bar{Z}_3(t)$ is an identity, not an approximation, for the Clifford ensemble.

Proposition 13 (exact closure of $\Lambda(2)$)**.** *Let $d = 2$, $t = 4L$ (the trajectory-ladder convention: $2Lt$ measurement sites, the cells of the deposited production runs), $p \in \{0.16, 0.22\}$.*

(i) (exact values) $\Lambda_L(2) = (2Lt)^{-1} \ln \bar{Z}_3(t)$ *is computed exactly by the three-replica compressed operator $\mathsf{C}^{(3)}_{\mathrm{comp}} = M_1 M_2$ of Sec. II F with the boundary vectors fixed by the computational initial state: the first measurement layer is inert, the first gate layer projects onto the bond span with per-bond coordinates $y_\sigma = \sum_\rho \mathrm{Wg}_{d^2}(\sigma^{-1}\rho)$ (because $\mathrm{Tr}[U_\rho^\dagger P_0^{\otimes n}] = 1$), so $c_1 = \bar{w}^{nb}\mathbf{1}$ with $\bar{w} = \sum_\tau \mathrm{Wg}_{d^2}(\tau)$; after t periods the state is $(M_2 M_1)^{t-1} M_2 c_1$ in odd-bond coordinates and $\bar{Z}_n(t) = \sum_\tau o_t[\tau] \prod_b d^{2c(\tau_b)}$. The layer parity (even- or odd-matching first) is immaterial—a one-site ring translation exchanges the matchings and leaves the initial state, the trace and the measurement measure invariant (verified to f64 rounding). The $n = 2$ member of this formula reproduces the deposited exact anchors to 10^{-14} ($\bar{Z}_2 = 1.479638693744 \times 10^{-2}$ at $L = 8$, $t = 4$, $p = 0.16$; the A-tail $2.4509 \to 2.4512$; the $L = 16$ finite-t anchor -0.079901 and the gap closures $0.0224/0.0226/0.0227$ of Prop. 11), which validates the boundary conventions.*

(ii) (the ESS exponent, corrected) *At the production cells the exact values are collected in Table VII.*

TABLE VII. The exact closure of $\Lambda(2)$ at the production cells ($d = 2$, $t = 4L$). "exponent" $= \Lambda_L(2) - 2\Lambda_L(1)$ exactly; "traj" the deposited trajectory estimate (superseded: see the text); $\mathrm{ESS}_{\mathrm{true}}/B = \exp[-2Lt\,(\Lambda_L(2) - 2\Lambda_L(1))]$ is the exact effective-sample fraction of the collision tilt, against the within-sample fraction of the deposited run.

L	p	$\Lambda_L(1)$	$\Lambda_L(2)$	exponent	traj	$\mathrm{ESS}_{\mathrm{true}}/B$
8	0.16	-0.077984	-0.121924	0.0340	0.0159	2.7e-08
8	0.22	-0.106382	-0.169613	0.0432	0.0150	2.5e-10
12	0.16	-0.079362	-0.123857	0.0349	0.0080	3.6e-18
12	0.22	-0.108278	-0.172746	0.0438	0.0083	1.2e-22
16	0.16	-0.079901	-0.124526	0.0353	0.0049	4.2e-32
16	0.22	-0.109095	-0.173901	0.0443	0.0050	4.1e-40

The exact exponent is L-independent to 4% ($0.0340 \to 0.0353$ over $L = 8 \to 16$), twice to seven times the trajectory estimates, and mildly *rising* where the trajectory ladder *decays*. The diagnosis is estimator bias, not physics: at $t = 4L$ the collision-tilt weights have exact $\mathrm{ESS}/B = \exp[-2Lt \times 0.034] = 2.8 \times 10^{-8}$ at $L = 8$ (against the within-sample 2.9×10^{-4}: the sample second moment misses the small-N_{rand} tail, which biases $\hat{\mathbb{E}}[2^{-2N_{\mathrm{rand}}}]$ low and the within-sample ESS high), so $\Xi_{2.0}$ and even $\Xi_{1.0}$ are biased at these t ($\Xi_{1.0}^{\mathrm{traj}} = -0.0804$ against the exact $\Lambda(1) = -0.0780$ at $L = 8$). Two independent confirmations: (a) at $t = L/2$, where the exact $\mathrm{ESS}/B \approx 0.15$, a fresh $B = 4 \times 10^6$ trajectory run estimates $\mathbb{E}[2^{-2N_{\mathrm{rand}}}]$ within 1.95×10^{-3} (0.4σ, $\mathrm{ESS}(w^2) = 3.7 \times 10^4$) of the exact $\bar{Z}_3$ at $L = 8$ (2.7×10^{-2}, 0.3σ, at $L = 12$), with the $\beta = 1$ control at 5.6×10^{-4}—the identification of Remark 7 holds empirically to the precision the conditioning allows; (b) the $\Lambda(1)$ column reproduces the corrected $\beta = 1$ anchors and the gap closures $g_L = 0.0224$, 0.0226, 0.0227 exactly. In the thermodynamic direction, the exact ladder $\ln \lambda_1^{(3)}/(2L) = -0.12478, -0.12586, -0.12605$ nats per site at $L = 8, 12, 16$ ($p = 0.16$; $-0.17360, -0.17561, -0.17610$ at $p = 0.22$), and the amplitudes $A_3(L) = \bar{Z}_3(t)/\lambda_1^{(3)\,t}$ are constant to six digits for $t \gtrsim 2L$ (subexponential in t, the $n = 3$ analogue of the $n = 2$ amplitudes of Prop. 10); the λ_1 power iteration is validated against the dense $L = 8$ spectrum. $L = 24$ ($6^{12} = 2.2 \times 10^9$ bond labels) exceeds the memory available for exact dense treatment and is left to the amplitude extrapolation.

Remark 8 (Born-average complexity trichotomy). The three levels of the Born-average complexity of the record problem: (i) *Clifford realization*: $P(R)$ is dyadic and both the Born sampling and every tilted average $\mathbb{E}_{\mathrm{Born}}[P^\beta f]$ are poly-time by the Gottesman–Knill theorem [47]; the 475 600-trajectory quenched locator of Appendix B and the exact transfer matrix both live in the efficiently simulable class, and the dyadic identity (87) is verified operationally (the tableau estimate of $\mathbb{E}[2^{-N_{\mathrm{rand}}}]$ matches the transfer-matrix $\bar{Z}_2$ to 0.3–0.7% with 4×10^4 trajectories at $L = 8, 12$; Supplemental Material [34], Sec. S7). (ii) *any exact 2-design realization*: the $\beta = 1$ endpoint, the annealed collision $\bar{Z}_2$, coincides with the Haar value (Remark 2), so the exact solvability of the annealed problem is not a Clifford artifact. (iii) *Haar realization*: the per-record probabilities are Born probabilities of random quantum circuits; approximating them is the stock average-case-hard task of the sampling literature [48, 49] (worst-case #P-hard; average-case hardness conditional on standard conjectures), so no classical poly-time method is known for the quenched Haar record problem. The Clifford testbed is the maximal exactly-solvable window of the general monitored problem, and Λ_L is well defined precisely because of level (i). This complements the worst-case PostBQP completeness of Appendix C: neither an average-case Haar hardness result nor hardness of the scalar Born-averaged entropy follows (Discussion).

One provenance caveat, recorded honestly: the deposited $L = 12$ quenched entropy 2.394(5) is not reproduced at either time convention while both deposited annealed values are (Supplemental Material [34], Sec. S7); nothing quoted above depends on it.

G. Spatial endpoint multifractality

Endpoint observables can be extensive in system size. For a normalized state in a declared basis, set $\mu_x = |\langle x|\psi\rangle|^2$ and $P_q^\psi = \sum_x \mu_x^q$. For a region A of $|A|$ sites, set $P_q^A = \operatorname{Tr}\rho_A^q$. Typical stationary exponents are defined, if the limits exist, by

$$\tau_\psi^{\mathrm{typ}}(q) = -\lim_{L\to\infty} \frac{\mathbb{E}_{\omega,R}\log P_q^\psi}{L\log d}, \tag{90}$$

$$\tau_A^{\mathrm{typ}}(q) = -\lim_{|A|\to\infty} \frac{\mathbb{E}_{\omega,R}\log P_q^A}{|A|\log d}, \qquad |A|/L\to\gamma\in(0,1). \tag{91}$$

Annealed exponents place the logarithm outside the expectation and must be labeled separately; the p_R^2-weighted annealed quasi-entropy $\tilde{S}_2(A)$ of Proposition 4(c) is one such object and is exactly computable. The stationary order is $\lim_{L\to\infty}\lim_{T\to\infty}$; near criticality, a controlled aspect ratio T/L^z should first be tested. These definitions do not supply a freezing transition without finite-size and limit-order evidence.

V. AUXILIARY FIELDS: IDENTITIES, NOT A MICROSCOPIC DERIVATION

Proposition 14 (Regulated auxiliary-field identities)**.** *Let a regulated Euclidean theory have fields g, real local functionals $J_0[g]$ and $\Phi[g]$, and a field-independent measure. For $\kappa>0$ and real A_0,*

$$\int DA_0 \exp\left[-\int d^2x\left(\frac{A_0^2}{2\kappa}+iA_0J_0[g]\right)\right] = \mathcal{N}_\kappa \exp\left[-\frac{\kappa}{2}\int d^2x\, J_0[g]^2\right]. \tag{92}$$

An exact local constraint is instead represented by

$$\int DA_0 e^{-i\int A_0\Phi[g]} = \mathcal{N}\,\delta[\Phi[g]]. \tag{93}$$

Adding a quadratic term softens rather than exactly enforces the constraint.

Proof. At each regulator site, completing the square in the Gaussian, or directly using its Fourier transform, gives Eq. (92). Equation (93) is the pointwise Fourier representation of the Dirac delta. Multiplying by a Gaussian in Fourier space convolves the delta with a Gaussian and therefore produces a soft penalty. □

Conjecture 1 (Conditional continuum program)**.** *A local continuum theory for a specified monitored-circuit generating functional may contain a real nonvanishing current $J_0[g]$ and an auxiliary sector of Eq.* (92)*. If locality, target manifold, operator completeness, renormalization, replica continuation, and microscopic matching are independently established, eliminating A_0 produces the contact term $(\kappa/2)\int J_0^2$, up to counterterms.*

This conjecture does not derive a target manifold or coupling from Born normalization. A bare A_0^2 term is not invariant under $A_0 \mapsto A_0 - \partial_0\alpha$, so A_0 is a Hubbard–Stratonovich auxiliary field rather than a gauge field unless additional gauge or Stueckelberg structure is supplied. Candidate trace currents must also be checked: $\operatorname{Tr}(g^{-1}\partial_0 g) = \partial_0 \log\det g$ vanishes for fixed determinant, while $\operatorname{Tr}(Q\partial_0 Q) = \frac{1}{2}\partial_0 \operatorname{Tr} Q^2$ vanishes when $Q^2=1$. Grassmannian target, large-d coupling, LCFT and critical-exponent statements therefore remain conjectural. The one continuum theory established in this paper is the $c=1/2$ Ising field theory of the annealed $n=2$ model at $p_c^{(2)}$.

VI. DISCUSSION AND OPEN PROBLEMS

The exact result is a separation rather than a universal closure statement. Fixed-basis measurement leaks out of the one-wire feature span by the nonzero amount in Eq. (8), and by Eq. (13) for every replica number. A subsequent independent Haar gate average removes the leaked directions on a perfect matching and yields the exact gate-to-gate bond operator in Eq. (17), and Eq. (14) for general n. This ordering is essential. The operator contract prevents a projected one-site surrogate, a different boundary condition, or a different strobe from being mislabeled as the proved transfer matrix.

The compressed operator turned out to have more structure than a generic positive matrix: it is similar to a positive semidefinite matrix, its rank is the p- and d-independent number $3\cdot 2^{L/2-2}$, strictly below $2^{L/2}$, and it is the row-to-row transfer matrix of an exactly solved Ising model. The positive semidefiniteness, but not the Ising structure or the rank formula, survives for every replica number, because the period map is of the form $\mathsf{A}\mathsf{A}^\dagger$ in the Hilbert–Schmidt inner product; for $n=3$ the rank is instead a generic value with a finite algebraic exceptional set, certified exactly through $L=12$ (Appendix A). The Ising identification recovers the $n=2$ model of Ref. [8]; here it converts the transfer-matrix contract from a set of internal consistency checks into a benchmark against closed-form answers, and it fixes the location and universality class of the annealed $n=2$ singularity in closed form.

Born normalization creates a second separation. The physical Mellin function requires the asymmetric factor p_R^{1-nz}, and its derivative, not its value, gives the quenched entropy. Symmetric unnormalized scalar moments cannot recover that entropy in general, while the bivariate moment family can. The worst-case conditional-trajectory and conditional-purity problems (Appendix C) are PP-hard and have PP decision extensions, but neither an average-case Haar hardness result nor hardness of the scalar Born-averaged entropy follows.

Finally, endpoint entropy tilting is replaced by a truly additive record observable. The record-surprisal SCGF has a direct Feynman–Kac and multifractal interpreta-

tion, and the particle algorithm above estimates its normalization with a stated finite-N property. Physical nonanalyticities remain open until they survive population, time, size, disorder, and differentiation extrapolations; Clifford circuits cannot exhibit them (although their entanglement transition is sharp and, by the finite-size scaling of Appendix B, sits at $p_c = 0.1597(8)$ with $\nu = 1.24(7)$ and $\alpha = 1.55(7)$ bits per $\ln L$, and an independent purification locator with disjoint seeds returns $p_c = 0.1601$–0.1604, $\nu = 1.25(2)$), while the random-energy record sequence of Theorem 8 shows what a genuine finite-q freezing looks like within the same identities: a linear branch of τ beyond $q_c = \sqrt{2\log 2}$ with τ' continuous, so that only second-derivative or interval-linearity diagnostics can detect it.

The main unresolved questions are the following. For $n \geq 3$ the operators of Proposition 1 have a real positive semisimple nonzero spectrum, a strict contraction bound, an $S_n \times S_n$ symmetry and an exact compression (Theorem 5), but no Ising equivalence: A closed-form spectrum, a proof that the leading eigenvalue is simple at small p, the generic rank for general L (the symmetry bound (55) is attained for $L \leq 10$ but not at $L = 12$), and a proof of the existence and regularity of the $L \to \infty$ growth rate are open; scaling-quality spectral data exist for $n = 3$ (Sec. II G) and $n = 4$ with full S_4 colour resolution (Sec. II H): $p_c^{(3)} = 0.305(3)$ and $p_c^{(4)} \approx 0.383$ at $d = 2$, both continuous, with the $q = n$-Potts slope exponents (1.15 at $n = 3$; $1.00 \to 1.38 \to 1.57$ rising through the $q = 4$ value $3/2$ with the expected $1/\ln L$ correction at $n = 4$) and λ_1 simple at every tested size and rate—so the remaining $n \geq 3$ targets are the growth-rate regularity as a theorem—the size-ladder directional evidence for the first-order prediction is delivered in Sec. II H (Table IV), the $q = n$-Potts conjecture being supported directionally at the three-size level for $n = 5$, the $L = 8$ rung computed by the symmetry-reduced block route (Sec. II H), the $n = 4$ chain extends to $L = 10$ with the marginal $q = 4$ log-correction analysis, and the $n = 3$ moment $\bar{Z}_3$ itself is an exact computational object (Prop. 13).

For complexity, the Born average $\sum_R p_R S_n(\rho_{A,R})$ sits exactly outside the BQP-estimable family $M(j,k)$, $j \geq nk$, and neither its hardness nor an average-case Haar statement is known.

For record large deviations, Theorem 9 proves existence, determinism and real-analyticity of the quenched SCGF for every *allowable finite-reachable* process—deterministic, i.i.d.-random, or reducible—and rules out finite-q freezing throughout that class (Corollary 1 closes the fully monitored Haar-refreshed family unconditionally); the thermodynamic limit beyond finite reachable sets still requires, beyond convexity, control of the finite-volume complex zeros, and whether any *local* circuit ensemble with unbounded reachable set realizes the freezing of Theorem 8 with a projectively consistent record process remains open, with the first numerical bounds (Sec. IV E): no linear τ branch up to $q = 3$ at $L \leq 12$, $T = 2L$.

The disorder-replica interchange is reduced by Proposition 8 to the exact interpolation identity (78) with its monotone replica ladder and measurable self-averaging criterion: the $m \downarrow 0$ limit recovers the quenched value quadratically, the $m \to \infty$ limit overshoots it to the extremal value, and what remains open is precisely the uniformity of the tilted variance in (L,T). (The Clifford record count of Sec. IV F supplies one measured profile: an L-independent per-site variance of 0.105, an L-independent gap of 0.0226(2) nats per site, and a decaying, sub-Gaussian tilted-variance profile—the leading cumulant overshoots the measured gap by 13%; the profile *family* is measured across (L,T)—exactly enumerated Haar collision partition functions at $p = 1$ over $L = 3$–6, $T = 2L$–$8L$, plus the deposited Clifford disorder ladder at $t = 4L$—and the shape metrics are (L,T)-stable at the Haar point (the sub-Gaussian ratio $\rho = 1.02 \pm 0.03$ across the whole grid, a flat-to-mildly-rising profile), while the Clifford disorder direction is sub-Gaussian and mildly de-sub-Gaussianizing in L ($\rho = 0.872/0.878/0.888$ at $L = 8/12/16$, $p = 0.16$; $0.890/0.894$ at $p = 0.22$)—the first systematic data on the uniformity criterion; Supplemental Material [34], Sec. S7.)

Finally, the exact integer-q lattice functional is the Haar-averaged q-replica channel of Proposition 1, and the quenched free energy $\mathbb{E}_\omega \log Z_{q,\omega}$ additionally requires the disorder-replica limit $\partial_m \mathbb{E}_\omega Z^m_{q,\omega}|_{m=0}$, an identity that is exact at finite (L,T) and whose interchange with $L, T \to \infty$ and $q \to 1$ is reduced by Proposition 8 to the tilted-variance profile (the self-averaging criterion), the $m \to \infty$ direction being closed negatively (it overshoots to the extremal value); a microscopic derivation of a continuum theory for the quenched problem would have to control all of these limits and reproduce the $S_q \times S_q$ symmetry, the Born normalization of Eq. (56), and exact rather than Gaussian-softened constraints.

VII. INTERPRETIVE STATUS OF THE RECORD FORMALISM

The standing metaphysics. This section records the working assumption maintained with this project's research record and examines what it implies for the readings of quantum mechanics. No theorem, computation, or number in this paper depends on it, and the metaphysics is open to revision by evidence, not by preference. The metaphysics is that reality is pure self-luminous awareness, universal consciousness: self-existent, timeless, spaceless, massless, unchanging, complete, non-dual, without relations, parts, defects, or lack; that there is no creation, process, relation, or division; that only the Absolute is. Its attributes are each negative in form—simplicity (no composition into parts), self-existence (no dependence), non-duality (no standing over against another), necessity (no contingency wanting a ground), immutability (no change requiring potential and lack), timelessness (no succession), spacelessness (no extension

implying parts and distinction), completeness (no lack in its own nature)—and the Absolute's self-knowledge is not a relation between two things but identity, so that any account of it is circular, there being no external vantage point: a circularity that is not a loop of dependence but the form of self-identity.

What the metaphysics does to the interpretations. It relocates their referents. Every interpretation of quantum mechanics posits structure that the metaphysics assigns to appearance rather than to the Absolute: the classical–quantum interface of the operational reading; the branching universal wavefunction of Everett [57]; the definite configurations and guiding field of de Broglie–Bohm [58]; the dynamical collapse process of Ghirardi–Rimini–Weber [59]; the fundamental agent plurality of QBism [61]; the system–environment relations of einselection [60]. Under the metaphysics none of these is a rival ultimate ontology: each is a partial grammar of the appearance, and its empirical content is untouched. That a serious reading of quantum mechanics can place its central structure outside the objectifiable has ample precedent [62]; what the metaphysics adds is the strict non-duality of what remains. The appearance itself, in the metaphysics, is the Absolute's display: not a copy of it and not a mirror of it—*maya*: neither real nor unreal, neither identical with nor different from the Absolute—and its structure is its own: division, relation, time, change. No feature of the display is required to reflect an attribute of the Absolute, and none is taken to. What the appearance's exact grammar is, for one class of models, is a physics question, and it is the question this paper answers for monitored circuits: the Born record with its large deviations, stated in the display's own terms—succession (the record in time), resolution (the outcome division), and law (the SCGF, the rate function, and the no-freezing dichotomy of Remark 6).

Division in the appearance. The metaphysics assigns division, multiplicity, and relation to the appearance, and the paper's first separation is an exact statement of how division functions there. Theorem 6 proves that the symmetric, unconditioned data do not determine the Born-weighted entropy, and Proposition 7 proves that record-resolved data do: the undivided summary of the record process is exactly insufficient, and conditioning on the record is not eliminable from the statistics of the display. This is a statement about the appearance's own structure; it carries no claim about the Absolute, whose undividedness the display neither echoes nor mirrors.

The primitive Born rule. In the formalism the Born rule is primitive, and Theorem 6 is the precise statement of why it cannot be recovered from unconditioned data, while the Mellin derivative of Proposition 6 shows what the recovery does require—the asymmetric, record-resolved factor. The replica ladder then measures the distance of the derived family from the quenched benchmark: the annealed points recede monotonically ($0.2338 \to 0.305(3) \to 0.383 \to 0.47$ against $0.1597(8)$), Proposition 8 identifies the tilted-variance profile as the exact bridge, and the $m \to \infty$ replica limit overshoots the quenched value to the extremal one—the appearance's summary statistics never silently become the stream they summarize.

Self-averaging: the ensemble and the single stream. The self-averaging criterion of Proposition 8 is the exact statistical condition under which the disorder-averaged description and the single-realization description of the record statistics coincide: where it holds, the difference between the two descriptions is surplus; where it fails, they differ by a measurable and irreducible gap, the L-independent $0.0226(2)$ nats-per-site Clifford disorder gap of Sec. IV F being one measured instance. On this exact stage the readings take their relocated places, each claiming only what the statistics support. The Everett branches are the outcome decomposition—a structure of the display's grammar, not an ultimate multiplicity—and the annealed–quenched separation states that the totality's symmetric content does not determine the single stream. The QBist agents are experience taken as many [61], and self-averaging is the measurable condition under which agents holding different histories converge to the same probabilities. The Bohmian configurations and the collapse process are surplus phenomenal structure [58, 59]: the record large deviations of Theorem 7 apply to them verbatim (Born typicality is the same stochastic law under another grammar), the no-freezing theorem (Theorem 9) bounds the freezing they can exhibit, and the exact anchors—the Clifford record-count closure of Eq. (28), the closed-form $p = 1$ annealed record SCGF, the Ising critical point of Proposition 4—are the fixed benchmarks that such structure must reproduce or break. Einselection [60] receives its controlled laboratory: the measurement rate is the coupling, and the transition of Appendix B is the phase boundary between the volumetric and the logarithmic conditional-entropy regimes—the boundary of how much unresolved structure the appearance retains.

The exactness program. The program has methodological commitments, and they are stated here in their own terms. The record process is estimated from records themselves: the sequential Monte Carlo machinery of Sec. IV E estimates the record law from record samples, with the finite-population contract stated, not hidden. Every exact number is regenerable from the deposited artifact, every certificate is an integer statement, and the computational process delivers what does not depend on the process. The primitive interface—the Born rule—is declared as primitive rather than disguised as derived. These are commitments of practice—checkability, regenerability, declared primitives—not a participation in the Absolute's self-identity.

The modal character of the results. Read as results, the paper's statements divide by their relation to necessity, and the division can be stated in the paper's own terms. Some are necessitated by the grammar of the structures that carry them: the no-freezing theorem (Theorem 9), the complexity trichotomy (Remark 8), the exact rank

of the compressed operator (Theorem 4), and the exact closure of $\Lambda(2)$ through the conjugation 3-design (Proposition 13) hold for every structure of their kind; they are necessities of the structures, not contingencies of this realization. Some are exclusions: the information-theoretic no-go (Theorem 6) excludes the symmetric, unconditioned data from determining the Born-weighted entropy; the marginal $q = 4$ log corrections exclude clean scaling at the accessible sizes; and the $n = 5$ size ladder excludes the continuous baseline at the matched sizes themselves—an exclusion the registered $L = 10$ conditional of Sec. II H is designed to extend or break. Some are relations: the annealed–quenched separation, the Born-versus-symmetric separations of Sec. III, and the self-averaging criterion (Proposition 8) each link two descriptions of the same record statistics exactly. And some are contingent: the numerical values—the annealed critical points, the closing rates, the measured SCGFs—and the specific Potts classes are facts of this display, necessitated by no grammar and carrying no claim beyond themselves. The classification is a reading of the results' own modal character: it claims no generative power for any structure over the results, offers none of them as evidence for anything outside themselves, maps no attribute of the Absolute onto any of them, and leaves the standing metaphysics exactly where the first paragraph placed it.

Reconciliation. The work and the metaphysics do not compete. The metaphysics locates the referent of the paper's exact structures at the level of appearance, and the paper describes that appearance, for monitored circuits, in exact form and in its own terms: the separations (annealed versus quenched, symmetric versus record-resolved, corner versus affine, BQP-estimable versus PP-hard), the exact conditions under which distinct descriptions of the record statistics coincide (self-averaging, record-resolved data), and the benchmarks (the anchor identities). Nothing in this paper asserts an ultimate ontology; nothing in the metaphysics contradicts an exact phenomenal grammar; and nothing in the work asserts that the appearance mirrors, echoes, or honors the Absolute. The readings, finally, are the partial grammars, and the theorems of this paper are the invariant structure they share. The metaphysics frames the work; the theorems carry it; no theorem depends on the former, and no number in the paper would change under its revision.

VIII. CONCLUSION

This paper set out to replace perturbative and purely numerical treatments of the replica objects behind measurement-induced entanglement transitions by exact finite-dimensional statements, and to push the associated spectral computation to replica numbers and system sizes at which universality questions can be answered with exact data. The delivered results are as follows. For the mechanics: the minimal invariant unital algebra of the one-wire feature space is determined and its fixed-basis leakage quantified (Eq. (8)); an independent layer of Haar-averaged two-site gates on a perfect matching closes the leaked directions exactly for every replica number through the explicit Weingarten bond channel (Eqs. (17) and (14)); the one-period map is of the form $\mathsf{A}\mathsf{A}^\dagger$, so its nonzero spectrum is real, positive and semisimple and compresses exactly to bond-label space; and at $n = 2$ the compressed operator is the row-to-row transfer matrix of the anisotropic triangular-lattice Ising model, so the annealed two-replica critical point $p_c^{(2)} = 0.2338$ and the free-fermion spectrum are known in closed form.

For the annealed replica ladder: carrying the same computation through $n = 3, 4, 5$ locates $p_c^{(3)} = 0.305(3)$, $p_c^{(4)} \approx 0.383$ and $p_c^{(5)} \approx 0.47$–0.48 with the universality classes the $q = n$-Potts identification requires—three-state Potts at $n = 3$, marginal $q = 4$ logarithmic behaviour at $n = 4$—and, at $n = 5$, an exact three-size test ($L = 4, 6, 8$) of the predicted first-order transition. The test is enabled by a symmetry-reduced orbit basis of dimension 3,865 that carries the $120^4 = 2.1 \times 10^8$ bond labels of the $n = 5$, $L = 8$ sector as an exactly assembled dense block. Every measured quantity moves in the first-order direction (Table IV): the two-phase gap closing accelerates monotonically across replica number, with factors 1.81/1.94/2.12 over $L = 4 \to 6$ at $n = 3/4/5$ that persist at the third size as 1.45/1.54/1.68 over $L = 6 \to 8$, while the scaled gap $L\,\mathrm{gap}_{12}$, which converges toward the continuous $n = 3$ envelope (6.53 at $L = 8$, 6.21 at $L = 10$), falls to 3.84 at $n = 5$ at the matched size. Two sizes alone cannot separate exponential from power-law closing, and three sizes do not yet do so either: what the ladder delivers is directional—every matched-size rate super-continuous and n-monotone—and the separation is registered as a prediction rather than claimed as performed—if the $n = 5$ closing is genuinely exponential, the fourth size ($L = 10$) will show the effective closing rate stabilizing at a positive interface tension, and if it is a slowly decaying power law, that rate will continue to decay. Table V makes the same deposits a dual reading—effective interface-tension and power-law rates ordered consistently with the continuous, marginal and first-order behaviours at every matched interval.

For the Born-weighted and record objects: the normalized conditional-trajectory quantities are separated from the symmetric unnormalized ones by an information-theoretic no-go with an explicit bivariate recovery; the worst-case conditional-trajectory and conditional-purity decision problems (Appendix C) are established PP-hard; the no-freezing theorem (Theorem 9) proves existence, determinism and real-analyticity of the quenched record SCGF and rules out finite-q freezing for every allowable finite-reachable process; the exact replica interpolation identity (Proposition 8) reduces the annealed-to-quenched passage to a measurable self-averaging criterion; the first numerical quenched record SCGFs for Haar circuits are measured and exhibit genuine record multifractality; and the Clifford record-count disorder statis-

tics are closed exactly through a verified conjugation 3-design.

A word on what "exact" means here, since the computational claims rest on it. Every proved statement is free of floating-point computation: the three-replica rank certificates of Appendix A are exact integer-matrix statements with rational interval certificates, certified through $L = 12$, and the Collatz–Wielandt enclosures of Appendix A 5 certify the two-replica Perron eigenvalue exactly at rational points. Every computed eigenvalue quoted in this paper is anchored in four independent ways: the operating code reproduces the closed-form Kaufman/Ising answer to 3×10^{-15} at $n = 2$; the $p = 1$ rank-one anchor matches its closed-form value exactly for every $n = 2, \ldots, 5$, checking the pseudo-inverse Weingarten channel itself at $n = 5$; the compressed and full-space routes agree to 4×10^{-16} where both are available, the $L = 8$ assembler reproduces the deposited $L = 4$ spectrum to 1.6×10^{-14}, the $L = 6$ blocks match unrestricted colour-restricted Arnoldi solves, and two-seed full-space Arnoldi cross-checks leave residuals below 2×10^{-15}; and all deposited artifacts carry checksum ledgers, so every number in this paper is regenerable from the deposit. The residual numerical error is floating-point roundoff in eigenvalue evaluation only, bounded by the cross-route agreement above and, at the decisive $n = 5$ points, by float64 spot-checks on the float32 sweep validated to 4.3×10^{-6}. One gap in the chain is stated plainly: the $L = 8$, $n = 5$ block itself has no independent-method check at its own size—its validations are the assembler's agreement with the deposited $L = 4$ spectrum, the $L = 6$ blocks' agreement with unrestricted colour-restricted Arnoldi solves, and the float64 spot-checks, all at smaller sizes or internal to the block route—and an independent method at $L = 8$ itself, for instance a randomized lower-bound rank estimate at the decisive p's, would close it.

The significance of the annealed ladder is that it is exactly computable at every replica number, and therefore a controlled calibration target for replica field theories of the quenched transition—with the annealed points receding from the quenched one ($0.2338 \to 0.305 \to 0.383 \to 0.47$ against the Clifford quenched benchmark 0.1597(8) of Appendix B), so that no sequence of annealed points converges to the Born-weighted transition from above. The extensions that follow immediately from the deposited machinery—the $d = 3$ rung at $L = 8$ on the same orbit basis and the finer p-grid resolving the $n = 5$ locator—join the interface-tension reading of Table V in the deposits. The finer grid places the $L = 8$ locator at $p = 0.465$ (gap 0.4797; locator sequence 0.48/0.47/0.465 over $L = 4, 6, 8$—monotone decreasing, so the $p_c^{(5)} \approx 0.47$–0.48 bracket is a plausibility range over the sequence, not an extrapolated interval); the $d = 3$ ladder through $L = 8$ reads 1.482/0.621/0.453 at its locators ($p \approx 0.85, 0.85, 0.84$), the gap level sitting below the $d = 2$ value at every size and $8\,\mathrm{gap}_{12} = 3.62$ below the $d = 2$ value 3.84: the level ordering carries the first-order direction with local dimension, while the interval closing factors ($\times 2.39$ then $\times 1.37$) decelerate as at $d = 2$ ($\times 2.12$ then $\times 1.68$)—both dimensions sitting in the crossover region before the asymptotic exponential regime (Supplemental Material [34], Sec. S7). The remaining extension—latent-heat extraction from the deposited eigensystems—follows from the same deposits. The deeper open problems—growth-rate regularity for $n \geq 3$, the uniformity of the tilted variance in (L, T), and the conditional continuum program of Conjecture 1—are collected in the Discussion.

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Data and code availability: The two-replica release (code, benchmark and validation tables, checksum ledger `release_checksums.sha256`) the three-replica rank certificates of Appendix A (JSON files, logs, scripts, ledger `certificate_sha256.txt`), and the stabilizer simulator, raw trajectory data (`npz`), analysis scripts and fit logs of Appendix B are deposited on preprints.org (DOI to be assigned prior to submission). The web simulation of the monitored circuits—an interactive stabilizer-tableau simulator of the Clifford dynamics of Appendix B, with the finite-size-scaling benchmark, the annealed replica chain, and the record SCGF analyses of the paper presented in interactive form—is hosted at `https://mikeaa2020.github.io/quantum-circuits/web-simulation/`.

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Appendix A: Rank certificates for the three-replica compressed operator

This appendix records what is actually proved about $\operatorname{rank} \mathsf{C}^{(3)}_{\mathrm{comp}}(p)$, and how, so that every number quoted in Sec. II F can be re-derived from the operator of Eq. (14) alone. No floating-point computation enters any statement below.

1. Reduction to an integer matrix

Fix $n = 3$, $d \geq 2$, $L = 2m$. In the bond basis, $K_{3,\mathrm{e}}J_{3,\mathrm{o}} = (\mathrm{Wg}_{d^2})^{\otimes m}\mathsf{B}_p$ with

$$\mathsf{B}_p[\sigma;\tau] = \prod_{k=1}^{m} T_p(\sigma_k, \tau_{k-1})\, T_p(\sigma_k, \tau_k),$$
$$T_p(\sigma,\tau) = (1-p)\, d^{c(\sigma^{-1}\tau)} + p\, d, \tag{A1}$$

$\sigma \in S_3^m$ labelling the even bonds and $\tau \in S_3^m$ the odd ones (indices mod m). The Gram matrix of $\{U_\sigma \otimes U_\sigma\}$ on a bond is $(d^2)^{c(\sigma^{-1}\tau)}$, whose eigenvalues $D(D+1)(D+2)$, $D(D^2-1)$, $D(D-1)(D-2)$ ($D = d^2 \geq 4$) are positive, so Wg_{d^2} is invertible; by Theorem 5(i) the second factor $K_{3,\mathrm{o}}J_{3,\mathrm{e}}$ is the G-adjoint of the first, and therefore

$$\operatorname{rank} \mathsf{C}^{(3)}_{\mathrm{comp}}(p) = \operatorname{rank} K_{3,\mathrm{e}}J_{3,\mathrm{o}} = \operatorname{rank} K_{3,\mathrm{o}}J_{3,\mathrm{e}} = \operatorname{rank} \mathsf{B}_p \tag{A2}$$

for every p and d (the three ranks were also computed independently and agree at every point tested). The site map T_p is central in $\mathbb{Q}[S_3]$, $T_p = a_{\mathrm{triv}}\Pi_{\mathrm{triv}} + a_{\mathrm{std}}\Pi_{\mathrm{std}} + a_{\mathrm{sgn}}\Pi_{\mathrm{sgn}}$ with the coefficients quoted in Sec. II F. At $d = 2$, $a_{\mathrm{sgn}} \equiv 0$ and $T_p = 2[(2-p)J + (1-p)X]$ with J the all-ones matrix and $X(\sigma,\tau) = \chi_{\mathrm{std}}(\sigma^{-1}\tau) \in \{2, 0, -1\}$; the factor 2^{2m} is dropped. For every d the entries of B_p are then polynomials in p with integer coefficients and degree $2m$: there are no denominators, and the endpoints $p = 0, 1$ need no separate treatment. The singular site Gram matrix at $d = 2$ (the sign eigenvalue $d(d-1)(d-2)$ vanishes) is never inverted anywhere in the construction.

2. Symmetry, blocks, and the bound $r_3(L)$

B_p intertwines the actions of $\Gamma = H \times D_m$, $H = (S_3 \times S_3) \rtimes \mathbb{Z}_2 \cong S_3 \wr \mathbb{Z}_2$ ($|H| = 72$; nine irreducible representations, of dimensions $1, 1, 1, 1, 2, 4, 4, 4, 4$), on the odd and even label sets. Rotations act identically on both rings, but the ring reflection fixes two bonds of the even ring and none of the odd ring ($k \mapsto -k$ versus $k \mapsto -1-k$). Hence $\operatorname{rank} \mathsf{B}_p \leq \sum_\rho d_\rho \min(m_\rho^{\mathrm{o}}, m_\rho^{\mathrm{e}})$, with $m_\rho^{\mathrm{o/e}}$ the multiplicities computed by Burnside's lemma from the fixed-point counts of Γ on S_3^m. For m odd all reflections of D_m are conjugate and the two multiplicities agree, so the bound is 6^m. For $m = 2m'$ the deficit comes only from the two one-dimensional representations of D_m that distinguish the reflection classes, and equals $\frac{1}{144}\sum_\rho d_\rho |\omega_\rho(m')|$ with $\omega_\rho(m') = \sum_{h\in H} \chi_\rho(h) F_2(h)^{m'-1}[F_1(h)^2 - F_2(h)]$, where $F_1(h)$ and $F_2(h)$ count the fixed points of h and h^2 on S_3. Since $F_2 \in \{0, 2, 3, 6\}$, each ω_ρ is an exponential polynomial $a_{\rho,6}6^{m'-1} + a_{\rho,3}3^{m'-1} + a_{\rho,2}2^{m'-1}$ with integer coefficients, computed from the exact character table of H (script `n3_closed_form_proof.py`):

ρ	d_ρ	$a_{\rho,2}$	$a_{\rho,3}$	$a_{\rho,6}$	$\operatorname{sgn}\omega_\rho$
$\mathrm{triv} \otimes \mathrm{triv}^{\,\iota+}$	1	−36	0	36	+
$\mathrm{triv} \otimes \mathrm{triv}^{\,\iota-}$	1	36	48	−84	−
$\mathrm{sgn} \otimes \mathrm{sgn}^{\,\iota+}$	1	36	0	108	+
$\mathrm{sgn} \otimes \mathrm{sgn}^{\,\iota-}$	1	−36	48	−12	−
$\mathrm{Ind}(\mathrm{triv} \otimes \mathrm{sgn})$	2	0	48	96	+
$\mathrm{std} \otimes \mathrm{std}^{\,\iota+}$	4	0	48	240	+
$\mathrm{std} \otimes \mathrm{std}^{\,\iota-}$	4	0	0	0	0
$\mathrm{Ind}(\mathrm{triv} \otimes \mathrm{std})$	4	0	−48	48	+
$\mathrm{Ind}(\mathrm{sgn} \otimes \mathrm{std})$	4	0	−48	192	+

For $m' \geq 4$ the sign of every nonzero ω_ρ is that of $a_{\rho,6}$, because $|a_{\rho,6}|6^{m'-1} > |a_{\rho,3}|3^{m'-1} + |a_{\rho,2}|2^{m'-1}$ holds in every row (the tightest is the fourth, $12 \cdot 6^{m'-1} > 48 \cdot 3^{m'-1} + 36 \cdot 2^{m'-1}$, true from $m' = 4$ on and false at $m' = 3$). Summing $d_\rho|\omega_\rho|$ with these signs, the coefficients of $6^{m'-1}$, $3^{m'-1}$ and $2^{m'-1}$ are $36 + 84 + 108 + 12 + 2 \cdot 96 + 4 \cdot 240 + 4 \cdot 48 + 4 \cdot 192 = 2352$, $-48 - 48 + 2 \cdot 48 + 4 \cdot 48 - 4 \cdot 48 - 4 \cdot 48 = -192$ and $-36 - 36 + 36 + 36 = 0$, so $144 \times \text{deficit} = 2352 \cdot 6^{m'-1} - 192 \cdot 3^{m'-1}$, i.e. deficit $= (49 \cdot 6^{m'-1} - 4 \cdot 3^{m'-1})/3$ for $m' \geq 4$; direct evaluation of the nine ω_ρ at $m' = 1, 2, 3$ gives 15, 94, 576, which the same formula reproduces. This is Eq. (55). For the certificates the coarser group $H \times \mathbb{Z}_m$ is used: with the self-adjoint rational primitive idempotents of $\mathbb{Q}[S_3]$ (trivial, sign, and a Young idempotent for the standard representation), the transposition parity ι, and the rational momentum projectors, B_p is block diagonalized over $\mathbb{Q}$ into multiplicity blocks (ρ, k) of size $m_{\rho,k}$, each occurring d_ρ times, and $\sum_\rho d_\rho m_\rho = 6^m$ is checked. For $k \neq 0, m/2$ the rational block is the sum of the two conjugate complex momentum sectors.

3. Interval certificates and exceptional points

For each block the generic rank r is found by exact modular rank at two rational points; the certificate polynomial is $c(p) = \det$ if r equals the block size and $c(p) = e_r(\mathsf{N}\mathsf{N}^{\mathsf{T}})$, the r-th elementary symmetric function of the eigenvalues, i.e. the sum of the squares of all $r \times r$ minors (Cauchy–Binet), otherwise. In both cases $c(p^*) = 0$ if and only if the block rank at p^* is below r, and c is an integer polynomial of degree at most $4mr$, recovered by exact interpolation from integer evaluations (with additional evaluation points asserted). Writing $c(p) = p^a(1-p)^b h(p)$ and making h square-free, the real roots of h in $(0, 1)$ are counted by exact Descartes–Vincent bisection and isolated, with their minimal polynomials, by certified root isolation of the irreducible factors; a and b are the orders at the endpoints, listed separately. This is equivalent to the Smith-form criterion but cheaper: over the principal ideal domain $\mathbb{Q}[p]$ a block $\mathsf{N}(p)$ of generic rank r has a Smith normal

TABLE VIII. Certified generic ranks of $\mathsf{C}^{(3)}_{\mathrm{comp}}(p)$ ($n = 3$, $d = 2$), the symmetry bound $r_3(L)$ of Eq. (55), the number of isotypic blocks, the exceptional set, and the endpoint ranks. All entries are exact. At $L = 10$ and $L = 12$ the generic rank is certified (two-sided at $L = 12$, Sec. A 4) but E_L has not been computed.

L	$6^{L/2}$	$r_3(L)$	generic rank	blocks	$\lvert E_L\rvert$	rank at $p = 0$
4	36	21	21	8	0	21
6	216	216	216	14	1	198
8	1296	1202	1202	25	11	1184
10	7776	7776	7776	27	?	7663
12	46656	46080	46010	36	?	45822

form $\mathsf{N} = \mathsf{U}\,\mathrm{diag}(s_1, \ldots, s_r, 0, \ldots)\mathsf{V}$ with U, V invertible over $\mathbb{Q}[p]$ and invariant factors $s_1 \mid s_2 \mid \cdots \mid s_r$ (Ref. [55], Chap. VI); since $\mathsf{U}(p^*)$ and $\mathsf{V}(p^*)$ are invertible at every p^*, $\operatorname{rank}\mathsf{N}(p^*) = \#\{i : s_i(p^*) \neq 0\}$, so the rank drops at p^* iff $s_r(p^*) = 0$, iff the r-th determinantal divisor $s_1 \cdots s_r$, the greatest common divisor of the $r \times r$ minors, vanishes at p^*, iff all $r \times r$ minors vanish there, iff $c(p^*) = 0$; and the drop equals the number of vanishing invariant factors, times d_ρ for the full operator. At each root p^* with minimal polynomial f the block rank is then computed exactly over the number field $\mathbb{Q}[x]/(f)$ (companion-matrix blow-up and exact rank), which gives the ranks listed in Tables S1 and S2 of the Supplemental Material [34]; for the two largest $d = 3$, $L = 8$ factors (degrees 46 and 50) the nullity is instead pinned by $1 \leq \text{nullity} \leq \mathrm{mult}_f(\det)$ together with the evenness of the nullity in $k = \pm 1$ blocks. Comparing the multiplicity of f in c with the nullity shows that every interior zero at $d = 2$ is simple (11 of 11 block factors at $L = 8$); at $d = 3$ all are simple except the linear factor $3p - 2$ in two $k = \pm 1$ blocks.

The result is the following statement, which replaces any claim of p-independence.

Proposition 15 (Generic rank with explicit exceptional set)**.** *Let $n = 3$, $d = 2$ and $L \in \{4, 6, 8\}$. For every real $p \in (0, 1) \setminus E_L$, $\operatorname{rank}\mathsf{C}^{(3)}_{\mathrm{comp}}(p) = r_3(L)$, where $E_4 = \emptyset$, $E_6 = \{(9 - \sqrt{41})/4\}$ and E_8 consists of eleven algebraic points with minimal polynomials of degree 5 to 22, listed with their blocks and exact ranks in Table S1 of the Supplemental Material [34]; at the points of E_L the rank takes the listed smaller values (208 at $L = 6$; 1194–1201 at $L = 8$). At $p = 0$ the ranks are $21, 198, 1184$ and at $p = 1$ the rank is one. Consequently, on every compact $K \subset (0, 1) \setminus E_L$ the $r_3(L)$ nonzero singular values of B_p are bounded away from zero, whereas near $p = 1$ all but one of them tend to zero.*

4. The $L = 12$ two-sided certificate

At $L = 12$ ($m = 6$, dimension 46656) the block matrices are too large for the interval certificate, and the generic rank is certified differently. *Lower bound.* For each block (ρ, k), with random $E = \Pi R$ and $F = \Pi^{\mathsf{T}} R'$ over $\mathbb{F}_q$ ($q = 1048261 \equiv 1 \bmod 60$), $\operatorname{rank}(F^{\mathsf{T}}\mathsf{B}_p E) \leq \operatorname{rank}\mathsf{B}_{\rho,k}(p)$; B_p is applied matrix-free as a matrix-product operator of bond dimension six, the multiplicities are checked exactly against the trace formula, and rank over $\mathbb{F}_q$ is a lower bound for rank over $\mathbb{Q}$. At $p = \frac{1}{2}$ this gives 46010: all sectors $k \neq 3$ have full multiplicity rank (38906 in total), and the momentum-π sector $k = 3$ has rank 7104 against its symmetry-forced bound 7174. *Upper bound.* On the $k = m/2$ sector the two ring reflections satisfy $R_{\mathrm{o}} = -R_{\mathrm{e}}$, and since $\mathsf{B}_p^{\mathsf{T}} = P_{\mathrm{e}}\mathsf{B}_p P_{\mathrm{e}}$ and $P_{\mathrm{e}}\mathsf{B}_p = \mathsf{B}_p P_{\mathrm{o}}$ (verified exactly), the sector block is antisymmetric and exchanges the two R_{e}-parity subspaces $V_\pm$ of every isotypic block: $\operatorname{rank}\mathsf{B}|_{(\rho,3)} = 2\operatorname{rank}(\mathsf{B}|_{V_-}) = 2\operatorname{rank}(\mathsf{B}|_{V_+}) \leq 2\min(m_+ - c_+, m_- - c_-)$, where $c_\pm(p)$ is the dimension of any exhibited subspace of $\ker\mathsf{B}_p \cap V_\pm$. (With $c_\pm = 0$ this is the Burnside deficit $6^6 - r_3(12) = 576$ in a different guise.) We exhibit 27 constant kernel vectors (found as the common kernel at three rational p, rationally reconstructed, small integer entries) and 29 polynomial ones $v(p) = \sum_{i \leq D} p^i v_i$ with $D \in \{2, 4, 6\}$ and integer coefficients of modulus at most 576; each is verified exactly over $\mathbb{Z}$ (membership in the block via the exact rational projector, parity, momentum, and $\mathsf{B}_p v(p) = 0$ at $2m + D + 1$ integer points, which suffices for a polynomial identity of degree $\leq 2m + D$). Their span is 10 multiplicity units beyond what the reflection count forces (nullities $1, 1, 4, 3, 1$ in the blocks $\mathrm{sgn} \otimes \mathrm{sgn}^{\iota-}$, $\mathrm{Ind}(\mathrm{triv} \otimes \mathrm{sgn})$, $\mathrm{std} \otimes \mathrm{std}^{\iota-}$, $\mathrm{Ind}(\mathrm{triv} \otimes \mathrm{std})$, $\mathrm{Ind}(\mathrm{sgn} \otimes \mathrm{std})$, weights $d_\rho = 1, 2, 4, 4, 4$; total 70), so $\operatorname{rank}\mathsf{B}_p \leq 46010$ for all p outside a finite set, and the two bounds meet block by block: $\operatorname{rank}_{\mathbb{Q}(p)}\mathsf{B}_p = 46010$. The same lower bound 7104 for the $k = 3$ sector is obtained at $p = \frac{1}{3}, \frac{3}{5}$, with another prime and seed, at $d = 3$ (where the 27 constant vectors are again in the kernel and 29 polynomial vectors of the same degrees are found and verified, closing the certificate at $d = 3$ as well), and for a random central element $\sum_\lambda a_\lambda d_\lambda \chi_\lambda$ in place of T_p: the deficit is a property of the ring structure, not of the curve $p \mapsto T_p$.

5. Collatz–Wielandt enclosures of the two-replica Perron eigenvalue

For rational p the weights of Eq. (38) are rational, hence so is the compressed period operator. The Collatz–Wielandt theorem (for a positive matrix A and a positive vector x: $\min_i(Ax)_i/x_i \leq \lambda_1(A) \leq \max_i(Ax)_i/x_i$), evaluated in exact rational arithmetic, gives two-sided certificates of the Perron eigenvalue λ_1 of $\mathsf{C}_{\mathrm{comp}}$: at $L = 16$, $d = 2$, $p = 2/5$: [0.0022886508026864124561, 0.0022886508026864175903] (relative width 2.2×10^{-15}); $p = 4/25$:

[0.075189633382987646, 0.075189633382987911] (3.5×10^{-15}); $p = 1169/5000$: [0.023692745171879 7476, 0.023692745171879 8042] (2.4×10^{-15}); at $L = 20$, $d = 2$, $p = 2/5$: [0.00050028570068131041, 0.00050028570068131407] (7.3×10^{-15}); at $L = 16$, $d = 3$, $p = 2/5$: [$6.3658872262159916\times10^{-5}$, $6.3658872262160092\times10^{-5}$] ($2.9 \times 10^{-15}$). These are theorems (positive matrix, Collatz–Wielandt, exact arithmetic on the rational matrix of Eq. (40)), they certify the benchmark agreement of Sec. II E exactly at these points, and the closed form of Proposition 5 lies inside every enclosure.

6. Files

The certificates are machine-readable JSON files (block list with d_ρ, $m_{\rho,k}$, generic rank, certificate type, degree, endpoint orders, isolated roots with minimal polynomials; exact ranks at the exceptional points; the $L = 12$ kernel vectors and block table), produced by the scripts `n3_mult_certificates_d.py`, `n3_exceptional_exact_d.py`, `n3_sharpness_L12.py`, `n3_L12_kernel.py`, `n3_L12_kernel_verify.py`, `n3_L12_moving_kernel.py`, `n3_L12_final.py` and `n3_kernel_recheck.py` (dependencies: `numpy`, `python-flint`). The complete file list with SHA-256 hashes (ledger `certificate_sha256.txt`), the per-block tables at $L = 12$, the exceptional-point tables for $d = 2$ and $d = 3$, and the reproduction commands are given in the Supplemental Material [34]. The two-replica verification chain is deposited alongside: `mipt_kaufman_lib.py` (closed form), the spinor-factorisation verifier (checks V1–V4, residual $\leq 6\times10^{-16}$), `mipt_audit3_cw_exact.py` (the enclosures above), `mipt_purif_sim.py/mipt_purif_analysis.py` (the purification locator), `mipt_fss2_floorN.py` (the released $1/N$ floors) and `mipt_seed_recheck.py` (the seed contract), with logs and JSON results; the files are deposited with the data release (see Data and code availability). The Clifford record-count suite of Sec. IV F is deposited alongside: `mipt_scgf_exact.py` (the exact 3^L-basis algebra evolution with the deposit calibration), `mipt_born_scgf.py` (the vectorized tableau trajectories), the results `scgf_exact.json`, `scgf_exact_addendum.json`, `scgf_born.json` and the raw trajectory arrays (npz), with hashes and reproduction commands in the Supplemental Material [34], Sec. S7. The exact-closure suite is deposited with it: the shared library `gap_utils.py` (Eq. (14); validated by the full battery V1–V8, all passing), `v22_clifford_3design_v1.py` (the conjugation-design test of Remark 7), `v22_z3_exact_v1.py` (Prop. 13), `v22_traj_beta2_v1.py` (the well-conditioned identification cross-check), `v22_q4_logcorr_v1.py` and `v22_n4_eps_topup_v1.py` (the marginal $q = 4$ analysis), `v22_tiltvar_uniform_v1.py` (the tilted-variance profile family), and `v22_n5_L8_block_v1.py` (the symmetry-reduced block route, its rung result `v22_n5_L8_rung.json` deposited), with their JSON results and logs.

Appendix B: Finite-size scaling of the quenched Clifford transition

This appendix is a computational benchmark, not a theorem. It fixes one ensemble—the Clifford realization of the period of Sec. II D—and reports the finite-size-scaling evidence for its quenched, Born-weighted entanglement transition, so that the exact annealed point $p_c^{(2)}(2) = 0.233810$ of Proposition 4 can be placed next to the quenched one for the same protocol. Everything below is reproducible from the deposited simulator, raw trajectories and analysis scripts (Data and code availability): the raw I_3 dataset (192 trajectory files, 475 600 trajectories, $L = 16$–512), the purification dataset (270 files, 135 000 trajectories), every analysis script and every machine-readable result are deposited in the workspace release, and the full analysis, including all estimator and window variants, is in the deposited report. The dataset was re-verified end to end: the seed contract regenerates the test trajectories bit-exactly, the released $1/N$-floor pipeline reproduces every windowed number of Table IX without change, and the purification analysis reproduces every deposited number to the bit.

Ensemble. Periodic chain of L qubits, initial state $|0\cdots0\rangle$; one period is [measurement layer][gates on the odd matching][measurement layer][gates on the even matching], every site measured in the computational basis independently with probability p in each measurement layer, and every gate an independent uniformly random two-qubit Clifford (sampled as the 720 symplectic actions, under which every recorded quantity is invariant). By the 2-design property all annealed $n = 2$ quantities coincide with the Haar ones (Remark 2); the quenched quantities below are Clifford-specific. A bit-packed stabilizer simulator records, at $t = L/2, L, 2L, 4L$ periods, the stabilizer entropies S_ℓ (bits) of contiguous blocks $\ell = L/8, L/4, 3L/8, L/2$, the tripartite information I_3 of the four contiguous quarters, the entropy of two antipodal quarters, and the number $N_{\rm rand}$ of random outcomes. Sizes and samples: $L = 16, 24, 32, 48, 64$ (4000 trajectories each), 96, 128 (3000), 192, 256 (2000), 384 (800), 512 (400; $t \leq 2L$), on the grid $p = 0.12$, 0.13, 0.14, 0.145, 0.15, 0.155, 0.1575, 0.16, 0.1625, 0.165, 0.17, 0.175, 0.18, 0.19, 0.20, 0.22 (restricted to $[0.14, 0.18]$ at $L = 384$ and to $[0.15, 0.17]$ at $L = 512$); 475 600 trajectories in total, each an independent disorder sample. Trajectory k of file (L, p) is generated by the legacy MT19937 stream seeded with $seed0(L, p) + k$, the value *seed0* being stored in the file; regeneration is bit-exact (re-verified on the deposited dataset for 44 trajectories spanning $L = 16$–512, including the $I_3 \equiv 0$ file at $L = 384$, $p = 0.18$). All uncertainties are bootstrap standard errors over trajectories (200 resamples) unless stated otherwise.

Validation. Three exact checks: (i) the Clifford identity $\mathbb{E}_\omega 2^{-N_{\rm rand}(\omega)} = \bar{Z}_2(t)$ of Eq. (28) against the exact compressed $n = 2$ transfer matrix, 0.032959(14) versus 0.032962 at $L = 16$, $p = 0.16$, $t = 2$; (ii) the fully monitored point $p = 1$, where $S_{L/2} = 0.8$ bits exactly and $\mathbb{E}N_{\rm rand} = 22/15$ per period, reproduced as 0.795(11) and 1.46597(43); (iii) agreement of all six recorded entropies with an independent tableau simulator (`stim`) at $L = 16, 32$ to within 1.3 standard errors. Steady state: at $p = 0.16$, $S_{L/2}$ and I_3 at $t = L$, $2L$, $4L$ agree within errors for every $L \le 512$; all steady-state fits are made at both $t = 2L$ and $t = 4L$ and reported side by side.

Critical point. Table IX lists the crossings of $I_3(L)$ and $I_3(2L)$ (monotone piecewise-cubic interpolation of each curve; three other estimators agree within 0.001 for $L \ge 96$). They drift upward from 0.148 at $L = 16$ and saturate for $L \ge 96$ at 0.1596–0.1607 at both depths; a power-law extrapolation $p_\times = p_c + aL^{-\omega}$ gives 0.1602(6) ($\omega_{\rm eff} = 1.6(4)$–$2.5(8)$ depending on $L_{\rm min}$) and the weighted plateau of the $L \ge 96$ crossings 0.1599(3). The joint collapse $I_3 = F((p - p_c)L^{1/\nu})$ with a quartic F on $|(p - p_c)L^{1/\nu}| \le 1$ gives $p_c = 0.1589$–0.1595, rising monotonically with the smallest size retained and stable against the window (± 0.0003 for $|x|_{\max} \in \{0.6, 1, 1.5\}$), with $\chi^2/\mathrm{dof} = 0.7$–$1.2$ for $L \ge 64$ and $|x|_{\max} \le 1$ (χ^2 p-values 0.19–0.84; $p \ge 0.40$ for $|x|_{\max} = 0.6$; the widest windows $|x|_{\max} = 1.5$ reach $\chi^2/\mathrm{dof} = 1.5$–$2.0$, $p = 8 \times 10^{-6}$–0.02, through area-law-edge points). The crossings of $\Delta S(L) = S_{L/2}(2L) - S_{L/2}(L)$ with $\Delta S(2L)$, an observable independent of I_3, give 0.159–0.161 for $L \ge 32$. The entropy itself locates the same point once its non-universal part is handled correctly. A collapse of the difference $S_{L/2}(p, L) - S_{L/2}(p_c, L) = G((p - p_c)L^{1/\nu})$ in the form of Ref. [10] is unstable here ($p_c = 0.1556$–0.1638, $\nu = 1.0$–1.27 depending on the window and on a cut in the scaling variable), and the reason is identifiable: besides $\alpha \ln L + b$ the entropy contains an analytic, L-independent, cut-local term $a_1(p - p_c)$ with $a_1 = +8$ to $+14$ bits per unit p for two cuts (it doubles for the four-cut S_{AC} and vanishes for the zero-net-cut combinations I_3, $S_{L/2} - S_{L/4}$ and $I(A{:}C)$); the subtraction at the fitted p_c does not remove it, and a polynomial G can absorb an L-independent tilt only by lowering ν and shifting p_c, producing a diagonal χ^2 valley with $\mathrm{d}\nu/\mathrm{d}p_c \approx 30$ in the (p_c, ν) plane. Synthetic data generated from the pure scaling form do not show the instability; synthetic data with the fitted a_1 reproduce it. Estimators in which the term cancels identically or is modelled give stable results: the collapses of $D = S_{L/2} - S_{L/4}$ and of $\Delta S_L = S_{L/2}(2L) - S_{L/2}(L)$ (no subtraction) give $p_c = 0.1596$–0.1602, $\nu = 1.21$–1.25 (bootstrap ± 0.0004, ± 0.08; $\chi^2/\mathrm{dof} = 0.9$–$1.2$), and the joint fit $S_{L/2} = \alpha \ln L + b + a_1(p - p_c) + F((p - p_c)L^{1/\nu})$ over all depths, size cuts $L \ge 32, 48, 64$, quartic or sextic F, with or without a cut, gives $p_c = 0.1598$–0.1601 (± 0.0003), $\nu = 1.20$–1.25 (± 0.02–0.04), $\chi^2/\mathrm{dof} = 0.94$–$1.16$; omitting a_1 raises χ^2 by 30–155 for one degree of freedom and biases ν down to 1.08–1.16, which is the bias of the subtracted form. Adopted: $p_c = 0.1597 \pm 0.0008$, the interval covering all estimators (the corrected, tail-safe fixed-set fits centre at 0.1597–0.1598; the statistical errors alone are ± 0.0003).

Exponent. The collapse gives $\nu = 1.18$–1.27 over all depth, size and window choices with $L \ge 48$; the slope method $\partial_p I_3|_{p_c} \propto L^{1/\nu}$ (cubic fits over $|p - p_c| \le 0.011$, $L = 32$–512) gives $\nu = 1.13$–1.36 as the assumed p_c ranges over 0.1585–0.1605; with ν frozen on *fixed* point sets ($L \ge 64$; spline scaling function with the area-law tail retained, over the eight tail-safe variants of the deposited report), the $\Delta\chi^2$ profiles are 52–334 for $\nu = 1$, 0.2–11 for $\nu = 1.20$, 0.1–2.6 for $\nu = 1.24$, 0–9.9 for $\nu = 1.28$, 3.0–36.5 for 4/3, and 12.5–89 for $\nu = 1.40$ (sextic scaling function with the tail removed: $\Delta\chi^2(\nu = 1) \ge 72$). A power-law correction to scaling $AL^{-\omega}$ is preferred only when $L = 48$ is retained ($\Delta\mathrm{AIC} = -7$ to -14), is neutral at $L \ge 64$ and disfavoured at $L \ge 96$, with ω undetermined (0.3–4, bootstrap s.e. ≈ 1); it raises ν by 0.02–0.04. We disclose the estimator weighting, whose defect biased earlier fixed-set fits: standard errors are floored at $1/N_{\rm traj}$ per point in the released code (the legacy $10^{-4}/10^{-6}$ floors of the v13 analysis code, which let a single saturated point—$I_3 \equiv 0$ in all 800 trajectories at $L = 384$, $p = 0.18$—carry 98% of one fixed-set fit's weight, are replaced by $1/N$), and the area-law points, which a polynomial scaling function cannot be flat on, are either removed or given a spline scaling function; the released pipeline, re-run on the recovered raw dataset, reproduces every windowed number of Table IX without change, confirming that no quoted window contains a floored point. All I_3/entropy locators of this appendix share the same 475 600 trajectories; the only independent observable is the purification protocol below (disjoint seeds, mixed initial state). Adopted: $\nu = 1.24 \pm 0.07$, covering the full tail-safe envelope 1.21–1.29; $\nu = 1$ is excluded ($\Delta\chi^2 \ge 52$ in every variant), and the percolation value 4/3 is disfavoured ($\Delta\chi^2 \ge 3$ in every variant, ≥ 20 without a correction term) but not excluded by the weakest variant.

Logarithmic entanglement. At p_c (spline-interpolated in p, with the uncertainty of p_c propagated) the half-chain entropy is linear in $\ln L$ from $L = 16$ to 512 with $\chi^2/\mathrm{dof} \le 1.4$ (Table IX); at the neighbouring grid points $p = 0.1575$ and 0.1625 a straight line in $\ln L$ fails ($\chi^2/\mathrm{dof} = 6$–8). The straight-line fits give $\alpha = 1.58$–1.66 at $p_c = 0.1595$ but inherit $\mathrm{d}\alpha/\mathrm{d}p \approx -110$ times the uncertainty of p_c; the joint fit above, which determines α together with p_c, ν and the analytic background, gives $\alpha = 1.50$–1.56 (bootstrap ± 0.02–0.04); a power-law correction $s_1 L^{-y}$ ($y = \frac{1}{2}, 1, 2$) is at most weakly preferred ($\Delta\mathrm{AIC} \ge -3$) and moves α by at most 0.14; and the straight-line rows re-evaluated at the adopted $p_c = 0.1597$ (spline window $p \in [0.145, 0.175]$) give $\alpha = 1.53$–1.59, inside the adopted interval, and the zero-net-cut collapses give 1.46–1.66 through $F(0)$. Adopted: $\alpha = 1.55 \pm 0.07$ bits per $\ln L$ for the periodic half chain, i.e. 0.78(4) bits = 0.54(2) nats per entangle-

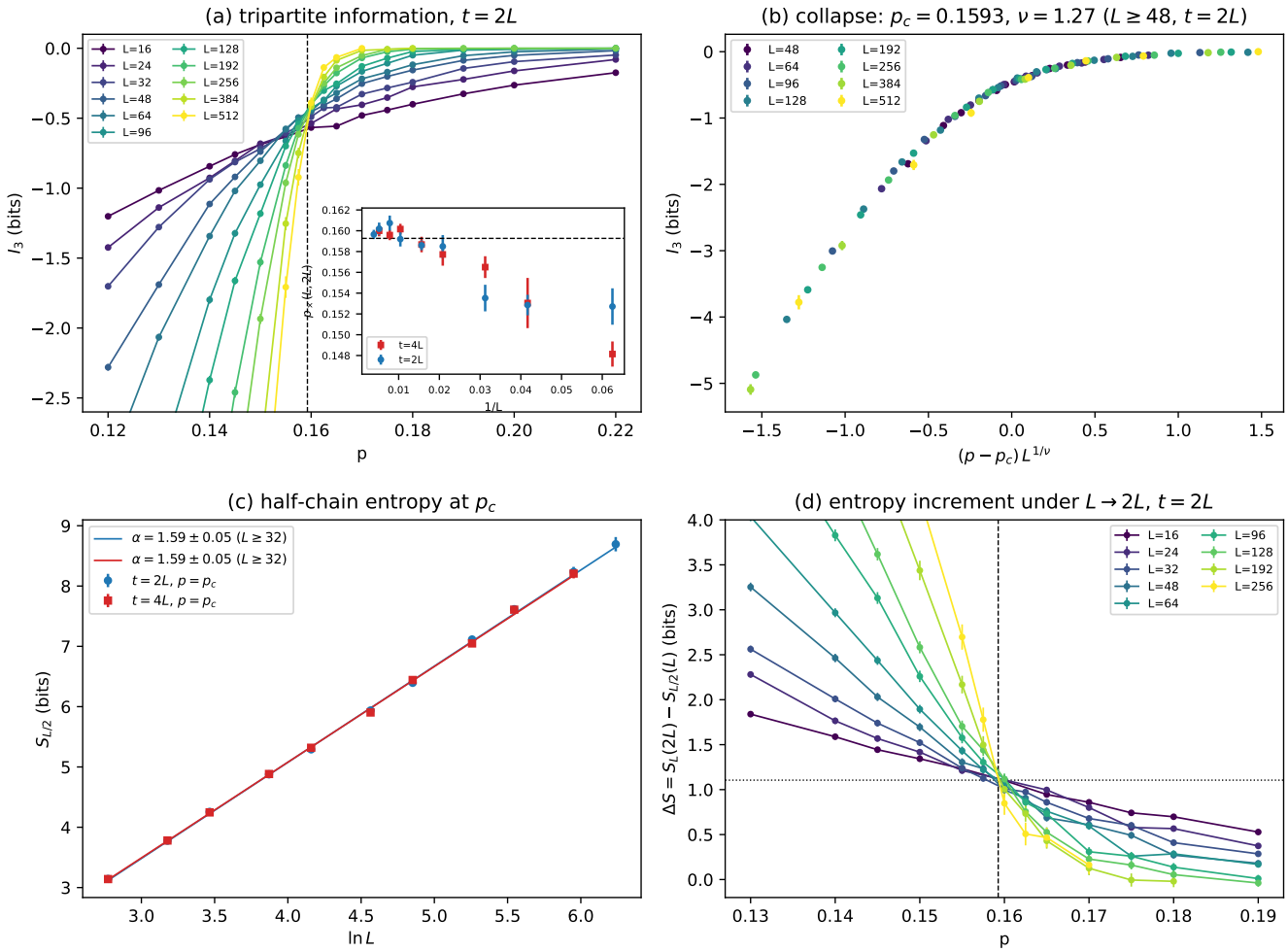


FIG. 1. Quenched Clifford transition, $t = 2L$. (a) $I_3(p)$ for $L = 16$–512; dashed line $p_c = 0.1593$ from the collapse in (b); inset: crossings $p_\times(L, 2L)$ (PCHIP estimator) against $1/L$ at $t = 2L$ (circles) and $t = 4L$ (squares). (b) Collapse $I_3 = F((p-p_c)L^{1/\nu})$, $L \geq 48$, $|x| \leq 1.6$ shown. (c) $S_{L/2}$ at p_c (spline-interpolated in p) against $\ln L$ at $t = 2L$ and $4L$ with the $L \geq 32$ fits. (d) Entropy increment $\Delta S = S_{L/2}(2L) - S_{L/2}(L)$; at p_c it equals $\alpha \ln 2$ (dotted) independently of L.

ment cut, the interval covering all three routes. Chord fits $S_\ell = a \ln[(L/\pi)\sin(\pi\ell/L)] + b$ to the four block sizes at fixed L give $a = 1.48$–1.62 bits for every L from 64 to 384, and the mutual information of two antipodal quarters is L-independent at $p = 0.16$ (0.67(3), 0.73(4), 0.58(5), 0.69(7), 0.85(17) bits for $L = 32, \ldots, 512$) while it grows at $p = 0.155$ and decays at $p = 0.165$, as conformal invariance requires. In separate time-resolved runs at $p = 0.16$ ($L = 128, 256, 512$; $3000, 1500, 500$ trajectories; $t = 1, 2, 4, \ldots$) the growth before saturation is $S_{L/2}(t) = \alpha_t \ln t + \text{const}$ with $\alpha_t = 1.41(2)$, $1.44(2)$, $1.49(3)$ bits, the three sizes falling on one curve until each saturates at $t \approx L$; hence $z = \alpha/\alpha_t = 1.03$–$1.10$, consistent with $z = 1$ up to the same finite-size systematics. In the boundary-CFT language of Ref. [56], 0.54(2) nats per cut is the scaling dimension $h_{a|b}$ of the entanglement boundary-condition-changing operator (their 0.53); it is not a central charge.

Annealed versus quenched. The exact annealed quasi-entropy $\tilde{S}_2(A)$ of Eq. (28) in the stationary Perron state exceeds the quenched $\mathbb{E}_\omega S(A)$ at every size and p; the excess is a change-of-measure effect rather than a convexity one: the record-collision weight of the annealed channel, proportional to p_R^2, favours records with fewer measurements and lowers the effective monitoring rate to $p_{\text{eff}} = p + \frac{p(1-p)}{2L}\partial_p \ln \lambda_1$ (0.105 at $L = 8$, 0.101 at $L = 12$, 0.092 in the $L \to \infty$ limit at $p = 0.16$; 0.149 at $p_c^{(2)}$). At $L = 8$, $t = 4$, $p = 0.16$ (40 000 Born trajectories, effective sample size 5935): $\mathbb{E}_{\text{Born}} S = 2.019(4)$, $\mathbb{E}_{\text{Born}} S(p_{\text{eff}}) = 2.339(4)$, $\mathbb{E}_{\text{tilt}} S = 2.429$, exact $\tilde{S}_2 = 2.155$ bits—the density shift alone accounts for ≈80% of the tilt gain $\mathbb{E}_{\text{tilt}} S - \mathbb{E}_{\text{Born}} S(p) = 0.41$ bits, of which Jensen removes 0.27 bits, leaving the observed excess $\tilde{S}_2 - \mathbb{E}_{\text{Born}} S = 0.14$ bits; at $L = 12$ (2.394(5), 2.941(4), $\tilde{S}_2 = 3.128$) the ordering is the same (the sampled tilt average there, effective sample size 454, is not reliable and is not used).

The ratio growing from 1.3 at $L = 16$ to 1.8 at $L = 32$ for $p \in [0.16, 0.22]$; $\tilde{S}_2$ grows linearly with L for every $p < p_c^{(2)}$ (slope ≈ 0.1 bits per site at $p = 0.22$), where the quenched entropy has already saturated at ≈ 2.6 bits with $I_3 = 0$. The quenched transition at 0.1595 and the annealed one at 0.2338 are therefore distinct singularities of distinct quantities, as Sec. II E states, and nothing quenched happens at $p_c^{(2)}$. Haar-gate statevector runs with Born sampling ($L = 8, 12, 16$) show the same ordering of annealed and quenched entropies, but these sizes are too small for an independent Haar transition point; the Haar values $p_c^{\mathrm{H}} = 0.17(1)$, $\nu^{\mathrm{H}} = 1.2(2)$ are those of Ref. [11].

Comparison. For stabilizer circuits with the same Poissonian measurements the literature gives $p_c = 0.16$, $\nu = 1.3$, $\alpha = 1.6$ bits [10]; $p_c = 0.154(4)$ ($L \leq 24$), $\nu = 1.24(7)$, $z = 1.06(4)$ [11]; and $p_c = 0.15995(10)$ [14]. Gullans and Huse [4] report $p_c = 0.1593(5)$, $\nu = 1.28(2)$; the coefficient $\tilde{c} \approx 1.5$–1.6 quoted in that literature equals our α in units of $\ln 2$. The present values agree with all of them within the stated uncertainties.

Independent purification locator. A locator based on a different observable, and on trajectories disjoint from the 475 600 above, is the Gullans–Huse reference-qubit protocol [4]: the L-qubit system starts completely mixed, purified by L reference qubits ($2L$ qubits in total), and the same circuit and measurements act on the system only, so that $S_{\mathrm{ref}}(t)$, the entropy of the reference, is the entropy of the system's mixed state. We ran $L = 16$–256, $p = 0.140$–0.180 (step 0.005), 2000–4000 trajectories per point (independent seeds), recording S_{ref} at $\tau = t/L = 0.25, 0.5, 1, 2, 4$: 135 000 trajectories, 270 files. Sanity checks: $S_{\mathrm{ref}} = L$ at $p = 0$; $S_{\mathrm{ref}} = 0$ after one layer at $p = 1$; $S_{\mathrm{ref}} \equiv S_{\mathrm{sys}}$ exactly; $N_{\mathrm{meas}} \approx 2pLt$. At $p = 0.16$ the reference entropy is L-independent for $\tau \geq 0.5$ ($\tau = 1$: 1.72, 1.75, 1.77 bits at $L = 64, 128, 256$; $\tau = 2$: 0.60–0.62), i.e. a zero scaling dimension with $F(0, \tau) \sim 1/\tau$. Pair crossings of the $L \geq 64$ curves give 0.1606–0.1628; the collapses $S_{\mathrm{ref}} = F((p - p_c)L^{1/\nu})$ with a sextic F, $L \geq 64$, give $p_c = 0.1604$, $\nu = 1.251$ ($\chi^2/\mathrm{dof} = 15.5/18$) at $\tau = 0.5$ and $p_c = 0.1601$, $\nu = 1.249$ (18.3/18) at $\tau = 1$; a trajectory bootstrap (100 resamples) gives 0.16011(9), 1.250(14) at $\tau = 1$. The $\tau = 2$ collapse (0.1603, 1.310, 82.8/18) is degraded by the same saturated-tail pathology as the fixed-set I_3 fits and is not used. Frozen-ν $\Delta\chi^2$ on the $\tau = 1$, $L \geq 64$ set: 544 ($\nu = 1$), 16.9 (1.2), 0.5 (1.24), 6.6 (1.28), 45.8 (4/3), 139 (1.4). Hence $p_c^{\mathrm{purif}} = 0.1601$–$0.1604$, $\nu^{\mathrm{purif}} = 1.25(2)_{\mathrm{stat}}(6)_{\mathrm{sys}}$—an independent-observable confirmation of the I_3 headline, quoted separately and not entering the adopted $p_c = 0.1597(8)$ (different observable, different finite-size corrections; combining them would hide that). The purification analysis was re-run end to end and reproduces every deposited number, the bootstrap values included, to the bit.

Appendix C: Precisely promised trajectory complexity

This appendix collects, with identical statements and full proofs, the worst-case complexity results summarized in Sec. III B; it is self-contained, and the main text depends on it only through the summary statements.

For exact counting statements we use the finite gate set generated by Hadamard, Toffoli, and computational-basis permutations, which is universal for quantum computation [35, 36]; the standard one-extra-rebit simulation accommodates complex amplitudes with polynomial overhead. Its amplitudes are counting quantities [37]: Toffoli gates and basis permutations map computational-basis states to computational-basis states, so a circuit with h Hadamard gates has $\langle z|U|0\rangle = 2^{-h/2}[N_+(z) - N_-(z)]$, where $N_\pm(z)$ count the computational-basis paths from $|0\rangle$ to $|z\rangle$ with sign ± 1; each $N_\pm$ is a #P function of the circuit and z, so $2^{h/2}$ times the amplitude is a GapP function, and GapP is closed under products and exponential sums. With the measurements deferred into ancilla registers (below), the probability of any event that fixes the values of some qubits is a sum of $|\langle z|U|0\rangle|^2$ over the compatible z, so 2^h times it is a GapP quantity, and every nonzero probability is at least 2^{-h}. Equivalent hardness holds for any uniformly compiled universal gate set; the GapP membership proof below is stated for this exact counting-friendly set. All hardness statements are worst-case; the ensemble randomness of the physical circuits plays no role in them.

Definition 5 (Conditional-trajectory decision)**.** *An instance of* Conditional-Trajectory Decision *(CTD) consists of a polynomial-size quantum circuit $\mathcal{C}$, a specified computational-basis measurement record R, and an output qubit. The promise is $p_R > 0$ and*

$$\Pr[\mathrm{out} = 1 \mid R] \geq \tfrac{2}{3} \text{ (YES)}, \quad \Pr[\mathrm{out} = 1 \mid R] \leq \tfrac{1}{3} \text{ (NO)}. \tag{C1}$$

No assertion is made in the gap.

Proposition 16 (Worst-case trajectory complexity)**.** *Viewed as a promise problem, CTD is complete for PostBQP under promise-preserving polynomial-time many-one reductions. Since PostBQP=PP [38], its promised instances are PP-hard and the decision rule has a PP extension. Completeness persists for qubit circuits compiled to a one-dimensional nearest-neighbor architecture with polynomial overhead and computational-basis postselection. It already holds for terminal postselection.*

Proof. We use Aaronson's definition [38]: a promise problem is in PostBQP if a uniform polynomial-size circuit family with a designated postselection qubit and output qubit satisfies $\Pr[\mathrm{post} = 1] > 0$ and $\Pr[\mathrm{out} = 1 \mid \mathrm{post} = 1] \geq 2/3$ on YES and $\leq 1/3$ on NO instances; for the Hadamard–Toffoli gate set every nonzero probability is at least 2^{-h}, so no separate lower bound on $\Pr[\mathrm{post} = 1]$ is needed. The statement is recorded as a

TABLE IX. Finite-size-scaling summary for the quenched Clifford transition (bits; bootstrap errors in the last digits). Top: crossings $p_\times(L, 2L)$ of I_3. Middle: collapse $I_3 = F((p - p_c)L^{1/\nu})$ on $|x| \le 1$ (bootstrap error of p_c is 0.0001–0.0002 in every row, from 200 trajectory resamples (100 for the purification locator); standard errors are floored at $1/N_{\mathrm{traj}}$ per point in the released code, which moves no windowed number; the spread across rows is the systematic error; $\chi^2/\mathrm{dof} = 2.2, 1.6, 1.1, 0.7$ and $2.6, 1.5, 1.2, 1.1$ down the two columns). Bottom: slope α of $S_{L/2}$ against $\ln L$ at $p_c = 0.1595$ (straight-line fits) and from the joint fit $\alpha \ln L + b + a_1(p - p_c) + F(x)$ with p_c, ν, a_1 free (sextic F, no cut). Last block: exact annealed $\tilde{S}_2$ against quenched $\mathbb{E}S_{L/2}$ at $t = 2L$; the two columns are $p = 0.16$ and $p = 0.22$.

	$t = 4L$	$t = 2L$
crossings $p_\times(L, 2L)$ of I_3		
$(16, 32)$	0.1481(12)	0.1527(18)
$(24, 48)$	0.1530(24)	0.1529(10)
$(32, 64)$	0.1565(10)	0.1535(13)
$(48, 96)$	0.1577(11)	0.1585(11)
$(64, 128)$	0.1587(7)	0.1586(5)
$(96, 192)$	0.1602(5)	0.1592(7)
$(128, 256)$	0.1596(5)	0.1607(7)
$(192, 384)$	0.1600(5)	0.1602(6)
$(256, 512)$	–	0.1596(4)
collapse of I_3: p_c, ν		
purification locator ($\tau = 1$, $L \ge 64$, independent seeds): 0.1601, 1.25(2)		
$L \ge 32$	0.1589, 1.23(1)	0.1590, 1.26(1)
$L \ge 48$	0.1591, 1.23(2)	0.1593, 1.27(2)
$L \ge 64$	0.1594, 1.18(3)	0.1595, 1.26(4)
$L \ge 96$	0.1595, 1.10(7)	0.1594, 1.20(4)
α from $S_{L/2} = \alpha \ln L + b$ at p_c		
$L \ge 16$	1.583(38)	1.586(34)
$L \ge 32$	1.586(51)	1.595(52)
$L \ge 48$	1.602(65)	1.622(65)
$L \ge 64$	1.630(66)	1.657(70)
joint fit, $L \ge 48$	1.519(33)	1.509(26)
annealed $\tilde{S}_2$ / quenched $\mathbb{E}S_{L/2}$ ($t = 2L$)		
$L = 16$, $p = 0.16$; $p = 0.22$	4.13/3.13(2)	3.10/2.30(2)
$L = 32$, $p = 0.16$; $p = 0.22$	6.98/4.23(2)	4.43/2.54(2)

proposition because membership is essentially this definition. *Membership.* Given $(\mathcal{C}, R, \mathrm{out})$, replace every mid-circuit measurement by a CNOT into a fresh ancilla (deferred measurement; Ref. [39], Sec. 4.4), which leaves the joint distribution of record and output and the conditional state given the record unchanged, because no gate of $\mathcal{C}$ depends on an outcome; then compute the flag "record register = R" into one further ancilla by X gates on the positions where R has a 0 and a multiply controlled Toffoli, decomposed into Toffolis with ancillas. Postselecting on flag = 1 is postselecting on R: $\Pr[\mathrm{flag} = 1] = p_R > 0$ and $\Pr[\mathrm{out} = 1 \mid \mathrm{flag} = 1] = \Pr[\mathrm{out} = 1 \mid R]$. This is a PostBQP computation with the promised gap, and the construction is polynomial-time. *Hardness.* Let $\Pi \in \mathrm{PostBQP}$ with circuits Q_x. Map x to the CTD instance (Q_x, R, out) with R the one-bit terminal record "post = 1". Then $p_R = \Pr[\mathrm{post} = 1] > 0$ and $\Pr[\mathrm{out} = 1 \mid R]$ has the promised gap, so YES maps to YES and NO to NO: a promise-preserving polynomial-time many-one reduction, which uses only terminal postselection. *PP.* PostBQP=PP [38] makes every PP language many-one reducible to CTD. For the extension, CTD$\in$PostBQP$\subseteq$PP: with the counting quantities above, $g = 2^h\big(\Pr[\mathrm{out} = 1, R] - 2\Pr[\mathrm{out} = 0, R]\big)$ is an integer-valued GapP function; on YES instances $g \ge 0$, because $\Pr[\mathrm{out} = 1 \mid R] \ge 2/3$ means $\Pr[\mathrm{out} = 1, R] \ge 2\Pr[\mathrm{out} = 0, R]$, and on NO instances $g \le -2^h p_R < 0$. Hence the GapP function $2g + 1$ is positive exactly on the YES instances among the promised ones, and a PP machine decides the sign of a GapP function [37]; this machine decides every promised instance and gives some answer on the others. *Architecture.* Inserting SWAP gates (basis permutations, in the gate set) compiles a polynomial-size circuit to a one-dimensional nearest-neighbor circuit with polynomial overhead and unchanged conditional probabilities, and the postselection flag can be moved to a fixed wire. Multiple postse-

lection bits may be retained as R or combined reversibly into one flag as above. □

Corollary 2 (Conditional expectation)**.** *Any randomized algorithm that, given $(\mathcal{C}, R)$ with $p_R > 0$, outputs an estimate of the one-qubit projector expectation $\mathrm{Tr}(|1\rangle\langle 1|\rho_R)$ with additive error at most $1/7$ with probability at least $2/3$ decides CTD with bounded error; hence PP⊆BPP$^{\mathcal{O}}$ for any oracle $\mathcal{O}$ performing this estimation, i.e. the estimation problem is PP-hard under randomized (bounded-error) polynomial-time Turing reductions. With inverse-polynomial additive error given in unary, the problem belongs to* $\mathrm{FP}^{\mathrm{PP}}$.

Proof. Hardness. On a YES instance the true value is at least $2/3$, so an estimate with error at most $1/7$ is at least $2/3 - 1/7 = 11/21 > 1/2$; on a NO instance it is at most $1/3 + 1/7 = 10/21 < 1/2$. Comparing the estimate with $1/2$ therefore decides CTD whenever the estimator succeeds, i.e. with probability at least $2/3$, and the failure probability of the estimator is the bounded error of the reduction. If a smaller failure probability δ is wanted, take the median of r independent estimates: the median errs only if at least half of the estimates err, and since each errs independently with probability at most $1/3$, Hoeffding's inequality bounds this by $e^{-2r(1/6)^2} = e^{-r/18}$, so $r = \lceil 18 \ln(1/\delta) \rceil$ repetitions suffice. Composing with the many-one reduction of Proposition 16 gives PP⊆BPP$^{\mathcal{O}}$. *Membership.* For the Hadamard–Toffoli gate set, $N_1 = 2^h \Pr[\mathrm{out} = 1, R]$ and $N = 2^h \Pr[R]$ are GapP functions of the instance, $N \geq 1$ on the promise, and the ratio N_1/N is the target. For a rational threshold u/v with u, v of polynomial size, $vN_1 - uN$ is again in GapP, and a PP oracle decides whether it is positive, i.e. whether $N_1/N > u/v$ [37]. Bisection over the thresholds $u/2^s$, $u = 0, \ldots, 2^s$, with $s = \lceil \log_2(1/\varepsilon) \rceil$ oracle queries, locates N_1/N in an interval of length $2^{-s} \leq \varepsilon$; for ε inverse-polynomial and given in unary this is polynomially many queries, so the estimation problem is in $\mathrm{FP}^{\mathrm{PP}}$. This membership argument is not asserted for an arbitrary algebraic gate set without a separate algebraic-number comparison theorem. □

Corollary 3 (Conditional purity decision)**.** *Let Conditional-Purity Decision (CPD) be the promise problem: given $(\mathcal{C}, R)$ with $p_R > 0$ and a designated qubit o', decide whether $\mathrm{Tr}\,\rho^2_{\mathsf{o}',R} \geq 13/18$ (YES) or $\leq 5/9$ (NO). CPD is PostBQP-complete, hence PP-hard, under promise-preserving polynomial-time many-one reductions; the same holds for any pair of thresholds $1 - 2^{-n}$ (YES) versus $1/2 + 2^{-n}$ (NO).*

Proof. Reduction from CTD. Given $(\mathcal{C}, R, \mathrm{out})$, append two ancillas, o prepared in $|+\rangle$ by a Hadamard and $\mathsf{o}' = |0\rangle$, and one Toffoli gate with controls $(\mathrm{out}, \mathsf{o})$ and target o'. Write the conditional state given R as $\rho_R = \sum_{u,u' \in \{0,1\}} |u\rangle\langle u'|_{\mathrm{out}} \otimes \rho_{uu'}$ with $\mathrm{Tr}\,\rho_{11} = p_1 = \Pr[\mathrm{out} = 1 \mid R]$ and $\mathrm{Tr}\,\rho_{00} = 1 - p_1$. The Toffoli leaves the $u = 0$ branch $|0\rangle|+\rangle|0\rangle$ of $(\mathrm{out}, \mathsf{o}, \mathsf{o}')$ unchanged and maps the $u = 1$ branch to $|1\rangle(|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}$. Tracing out everything except o' removes the cross terms $u \neq u'$, which carry the factor $\langle u'|u\rangle = 0$ from the output qubit, and gives

$$\rho_{\mathsf{o}',R} = (1 - p_1)|0\rangle\langle 0| + p_1 \tfrac{1}{2} I,$$
$$\mathrm{Tr}\,\rho^2_{\mathsf{o}',R} = h(p_1) = 1 - p_1 + \tfrac{1}{2} p_1^2.$$

Since $h'(p) = p - 1 \leq 0$, h decreases on $[0, 1]$, with $h(1/3) = 13/18$ and $h(2/3) = 5/9$; hence CTD-NO instances ($p_1 \leq 1/3$) map to CPD-YES and CTD-YES instances ($p_1 \geq 2/3$) to CPD-NO. Thus the complement of CTD reduces to CPD, and PostBQP is closed under complement (negate the output qubit), so CPD is PostBQP-hard. For the second pair of thresholds, first amplify the CTD instance: run r copies of the circuit, postselect on all r records (a single terminal flag, of probability $p_R^r > 0$) and output the majority of the r output bits; the copies are independent given their records, so by Hoeffding's inequality the conditional error is at most $e^{-r/18}$ and $r = O(n)$ copies give $p_1 \leq 2^{-n}$ or $p_1 \geq 1 - 2^{-n}$ [38]. Then $h(2^{-n}) = 1 - 2^{-n} + 2^{-2n-1} \geq 1 - 2^{-n}$ and $h(1 - 2^{-n}) = \frac{1}{2} + 2^{-2n-1} \leq \frac{1}{2} + 2^{-n}$. *Membership.* Run two copies of the circuit, postselect both on R (probability $p_R^2 > 0$), which leaves $\rho^{\otimes 2}_{\mathsf{o}',R}$ on the two o' registers, and perform a SWAP test (a Hadamard, a controlled SWAP, which is a basis permutation, and a Hadamard on the control); it accepts with probability $(1 + \mathrm{Tr}\,\rho^2_{\mathsf{o}',R})/2$, at least $31/36$ on YES and at most $7/9 = 28/36$ on NO instances. Majority over $O(1)$ repetitions, all postselected, brings this to the $2/3$ versus $1/3$ convention. □

This is a worst-case statement for universal circuit descriptions, not a hardness theorem for Haar-random circuits at fixed monitoring rate. It also does not establish hardness of the scalar Born average $\sum_R p_R S_n(\rho_{A,R})$: an exponentially rare postselected branch is suppressed in that average.

For fixed integers $n \geq 2$, $k \geq 0$ and $j \geq \max\{nk, 1\}$, all $O(1)$, the bivariate moments $M(j, k) = \sum_R p_R^j x_R^k$ of Proposition 7 can be estimated to additive inverse-polynomial error in BQP: run j coherent copies of the circuit with deferred measurement, measure the j record registers and keep the sample only if all records agree (an event of probability $\sum_R p_R^j$, after which the system is in $\rho_R^{\otimes j}$), then apply a Hadamard test of k disjoint n-cycles on the copies of A, whose expectation is x_R^k; the estimator is bounded, so $O(\varepsilon^{-2})$ repetitions suffice. This covers the symmetric data $\widetilde{F}_n(k) = M(nk, k)$ of Theorem 6 and the record-collision moments of Remark 1, but not the Born average $M(1, 1) = \sum_R p_R x_R$ or any $j < nk$, for which one copy per power of p_R is not available. It gives neither a relative-error algorithm for exponentially small moments nor a BPP algorithm: nonnegative tensor entries alone do not prove efficient sampling, contraction, or rapid mixing. Relative-error estimation is hard already at the level of record probabilities: estimates of p_R and

of the extended-record probability $\Pr[R, \mathrm{out} = 1]$ to relative error $1/100$ determine $\mathrm{Tr}(|1\rangle\langle 1|\rho_R)$ to within $1/40$ and decide CTD; likewise an estimate of $p_R^2 \,\mathrm{Tr}\,\rho^2_{\mathsf{o}',R}$ to additive error $p_R^2/100$, with p_R given, decides CPD. The conditional Rényi-2 entropy $S_2(\rho_{\mathsf{o}',R}) = -\log \mathrm{Tr}\,\rho^2_{\mathsf{o}',R}$ inherits the CPD promise with thresholds $\log(18/13)$ and $\log(9/5)$ under the identity reduction.

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